

Quantum Field Theory II

派蒙の素论派学习笔记

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Path integral method

1. Path integral in quantum mechanics

1.1 Functional method

Functional S maps a function to numbers, i.e.

$$S : x(t) \rightarrow S[x(t)]. \quad (1.1)$$

As we all know, the Hamilton operator in 1-dim is $H = \frac{p^2}{2m} + V(x)$, now let us consider the amplitude for motion of a particle from x_a to x_b . For this, we introduce the evolution operator $U(x_a, x_b; t)$ in Heisenburg picture, so we have

$$U(x_a, x_b; T) = \langle x_b | e^{-iHT/\hbar} | x_a \rangle, \quad (1.2)$$

where the Hamiltonian **is already quantized**, but in the following calculation, the Hamiltonian that in the path integral **is classical** due to the path integral itself is the process of quantization. **So we don't need to quantize anything inside of the integration.** So the path integral is directly defined from this operator. The basic idea is to cut the space and time into small pieces, and add them together, i.e.,

$$U(x_a, x_b; T) = \int \mathcal{D}x(t) e^{iS[x(t)]/\hbar}, \quad (1.3)$$

where $S[x(t)]$ is the action that depends on the paths. In QM, we can expand it as

$$S[x(t)] = \int_0^T dt L = \int_0^T dt \left[\frac{1}{2} m v^2 - V(x) \right]. \quad (1.4)$$

And the integration measure $\mathcal{D}x(t)$ accounts for all possible paths from x_a to x_b , i.e.,

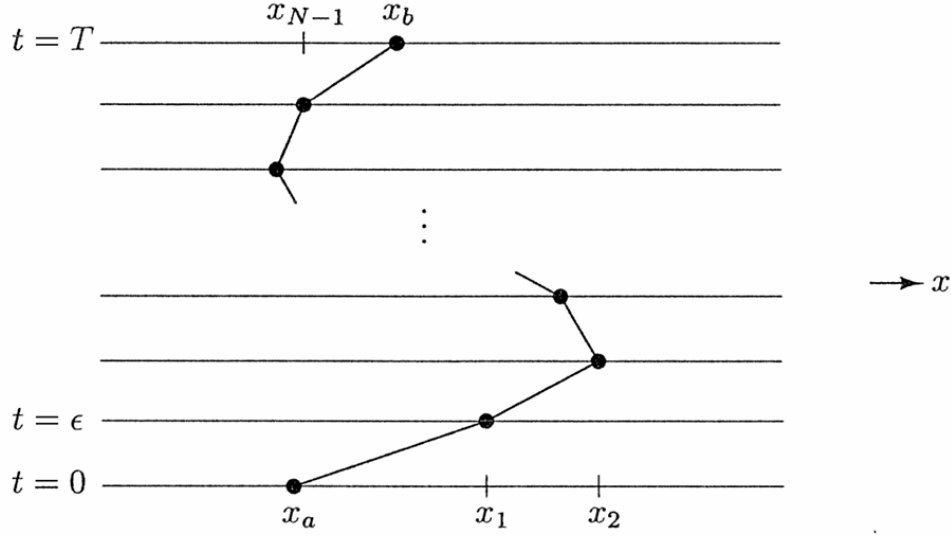
$$\sum_{\substack{\text{all paths from} \\ x_a \text{ to } x_b}} \longrightarrow \int \mathcal{D}x(t). \quad (1.5)$$

Now we need to prove that the definition we gave that can derive the same quantization result. Firstly, we discretize the time interval from 0 to T in steps $\Delta t = \varepsilon$:

$$S = \int_0^T dt \left[\frac{1}{2} m \dot{x}^2 - V(x) \right] = \sum_k \left\{ \frac{m}{2} \frac{(x_{k+1} - x_k)^2}{\varepsilon} - \varepsilon V \left(\frac{x_{k+1} + x_k}{2} \right) \right\}. \quad (1.6)$$

The path integral build up from N discrete time intervals

$$\mathcal{D}x(t) \equiv \frac{1}{c(\varepsilon)} \int \frac{dx_1}{c(\varepsilon)} \frac{dx_2}{c(\varepsilon)} \cdots \frac{dx_N}{c(\varepsilon)} = \frac{1}{c(\varepsilon)} \prod_{k=1}^N \int_{-\infty}^{+\infty} \frac{dx_k}{c(\varepsilon)}, \quad (1.7)$$



where $\frac{1}{c(\varepsilon)}$ is an undetermined normalization factor. Now we consider the evolution operator $U(x_a, x_b; T)$ evolve from $T - \varepsilon$ to T in the interval $[x_a, x_b]$, which is the last step in (1.7):

$$\begin{aligned} U(x_a, x_b; T) &= \langle x_b | e^{-iH[(T-\varepsilon)+\varepsilon]/\hbar} | x_a \rangle = \int dx' \langle x_b | e^{-iH\varepsilon/\hbar} | x' \rangle \langle x' | e^{-iH(T-\varepsilon)/\hbar} | x_a \rangle \\ &= \int_{-\infty}^{+\infty} \frac{dx'}{c(\varepsilon)} \exp \left[\frac{i}{\hbar} \frac{m(x_b - x')^2}{2\varepsilon} - \frac{i}{\hbar} \varepsilon V \left(\frac{x_b + x'}{2} \right) \right] U(x_a, x'; T - \varepsilon) \end{aligned} \quad (1.8)$$

Notice that we can expand $U(x_a, x'; T - \varepsilon)$ around x_b

$$U(x_a, x'; T - \varepsilon) = \left(1 + (x' - x_b) \frac{\partial}{\partial x_b} + \frac{1}{2} (x' - x_b)^2 \frac{\partial^2}{\partial x_b^2} + \dots \right) U(x_a, x_b; T - \varepsilon) \quad (1.9)$$

and for $\varepsilon \rightarrow 0$, the potential term can also expand as

$$\exp \left\{ \frac{i}{\hbar} \varepsilon V \left(\frac{x_b - x_{x'}}{2} \right) \right\} \stackrel{x' \rightarrow x_b}{=} e^{\frac{i}{\hbar} \varepsilon V(x_b)} = 1 - \frac{i}{\hbar} \varepsilon V(x_b) + \dots \quad (1.10)$$

Thus, (1.9) can be rewritten as

$$\begin{aligned} U(x_a, x_b; T) &= \int_{-\infty}^{+\infty} \frac{dx'}{c(\varepsilon)} \exp \left\{ \frac{i}{\hbar} \frac{m(x_b - x')^2}{\varepsilon} \right\} \left[1 - \frac{i\varepsilon}{\hbar} V(x_b) + \dots \right] \\ &\quad \times \left[1 + (x' - x_b) \frac{\partial}{\partial x_b} + \frac{1}{2} (x' - x_b)^2 \frac{\partial^2}{\partial x_b^2} + \dots \right] U(x_a, x_b; T - \varepsilon) \end{aligned} \quad (1.11)$$

To do this integration the Gauss integration formulas are very useful in the next:

$$\int_{-\infty}^{+\infty} d\xi e^{-b\xi^2} = \sqrt{\frac{\pi}{b}}; \quad (1.12)$$

$$\int_{-\infty}^{+\infty} d\xi \xi e^{-b\xi^2} = 0; \quad (1.13)$$

$$\int_{-\infty}^{+\infty} d\xi \xi^2 e^{-b\xi^2} = \frac{1}{2b} \sqrt{\frac{\pi}{b}}. \quad (1.14)$$

Thus,

$$U(x_a, x_b; T) = \frac{1}{c(\varepsilon)} \sqrt{\frac{\pi}{b}} \left[1 - \frac{i\varepsilon}{\hbar} V(x_b) + \frac{1}{4b} \frac{\partial^2}{\partial x_b^2} + \mathcal{O}(\varepsilon^2) \right] U(x_a, x_b; T - \varepsilon), \quad (1.15)$$

where $b = \frac{-im}{2\varepsilon\hbar}$. When $\varepsilon \rightarrow 0$,

$$\begin{aligned} U(x_a, x_b; T) &= \overbrace{\frac{1}{c(\varepsilon)} \sqrt{\frac{\pi}{b}}}^{=1} U(x_a, x_b; T - \varepsilon) \\ \Rightarrow c(\varepsilon) &= \sqrt{\frac{\pi}{b}}. \end{aligned} \quad (1.16)$$

So we have the Schrödinger equation for the evolution operator

$$i\hbar \frac{U(x_a, x_b; T) - U(x_a, x_b; T - \varepsilon)}{\varepsilon} = \left(V(x_b) - \frac{\hbar^2}{2m} \frac{\partial^2}{\partial x_b^2} \right) U(x_a, x_b; T - \varepsilon). \quad (1.17)$$

So we have derived the familiar differential equation with the canonical quantization did. Now we also need to consider the boundary condition. For $T \rightarrow 0$, canonical quantization requires $U(x_a, x_b; 0) = \langle x_b | \mathbb{1} | x_a \rangle \delta(x_a - x_b)$. For path integral, we firstly take a piece of time slice, since $T \rightarrow 0$ there is no difference for labeling the space interval, so we can just take $\delta x = x_b - x_a$:

$$\lim_{\varepsilon \rightarrow 0} \frac{1}{c(\varepsilon)} \exp \left[\frac{im}{\hbar} \frac{(x_b - x_a)^2}{\varepsilon} \right] = \delta(x_b - x_a) \quad (1.18)$$

Thus,

$$\lim_{\varepsilon \rightarrow 0} \int \mathcal{D}x(t) e^{iS[x(t)]/\hbar} = \lim_{\varepsilon \rightarrow 0} \frac{1}{c(\varepsilon)} \exp \left[\frac{im}{\hbar} \frac{(x_b - x_a)^2}{\varepsilon} \right] \rightarrow \delta(x_a - x_b) \quad (1.19)$$

- Path integral formulas quantize system through summation/integration over all possible paths; weighted by the phase $e^{-iS[x(t)]/\hbar}$.
- In the limit $\hbar \rightarrow 0$ (classical physics) path integral is dominated by classical path with $\delta S = 0$.

1.2 Generalization to quantum system with coordinates $\{q_i\}$ and conjugate momenta $\{p_i\}$

Now we consider the general case for a quantum system with Hamiltonian $H(q, p)$, and evolution operator $U(q_a, q_b; T) = \langle q_b | e^{-iHT} | q_a \rangle$. Split up $f(q, p) = e^{-iHT}$ in N time intervals $\Delta t = \varepsilon$

$$e^{-iHT} = \prod_{k=1}^N e^{-iH\varepsilon} = e^{-iH\varepsilon} \mathbb{1} e^{-iH\varepsilon} \mathbb{1} \dots \mathbb{1} e^{-iH\varepsilon}, \quad (1.20)$$

with $\mathbb{1} = \left(\prod_i \int dq_k^i \right) |q_k\rangle \langle q_k|$. At k -th position,

$$\langle q_{k+1} | f(q) | q_k \rangle \xrightarrow{\varepsilon=0} \langle q_{k+1} | 1 - i\varepsilon H + \mathcal{O}(\varepsilon^2) | q_k \rangle \quad (1.21)$$

For a function that its variable is only q , we have

$$\langle q_{k+1} | f(q) | q_k \rangle = f(q_k) \prod_i \delta(q_k^i - q_{k+1}^i) = f\left(\frac{q_k + q_{k+1}}{2}\right) \left(\prod_i \int \frac{dp_i}{2\pi} \right) \exp \left\{ i \sum_i p_k^i (q_{k+1}^i - q_k^i) \right\}, \quad (1.22)$$

and for a function that its variable is only p , we have

$$\langle q_{k+1} | f(p) | q_k \rangle = \left(\prod_i \int \frac{dp_k^i}{2\pi} \right) f(p_k) \exp \left\{ i \sum_i p_k^i (q_{k+1}^i - q_k^i) \right\}. \quad (1.23)$$

In general, we normally face the more complicated case that using the Weyl ordering is convenient to keep the properties we have in the above, i.e.,

$$\langle q_{k+1} | \frac{1}{4} (q^2 p^2 + 2qp^2 q + p^2 q^2) | q_k \rangle = \left(\frac{q_{k+1} + q_k}{2} \right)^2 \langle q_k + 1 | p^2 | q_k \rangle, \quad (1.24)$$

where the Weyl order is defined as

$$W(p, q) = \frac{1}{2} \{x, p\}. \quad (1.25)$$

Assume $H(p, q)$ is Weyl ordered, i.e.,

$$\langle q_{k+1} | H(p, q) | q_k \rangle = \left(\prod_i \int \frac{dp_k^i}{2\pi} \right) H \left(\frac{q_{k+1} + q_k}{2}, p_k \right) \exp \left\{ i \sum_i p_k^i (q_{k+1}^i - q_k^i) \right\}. \quad (1.26)$$

Thus

$$\langle q_{k+1} | e^{-iH\varepsilon} | q_k \rangle = \left(\prod_i \int \frac{dp_k^i}{2\pi} \right) \exp \left\{ -i\varepsilon H \left(\frac{q_{k+1} + q_k}{2}, p_k \right) \right\} \exp \left\{ i \sum_i p_k^i (q_{k+1}^i - q_k^i) \right\}. \quad (1.27)$$

Take $q_0 = q_a$ and $q_N = q_b$, the integration are

$$U(q_0, q_N; T) = \left(\prod_{i,k} \int dq_k^i \int \frac{dp_k^i}{2\pi} \right) \exp \left\{ i \sum_k \left[\sum_i p_k^i (q_{k+1}^i - q_k^i) - \varepsilon H \left(\frac{q_{k+1} + q_k}{2}, p_k \right) \right] \right\}. \quad (1.28)$$

In path integral notation

$$U(q_0, q_N; T) = \left(\prod_{i,k} \int \mathcal{D}q^i(t) \int \mathcal{D}p^i(t) \right) \exp \left\{ i \int_0^T dt \left[\sum_i p^i \dot{q}^i - H(p, q) \right] \right\}. \quad (1.29)$$

2. Path integral in the Scalar field

For a scalar field $\phi(\vec{x}, t)$, we can simply change the variables $\vec{q}_i(t) \mapsto \phi(\vec{x}, t)$, the evolution operator is then became:

$$U(\phi_a, \phi_b; T) = \langle \phi_b(\vec{x}) | e^{-i\hat{H}T} | \phi_a(\vec{x}) \rangle. \quad (1.30)$$

The action S and Hamiltonian H for the scalar field are defined as:

$$S = \int d^4x \left(\frac{1}{2} (\partial \cdot \phi)^2 - V(\phi) \right), \quad (1.31)$$

$$H = \int d^3x \left(\frac{1}{2} \pi^2 + \frac{1}{2} (\nabla \phi)^2 + V(\phi) \right). \quad (1.32)$$

The path integral form can then be simply seen from (1.29) as

$$\langle \phi_b(\vec{x}) | e^{-i\hat{H}T} | \phi_a(\vec{x}) \rangle = \int_{\phi(0, \vec{x})=\phi_a}^{\phi(T, \vec{x})=\phi_b} \mathcal{D}\phi(x) \int \mathcal{D}\pi(x) \exp \left\{ i \int_0^T dt \int d^3x \left(\pi \dot{\phi} - \frac{1}{2} \pi^2 - \frac{1}{2} (\nabla \phi)^2 - V(\phi) \right) \right\} \quad (1.33)$$

$$= \int \mathcal{D}\phi(x) \int \mathcal{D}(\pi(x) - \dot{\phi}) \exp \left\{ i \int_0^T dt \int d^3x \left[-\frac{1}{2} (-2\pi \dot{\phi} + \pi^2 + \dot{\phi}^2 - \dot{\phi}^2) - \frac{1}{2} (\nabla \phi)^2 - V(\phi) \right] \right\} \quad (1.34)$$

To cancel the $\int \mathcal{D}\pi(x)$, we change the variable $\mathcal{D}\pi(x) \rightarrow \mathcal{D}(\pi(x) - \partial_0\phi)$, i.e.,

$$\langle \phi_b(\vec{x}) | e^{-i\hat{H}T} | \phi_a(\vec{x}) \rangle = \int \mathcal{D}\phi(x) \int \mathcal{D}(\pi(x) - \dot{\phi}) \exp \left\{ i \int_0^T dt \int d^3x \left[-\frac{1}{2} (\pi - \dot{\phi})^2 + \frac{1}{2} (\partial\phi)^2 - V(\phi) \right] \right\} \quad (1.35)$$

$$= \int \mathcal{D}\phi(x) \int \mathcal{D}\tilde{\pi}(x) \exp \left\{ i \int_0^T dt \int d^3x \left[-\frac{1}{2} \tilde{\pi}(x)^2 + \frac{1}{2} (\partial\phi)^2 - V(\phi) \right] \right\}. \quad (1.36)$$

Thus

$$\langle \phi_b(\vec{x}) | e^{-i\hat{H}T} | \phi_a(\vec{x}) \rangle = \mathcal{C} \int \mathcal{D}\phi(x) \exp \left\{ i \int_0^T dt \int d^3x \left[\frac{1}{2} (\partial\phi)^2 - V(\phi) \right] \right\}, \quad (1.37)$$

where

$$\mathcal{C} = \int \mathcal{D}\tilde{\pi}(x) \exp \left\{ -i \int_0^T dt \int d^3x \frac{1}{2} \tilde{\pi}(x)^2 \right\} \quad (1.38)$$

So we can directly see it is

$$\langle \phi_b(\vec{x}) | e^{-i\hat{H}T} | \phi_a(\vec{x}) \rangle = \mathcal{C} \int_{\phi(0,\vec{x})=\phi_a}^{\phi(T,\vec{x})=\phi_b} \mathcal{D}\phi(x) \exp \left\{ i \int d^4x \mathcal{L} \right\} \quad (1.39)$$

For some reason that we will consider later, the normalization constant $\mathcal{C}$ is ignored in the next calculation.

2.1 Correlation Function

Now we consider the two-point correlation function. For that, we firstly consider the structure

$$\int_{\phi(-T,\vec{x})=\phi_a}^{\phi(T,\vec{x})=\phi_b} \mathcal{D}\phi(x) \phi(x_1) \phi(x_2) \exp \left\{ i \int_{-T}^T d^4x \mathcal{L} \right\}, \quad (1.40)$$

where we assume $x_2^0 > x_1^0$. Now the boundary condition changes from $-T$ to T . So we can separate the integration in three area $-T \rightarrow x_1^0$, $x_1^0 \rightarrow x_2^0$ and $x_2^0 \rightarrow T$ such that we can fix the field at x_1^0 and x_2^0 as $\phi_1(\vec{x})$ and $\phi_2(\vec{x})$ respectively. This operation means no matter how the field evolve from $-T$ to T , we can always separate the integration with 2 steps: First, fixing the field at these two time point and integrate the three area. Second, we integrate all possible situation of these two field.

$$\int \mathcal{D}\phi(x) = \int \mathcal{D}\phi_1(\vec{x}) \int \mathcal{D}\phi_2(\vec{x}) \int_{\substack{\phi(x_1^0,\vec{x})=\phi_1(\vec{x}) \\ \phi(x_2^0,\vec{x})=\phi_2(\vec{x})}} \mathcal{D}\phi(x). \quad (1.41)$$

So we have

$$\begin{aligned} & \int_{\phi(-T,\vec{x})=\phi_a}^{\phi(T,\vec{x})=\phi_b} \mathcal{D}\phi(x) \phi(x_1) \phi(x_2) \exp \left\{ i \int_{-T}^T d^4x \mathcal{L} \right\} \\ &= \int \mathcal{D}\phi_1(\vec{x}) \int \mathcal{D}\phi_2(\vec{x}) \int \mathcal{D}\phi(x) \phi(x_1) \phi(x_2) \exp \left\{ i \int_{-T}^T d^4x \mathcal{L} \right\}. \end{aligned} \quad (1.42)$$

Because we have fixed the time point x_1^0 and x_2^0 , so $\phi(x_1)$ and $\phi(x_2)$ can be directly picked from the integration $\int \mathcal{D}\phi$ and becomes $\phi_1(\vec{x}_1)$ and $\phi_2(\vec{x}_2)$, i.e.,

$$\begin{aligned} & \int_{\phi(-T, \vec{x})=\phi_a}^{\phi(T, \vec{x})=\phi_b} \mathcal{D}\phi(x) \phi(x_1) \phi(x_2) \exp \left\{ i \int_{-T}^T d^4x \mathcal{L} \right\} \\ &= \int \mathcal{D}\phi_1(\vec{x}) \int \mathcal{D}\phi_2(\vec{x}) \phi_1(\vec{x}_1) \phi_2(\vec{x}_2) \left[\int \mathcal{D}\phi(x) \exp \left\{ i \int_{-T}^T d^4x \mathcal{L} \right\} \right] \end{aligned} \quad (1.43)$$

$$= \int \mathcal{D}\phi_1(\vec{x}) \int \mathcal{D}\phi_2(\vec{x}) \phi_1(\vec{x}_1) \phi_2(\vec{x}_2) \left[\langle \phi_b | e^{-i\hat{H}(2T)} | \phi_a \rangle \right] \quad (1.44)$$

$$= \int \mathcal{D}\phi_1(\vec{x}) \int \mathcal{D}\phi_2(\vec{x}) \phi_1(\vec{x}_1) \phi_2(\vec{x}_2) \langle \phi_b | e^{-i\hat{H}(T-x_2^0)} | \phi_2 \rangle \langle \phi_2 | e^{-i\hat{H}(x_2^0-x_1^0)} | \phi_1 \rangle \langle \phi_1 | e^{-i\hat{H}(x_1^0+T)} | \phi_a \rangle. \quad (1.45)$$

Using $\hat{\phi}_S(\vec{x})|\phi_i\rangle = \phi_i(\vec{x})|\phi_i\rangle$ and $\int \mathcal{D}\phi_i |\phi_i\rangle \langle \phi_i| = \mathbb{1}$:

$$\int_{\phi(-T, \vec{x})=\phi_a}^{\phi(T, \vec{x})=\phi_b} \mathcal{D}\phi(x) \phi(x_1) \phi(x_2) \exp \left\{ i \int_{-T}^T d^4x \mathcal{L} \right\} = \langle \phi_b | e^{-i\hat{H}(T-x_2^0)} \hat{\phi}_S(\vec{x}_2) e^{-i\hat{H}(x_2^0-x_1^0)} \hat{\phi}_S(\vec{x}_1) e^{-i\hat{H}(x_1^0+T)} | \phi_a \rangle. \quad (1.46)$$

Defining the Heisenberg picture operator $\hat{\phi}_H(x) = e^{i\hat{H}x^0} \hat{\phi}_S(\vec{x}) e^{-i\hat{H}x^0}$:

$$\int_{\phi(-T, \vec{x})=\phi_a}^{\phi(T, \vec{x})=\phi_b} \mathcal{D}\phi(x) \phi(x_1) \phi(x_2) \exp \left\{ i \int_{-T}^T d^4x \mathcal{L} \right\} = \langle \phi_b | e^{-i\hat{H}T} \hat{\phi}_H(x_2) \hat{\phi}_H(x_1) e^{-i\hat{H}T} | \phi_a \rangle. \quad (1.47)$$

For a general time ordering $x_2^0 > x_1^0$, this becomes the time-ordered product:

$$\langle \phi_b | e^{-i\hat{H}T} T \{ \hat{\phi}_H(x_1) \hat{\phi}_H(x_2) \} e^{-i\hat{H}T} | \phi_a \rangle. \quad (1.48)$$

Using the ground state projection

$$\lim_{T \rightarrow \infty(1-i\epsilon)} e^{-i\hat{H}T} | \phi_a \rangle = \lim_{T \rightarrow \infty(1-i\epsilon)} \sum_n e^{-iE_n T} | n \rangle \langle n | \phi_a \rangle = \langle \Omega | \phi_a \rangle e^{-iE_0 \infty(1-i\epsilon)} | \Omega \rangle, \quad (1.49)$$

and similarly,

$$\lim_{T \rightarrow \infty(1-i\epsilon)} \langle \phi_b | e^{-i\hat{H}T} = \lim_{T \rightarrow \infty(1-i\epsilon)} \sum_n \langle \phi_b | n \rangle \langle n | e^{-iE_n T} = \langle \phi_b | \Omega \rangle \langle \Omega | e^{-iE_0 \infty(1-i\epsilon)}. \quad (1.50)$$

So we have

$$\langle \phi_b | e^{-i\hat{H}T} T \{ \hat{\phi}_H(x_1) \hat{\phi}_H(x_2) \} e^{-i\hat{H}T} | \phi_a \rangle = \langle \phi_b | \phi_a \rangle \langle \Omega | T \{ \hat{\phi}_H(x_1) \hat{\phi}_H(x_2) \} | \Omega \rangle. \quad (1.51)$$

Thus, we can finally reach the correlation function:

$$\langle \Omega | T \{ \hat{\phi}_H(x_1) \hat{\phi}_H(x_2) \} | \Omega \rangle = \frac{\langle \phi_b | e^{-i\hat{H}T} T \{ \hat{\phi}_H(x_1) \hat{\phi}_H(x_2) \} e^{-i\hat{H}T} | \phi_a \rangle}{\langle \phi_b | \phi_a \rangle}. \quad (1.52)$$

So in the end, we have

$$\langle \Omega | T \{ \hat{\phi}_H(x_1) \hat{\phi}_H(x_2) \} | \Omega \rangle = \lim_{T \rightarrow \infty(1-i\epsilon)} \frac{\int \mathcal{D}\phi \phi(x_1) \phi(x_2) e^{i \int_{-T}^T d^4x \mathcal{L}}}{\int \mathcal{D}\phi e^{i \int_{-T}^T d^4x \mathcal{L}}}. \quad (1.53)$$

2.2 Feynman Rules

Now we want to get the Feynman rules from (1.53), so we need to calculate the numerator and the denominator respectively. For that, we firstly consider the free action of the scalar field:

$$S_0 = \int d^4x \mathcal{L}_0 = \int d^4x \left(\frac{1}{2} (\partial_\mu \phi)^2 - \frac{1}{2} m^2 \phi^2 \right). \quad (1.54)$$

The general Gaussian integration formulas could be useful in the next:

$$\int d^n x e^{-x_i A_{ij} x_j} = \text{const.} \cdot (\det A)^{-1/2}; \quad (1.55)$$

$$\int \mathcal{D}\phi e^{\frac{i}{2} \int d^4x \phi (\partial^2 - m^2) \phi} = \text{const.} \cdot (\det(\partial^2 - m^2))^{-1/2}. \quad (1.56)$$

The idea is firstly discretizing the path integral and then take the limit to get it back to the continuous form. So we work in the lattice $\{x_i\}$, where we can define the discrete field to replace the continuous variable:

$$\mathcal{D}\phi = \prod_i d\phi(x_i). \quad (1.57)$$

And the field can be expressed by the Fourier series:

$$\phi(x_i) = \frac{1}{V} \sum_n e^{-ik_n \cdot x_i} \phi(k_n); \quad k_n^\mu = \frac{2\pi n^\mu}{L}, \quad (1.58)$$

where n^μ is integers and $|k^\mu| < \pi/\varepsilon$, ε is the distance between two close lattice points. $V = L^4$ is the volume of the space. Because of $\phi(x_n)$ is real, so

$$\begin{aligned} \phi^*(x_i) &= \frac{1}{V} \sum_n e^{+ik_n^* \cdot x_i} \phi^*(k_n) \\ &= \frac{1}{V} \sum_n e^{-i(-k_n) \cdot x_i} \phi(-k_n), \end{aligned} \quad (1.59)$$

which means $\phi^*(k_n) = \phi(-k_n)$. So the integral measurement becomes

$$\mathcal{D}\phi(x) = \prod_i d\phi(x_i) = |\det(M)| \prod_n d\phi(k_n) = \prod_{k_n^0 > 0} d\text{Re } \phi(k_n) d\text{Im } \phi(k_n), \quad (1.60)$$

where M is the Fourier transformation matrix, so the Jacobian determinant is 1. And the last equality is because $d\phi(k_n)$ is an area measurement in the complex plane, so this small piece of area measurement is $d\text{Re } \phi(k_n) d\text{Im } \phi(k_n)$. And the free action in momentum space can be rewritten as

$$S_0 = -\frac{1}{V} \sum_{k_n^0 > 0} \frac{1}{2} (m^2 - k_n^2) |\phi_n|^2 = -\frac{1}{V} \sum_{k_n^0 > 0} (m^2 - k_n^2) [(\text{Re } \phi_n)^2 + (\text{Im } \phi_n)^2]. \quad (1.61)$$

Therefore, we can together all these parts to express the denominator of (1.53), then using the Gaussian integration formula to calculate the integral:

$$\int \mathcal{D}\phi e^{iS_0} = \prod_{k_n^0 > 0} \left(\int d\text{Re } \phi_n \exp \left[-\frac{i}{V} (m^2 - k_n^2) (\text{Re } \phi_n)^2 \right] \right) \left(\int d\text{Im } \phi_n \exp \left[-\frac{i}{V} (m^2 - k_n^2) (\text{Im } \phi_n)^2 \right] \right) \quad (1.62)$$

$$= \prod_{k_n^0 > 0} \sqrt{\frac{-i\pi V}{m^2 - k_n^2}} \sqrt{\frac{-i\pi V}{m^2 - k_n^2}} \quad (1.63)$$

$$= \prod_{k_n^0 > 0} \sqrt{\frac{-i\pi V}{m^2 - k_n^2}} \sqrt{\frac{-i\pi V}{m^2 - (-k_n)^2}} \quad (1.64)$$

$$= \prod_{\text{all } k_n} \sqrt{\frac{-i\pi V}{m^2 - k_n^2}}, \quad (1.65)$$

where the last step is counting from the upper half plane to the whole complex k_n plane. And to avoid the divergence, we take $k^0 \rightarrow k^0(1 + i\epsilon)$ in the denominator. For the expression for the propagator, we still need to calculating $\phi(x_1)\phi(x_2)$ for the numerator of (1.53), so we have

$$\phi(x_1)\phi(x_2) = \frac{1}{V^2} \sum_m e^{-ik_m x_1} \phi_m \sum_l e^{-ik_l x_2} \phi_l. \quad (1.66)$$

So the numerator becomes

$$\begin{aligned} \int \mathcal{D}\phi \phi(x_1)\phi(x_2) e^{i \int_{-T}^T d^4x \mathcal{L}} &= \frac{1}{V^2} \sum_{m,l} e^{-i(k_m x_1 + k_l x_2)} \left(\prod_{n|k_n^0 > 0} \int d\text{Re } \phi_n d\text{Im } \phi_n \right) \\ &\quad \times (\text{Re } \phi_m + i \text{Im } \phi_m)(\text{Re } \phi_l + i \text{Im } \phi_l) \\ &\quad \times \exp \left\{ -\frac{i}{V} \sum_{n|k_n^0 > 0} (m^2 - k_n^2) [(\text{Re } \phi_n)^2 + (\text{Im } \phi_n)^2] \right\}. \end{aligned} \quad (1.67)$$

Notice that for $m \neq l$ and $m \neq -l$, the term

$$(\text{Re } \phi_m + i \text{Im } \phi_m)(\text{Re } \phi_l + i \text{Im } \phi_l) \exp \{ (m^2 - k_n^2) [(\text{Re } \phi_n)^2 + (\text{Im } \phi_n)^2] \} \quad (1.68)$$

is **(odd function) $\times$ (even function) = (odd function)**, so the integral is 0. And for $m = -l$, because of $\phi(-k) = \phi^*(k)$, which means $\phi_l = \phi_{-m} = \phi_m^*$, the production term is then

$$(\phi_m) \cdot (\phi_l) = \phi_m \cdot \phi_m^* = |\phi_m|^2 = (\text{Re } \phi_m)^2 + (\text{Im } \phi_m)^2. \quad (1.69)$$

So the integration is $\sim \int x^2 e^{-ax^2}$, which is non-zero. Thus only the terms with $k_m = -k_l$ contribute to the integral. Finally, we apply the Gaussian integration formula to (1.67), then we have

$$\int \mathcal{D}\phi \phi(x_1)\phi(x_2) e^{i \int_{-T}^T d^4x \mathcal{L}} = \frac{1}{V^2} \sum_m e^{-ik_m(x_1 - x_2)} \left(\prod_{k_n^0 > 0} \frac{-i\pi V}{m^2 - k_n^2} \right) \frac{-iV}{m^2 - k_m^2 - i\epsilon} \quad (1.70)$$

$$= \frac{1}{V^2} \sum_m e^{-ik_m(x_1 - x_2)} \left(\prod_{\text{all } k_n} \sqrt{\frac{-i\pi V}{m^2 - k_n^2}} \right) \frac{-iV}{m^2 - k_m^2 - i\epsilon} \quad (1.71)$$

Finally, we got the Feynman propagator through dividing (1.71) by (1.65):

$$\langle 0|T\{\phi(x_1)\phi(x_2)\}|0\rangle = \frac{\int \mathcal{D}\phi \phi(x_1)\phi(x_2)e^{i\int_{-T}^T d^4x \mathcal{L}}}{\int \mathcal{D}\phi e^{i\int_{-T}^T d^4x \mathcal{L}}} \quad (1.72)$$

$$= \frac{\frac{1}{V^2} \sum_m e^{-ik_m(x_1-x_2)} \left(\prod_{\text{all } k_n} \sqrt{\frac{-i\pi V}{m^2-k_n^2}} \right) \frac{-iV}{m^2-k_m^2-i\epsilon}}{\prod_{\text{all } k_n} \sqrt{\frac{-i\pi V}{m^2-k_n^2}}} \quad (1.73)$$

$$= \sum_m \frac{1}{V} \frac{i}{k_m^2 - m^2 - i\epsilon} e^{-ik_m(x_1-x_2)} \quad (1.74)$$

$$= \int \frac{d^4k}{(2\pi)^4} \frac{i}{k^2 - m^2 + i\epsilon} e^{-ik(x_1-x_2)} = D_F(x_1 - x_2) \quad (1.75)$$

2.3 Generating Functional

Now, we consider a more convenient and general method to calculate the Feynman rules. But it might be not very direct to understanding. First, we introduce $J(x)$ as a source in the outside to define the generating functional. For this, it's convenient to review some properties of the functional derivative:

$$\frac{\delta}{\delta J(x)} J(y) = \delta^{(4)}(x-y) \implies \frac{\delta}{\delta J(x)} \int d^4y J(y)\phi(y) = \phi(x). \quad (1.76)$$

And some basic examples of the functional derivative:

1. Exponential function:

$$\frac{\delta}{\delta J(x)} \exp \left\{ i \int d^4y J(y)\phi(y) \right\} = i\phi(x) \exp \left\{ i \int d^4y J(y)\phi(y) \right\}. \quad (1.77)$$

2. Source terms containing derivatives (obtained through integration by parts):

$$\frac{\delta}{\delta J(x)} \int d^4y \partial_\mu J(y) V^\mu(y) = -\frac{\delta}{\delta J(x)} \int d^4y J(y) \partial_\mu V^\mu(y) = -\partial_\mu V^\mu(x). \quad (1.78)$$

3. Functional composite differentiation (chain rule):

$$\frac{\delta F[\phi]}{\delta J(x)} = \int d^4z \frac{\delta F[\phi]}{\delta \phi(z)} \frac{\delta \phi(z)}{\delta J(x)} \quad \left(\text{Generalization of } \frac{\partial x_i(z_k)}{\partial y_j} = \sum_k \frac{\partial x_i}{\partial z_k} \frac{\partial z_k}{\partial y_j} \right). \quad (1.79)$$

Define the **Generating Functional** $Z[J]$ as:

$$Z[J] = \int \mathcal{D}\phi \exp \left\{ i \int d^4x (\mathcal{L} + J(x)\phi(x)) \right\}. \quad (1.80)$$

Using $Z[J]$, we can conveniently calculate the n -point correlation function:

$$\langle \Omega | T\{\phi(x_1)\phi(x_2)\} | \Omega \rangle = \frac{1}{Z[0]} \left(-i \frac{\delta}{\delta J(x_1)} \right) \left(-i \frac{\delta}{\delta J(x_2)} \right) Z[J] \Big|_{J=0} \quad (1.81)$$

2.4 Feynman Rules for the Free Theory

For a free scalar field, the Lagrangian is

$$\mathcal{L}_0 = \frac{1}{2} [(\partial_\mu \phi)(\partial^\mu \phi) + (-m^2 + i\epsilon)\phi^2] \quad (1.82)$$

$$= \frac{1}{2} [\partial_\mu(\phi \partial^\mu \phi) - (\phi \partial^2 \phi) + (-m^2 + i\epsilon)\phi^2] \quad (1.83)$$

$$= \frac{1}{2} \phi(-\partial^2 - m^2 + i\epsilon)\phi. \quad (1.84)$$

Define the operator $K_x = -\partial^2 - m^2 + i\epsilon$, and the action with source can be written as:

$$S_J = \int d^4x (\mathcal{L}_0 + J\phi) = \int d^4x \left(\frac{1}{2} \phi K_x \phi + J\phi \right) \quad (1.85)$$

$$= \int d^4x d^4y \left(\frac{1}{2} \phi(x) K(x, y) \phi(y) \right) + \int d^4x J(x) \phi(x), \quad (1.86)$$

where $K(x, y) = \delta^{(4)}(x - y) K_y$. Now the generating functional is:

$$Z_0[J] = \int \mathcal{D}\phi \exp \left\{ i \int d^4x \left(\frac{1}{2} \phi K \phi + J\phi \right) \right\}. \quad (1.87)$$

Now we change the variable $\phi \rightarrow \phi' = \phi + K^{-1}J$, so the measurement keeps unchanged $\mathcal{D}\phi' = \mathcal{D}\phi$:

$$Z_0[J] = \int \mathcal{D}\phi' \left| \det \frac{\delta\phi'}{\delta\phi} \right| \exp \left\{ i \int d^4x \left[\frac{1}{2} (\phi' - K^{-1}J) K (\phi' - K^{-1}J) + J(\phi' - K^{-1}J) \right] \right\} \quad (1.88)$$

$$= \left(\int \mathcal{D}\phi' \exp \left\{ \frac{i}{2} \int d^4x \phi' K \phi' \right\} \right) \exp \left\{ -\frac{i}{2} \int d^4x d^4y J(x) K^{-1}(x, y) J(y) \right\} \quad (1.89)$$

$$= Z_0[J = 0] \exp \left\{ -\frac{i}{2} \int d^4x d^4y J(x) K^{-1}(x, y) J(y) \right\}, \quad (1.90)$$

where $\left| \det \frac{\delta\phi'}{\delta\phi} \right| = 1$. And notice for

$$J(y) = K(x, y) (K^{-1}(x, y) J(y)) \quad (1.91)$$

$$= (-\partial_x^2 - m^2 + i\epsilon) K^{-1}(x, y) \delta^{(4)}(x - y) J(y) \quad (1.92)$$

$$= (-\partial_x^2 - m^2 + i\epsilon) K^{-1}(x, y) J(x). \quad (1.93)$$

Thus, K^{-1} satisfies the Green's function (except an $-i$ factor in the right hand side):

$$(-\partial_x^2 - m^2 + i\epsilon) K^{-1}(x, y) = \delta^{(4)}(x - y). \quad (1.94)$$

Take a sharp look! So we can directly realize that the Feynman propagator $-iD_F(x - y)$ also satisfies this equation, so we realize that $K^{-1}(x - y)$ is nothing but $-iD_F(x - y)$. So the **generating functional in free theory** can be then expressed as

$$Z_0[J] = Z_0[0] \exp \left\{ -\frac{1}{2} \int d^4x d^4y J(x) D_F(x - y) J(y) \right\}. \quad (1.95)$$

So the correlation function can be obtained from (1.81):

$$\langle 0 | T \{ \phi(x_1) \phi(x_2) \} | 0 \rangle = \frac{1}{Z_0[0]} \left(-i \frac{\delta}{\delta J(x_1)} \right) \left(-i \frac{\delta}{\delta J(x_2)} \right) Z_0[J] \Big|_{J=0} \quad (1.96)$$

$$= \left(-i \frac{\delta}{\delta J(x_1)} \right) \left(-i \frac{\delta}{\delta J(x_2)} \right) \exp \left\{ -\frac{1}{2} \int d^4x d^4y J(x) D_F(x-y) J(y) \right\} \Big|_{J=0} \quad (1.97)$$

$$= \left(-\frac{\delta}{\delta J(x_1)} \right) \left[-\frac{1}{2} \left(\int d^4y D_F(x_2-y) J(y) + \int d^4x J(x) D_F(x-x_2) \right) \right] \times \exp \left\{ -\frac{1}{2} \int d^4x d^4y J(x) D_F(x-y) J(y) \right\} \Big|_{J=0} \quad (1.98)$$

$$= \frac{\delta}{\delta J(x_1)} \left[\int d^4y D_F(y-x_2) J(y) \right] \frac{Z_0[J]}{Z_0[0]} \Big|_{J=0} \quad (1.99)$$

$$= D_F(x_1-x_2) \frac{Z_0[J]}{Z_0[0]} \Big|_{J=0} + \left[\int d^4y D_F(y-x_2) J(y) \right] \left[\int d^4y D_F(y-x_1) J(y) \right] \frac{Z_0[J]}{Z_0[0]} \Big|_{J=0} \quad (1.100)$$

$$= D_F(x_1-x_2). \quad (1.101)$$

Technically, the last step we do, which impose the boundary condition to cancel the second term is not very strict. In general, one can not impose the boundary condition before actually calculating the integration. More precisely, the reason why we have confidence to do so is based on the property that we can let $f \in L^2(\mathbb{R}^{1,3})$ be fixed, and define a functional

$$F_f : L^2(\mathbb{R}^{1,3}) \rightarrow \mathbb{C}, \quad F_f[J] := \int_{\mathbb{R}^{1,3}} f(x) J(x) dx. \quad (1.102)$$

Then F_f is a continuous linear functional on $L^2(\mathbb{R}^{1,3})$. Indeed, by the Cauchy–Schwarz inequality,

$$|F_f[J]| \leq \|f\|_{L^2} \|J\|_{L^2}, \quad (1.103)$$

so F_f is bounded and hence continuous. In particular, for any sequence (or net) $J_n \rightarrow 0$ in $L^2(\mathbb{R}^{1,3})$, one has

$$\lim_{n \rightarrow \infty} F_f[J_n] = F_f[0] = 0. \quad (1.104)$$

Therefore, the notation $F_f[J] \big|_{J=0}$ is to be understood as the evaluation of the functional F_f at the zero element of $L^2(\mathbb{R}^{1,3})$, i.e.

$$F_f[J] \big|_{J=0} := F_f(0). \quad (1.105)$$

Moreover, by continuity, this evaluation coincides with the limit along any sequence $J_n \rightarrow 0$ in the L^2 topology:

$$F_f(0) = \lim_{J \rightarrow 0} F_f[J], \quad (1.106)$$

where the limit is taken with respect to the L^2 norm.

2.5 Interaction theory: ϕ^4 theory as an example

For theories involving interactions, such as $\mathcal{L} = \mathcal{L}_0 - \frac{\lambda}{4!} \phi^4$ and $\mathcal{L}_0 = \frac{1}{2}(\partial_\mu \phi)^2 - \frac{1}{2}m^2 \phi^2$, the generating functional can be written as:

$$Z[J] = \int \mathcal{D}\phi \exp \left\{ i \int d^4x \left(-\frac{\lambda}{4!} \phi^4 + \mathcal{L}_0 + J(x) \phi(x) \right) \right\}. \quad (1.107)$$

Notice that

$$\begin{aligned}\frac{\delta}{\delta J(x)} Z_0[J] &= \frac{\delta}{\delta J(x)} \int \mathcal{D}\phi \exp \left\{ -i \int d^4x \mathcal{L}_0 + J(x)\phi(x) \right\} \\ &= \phi(x) \int \mathcal{D}\phi \exp \left\{ -i \int d^4x \mathcal{L}_0 + J(x)\phi(x) \right\} \\ &= \phi(x) Z_0[J].\end{aligned}\tag{1.108}$$

So we have

$$\left(-i \frac{\delta}{\delta J(x)} \right)^4 Z_0[J] = \phi^4 Z_0[J],\tag{1.109}$$

which means

$$Z[J] = \int \mathcal{D}\phi \left\{ 1 + \left(-i \int d^4x \frac{\lambda}{4!} \phi^4 \right) + \left(-i \int d^4x \frac{\lambda}{4!} \phi^4 \right)^2 + \dots \right\} \exp \left\{ -i \int d^4x [\mathcal{L}_0 + J(x)\phi(x)] \right\}\tag{1.110}$$

$$\begin{aligned}&= \int \mathcal{D}\phi \left\{ 1 + \left[-i \int d^4x \frac{\lambda}{4!} \left(-i \frac{\delta}{\delta J(x)} \right)^4 \right] + \left[-i \int d^4x \frac{\lambda}{4!} \left(-i \frac{\delta}{\delta J(x)} \right)^4 \right]^2 + \dots \right\} \\ &\quad \times \exp \left\{ -i \int d^4x [\mathcal{L}_0 + J(x)\phi(x)] \right\}\end{aligned}\tag{1.111}$$

$$= \exp \left\{ -i \int d^4x \frac{\lambda}{4!} \left(-i \frac{\delta}{\delta J(x)} \right)^4 \right\} \int \mathcal{D}\phi \exp \left\{ -i \int d^4x [\mathcal{L}_0 + J(x)\phi(x)] \right\}\tag{1.112}$$

$$= \exp \left\{ -i \int d^4x \frac{\lambda}{4!} \left(-i \frac{\delta}{\delta J(x)} \right)^4 \right\} Z_0[J],\tag{1.113}$$

where we expanded the exponential term of interaction part as the perturbed series, and using the derivative relation (1.109). So we link the interaction theory generating functional with the free theory generating functional:

$$Z[J] = \left(1 - \frac{i\lambda}{4!} \int d^4x \left(-i \frac{\delta}{\delta J(x)} \right)^4 + \dots \right) Z_0[J].\tag{1.114}$$

So the correlation function can be expanded by the free propagator as

$$\langle \Omega | T \{ \phi(x_1) \phi(x_2) \} | \Omega \rangle = \frac{1}{Z[0]} \left(-i \frac{\delta}{\delta J(x_1)} \right) \left(-i \frac{\delta}{\delta J(x_2)} \right) Z[J] \Big|_{J=0}\tag{1.115}$$

$$= \frac{1}{Z[0]} \left(-i \frac{\delta}{\delta J(x_1)} \right) \left(-i \frac{\delta}{\delta J(x_2)} \right) \left(1 - \frac{i\lambda}{4!} \int d^4x \left(-i \frac{\delta}{\delta J(x)} \right)^4 + \dots \right) Z_0[J] \Big|_{J=0}\tag{1.116}$$

$$= \frac{1}{Z[0]} \left[(-i)^2 \frac{\delta^2 Z_0[J]}{\delta J(x_1) \delta J(x_2)} + \left(-\frac{i\lambda}{4!} \right) \int d^4x (-i)^6 \frac{\delta^6 Z_0[J]}{\delta J(x)^4 \delta J(x_1) \delta J(x_2)} + \dots \right] \Big|_{J=0}\tag{1.117}$$

$$= \frac{1}{Z[0]} \langle 0 | T \{ \phi(x_1) \phi(x_2) \} | 0 \rangle + \frac{1}{Z[0]} \left(-i \frac{\lambda}{4!} \right) \int d^4x \langle 0 | T \{ \phi^4(x) \phi(x_1) \phi(x_2) \} | 0 \rangle + \dots\tag{1.118}$$

$$= \frac{1}{Z[0]} \langle 0 | T \left\{ \phi(x_1) \phi(x_2) \exp \left\{ i \int d^4x \left(-\frac{\lambda}{4!} \phi^4 \right) \right\} \right\} | 0 \rangle.\tag{1.119}$$

Comparing with the result we got in the canonical quantization

$$\langle \Omega | T \{ \phi(x_1) \phi(x_2) \} | \Omega \rangle = \frac{\langle 0 | T \{ \phi(x_1) \phi(x_2) \exp \{ i \int d^4x \left(-\frac{\lambda}{4!} \phi^4 \right) \} \} | 0 \rangle}{\langle 0 | T \{ \exp \{ i \int d^4x \left(-\frac{\lambda}{4!} \phi^4 \right) \} \} | 0 \rangle}, \quad (1.120)$$

we find in the interaction case, $Z[0]$ has different meaning comparing with $Z_0[0]$ in the free case, i.e.,

$$Z[0] = T \exp \left\{ i \int d^4x \left(-\frac{\lambda}{4!} \phi^4 \right) \right\}, \quad (1.121)$$

which included all the **Vacuum Bubbles**.

2.6 Other Type of Generating functional

We define the **Energy Functional** for the connected graphs $E[J]$ as

$$E[J] = i \ln Z[J]. \quad (1.122)$$

So we have

$$\frac{\delta E[J]}{\delta J(x)} = \frac{i}{Z[0]} \frac{\delta Z[J]}{\delta J(x)} = -\frac{\langle \Omega | \phi(x) | \Omega \rangle}{\langle \Omega | \Omega \rangle} = -\phi_{\text{classical}}(x), \quad (1.123)$$

where the second step is because $\delta Z[J] = \langle \Omega | i\phi(x) | \Omega \rangle$. And in physics, it is defined as the **Classical Field**. And the definition of $E[J]$ can be easily seen that it's very similar with the free energy in classical statistical mechanics with the partition function $Z[J]$. So using the Legendre transformation, we can define the **Effective Action**, which looks like the Gibbs free energy $\Gamma[\phi_{cl}]$ as

$$\Gamma[\phi_{cl}] = -E[J] - \int d^4x J(x) \phi_{cl}(x). \quad (1.124)$$

Base on this, we can see there is a symmetric relation:

$$\frac{\delta \Gamma[\phi_{cl}]}{\delta \phi_{cl}(x)} = -J(x); \quad \frac{\delta E[J]}{\delta J(x)} = -\phi_{cl}(x). \quad (1.125)$$

Taking another functional derivative with respect to $J(y)$ on both sides yields:

$$\Rightarrow \frac{\delta}{\delta J(y)} \frac{\delta}{\delta \Phi_{cl}(x)} \Gamma[\Phi_{cl}] = -\frac{\delta J(x)}{\delta J(y)} = -\delta(x - y). \quad (1.126)$$

Using the chain rule for functional differentiation $\frac{\delta}{\delta J(y)} = \int d^4z \frac{\delta \Phi_{cl}(z)}{\delta J(y)} \frac{\delta}{\delta \Phi_{cl}(z)}$, we obtain the following identity:

$$-\delta(x - y) = \int d^4z \frac{\delta \Phi_{cl}(z)}{\delta J(y)} \frac{\delta^2 \Gamma[\Phi_{cl}]}{\delta \Phi_{cl}(z) \delta \Phi_{cl}(x)}. \quad (1.127)$$

We can rewrite the first term as $\frac{\delta \Phi_{cl}(z)}{\delta J(y)} = \frac{\delta^2 W[J]}{\delta J(y) \delta J(z)}$, where $W[J]$ is the generating functional of connected Green's functions (denoted as $E[J]$ in the notes):

$$-\delta(x - y) = \int d^4z \underbrace{\frac{\delta^2 W[J]}{\delta J(y) \delta J(z)}}_{-iG(y-z)} \underbrace{\frac{\delta^2 \Gamma[\Phi_{cl}]}{\delta \Phi_{cl}(z) \delta \Phi_{cl}(x)}}_{iG^{-1}(z-x)}. \quad (1.128)$$

Here, the first part corresponds to the connected two-point function $-iG(y-z)$, and the second part corresponds to the inverse propagator $iG^{-1}(z-x)$. Thus, the equation simplifies to:

$$\delta(x-y) = \int d^4z [-iG(y-z)] [iG^{-1}(z-x)]. \quad (1.129)$$

The n -th derivative of $E[J]$ gives the connected correlation function

$$\left. \frac{\delta^n E[J]}{\delta J(x_1) \dots \delta J(x_n)} \right|_{J=0} = (i)^{n+1} \langle \Omega | T\{\phi(x_1) \dots \phi(x_n)\} | \Omega \rangle_{\text{connect}}. \quad (1.130)$$

And the n -th derivative of $\Gamma[\phi_{cl}]$ gives the correlation function for 1PI graphs

$$\frac{\delta^n \Gamma[\phi_{cl}]}{\delta \phi_{cl}(x_1) \dots \delta \phi_{cl}(x_n)} = -i \langle \Omega | T\{\phi(x_1) \dots \phi(x_n)\} | \Omega \rangle_{1PI}. \quad (1.131)$$

3. Quantization of EM Field in Path Integral Formalism

The classical action for the electromagnetic field is given by:

$$S[A] = -\frac{1}{4} \int d^4x F_{\mu\nu} F^{\mu\nu}, \quad (1.132)$$

where $F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$. Integrating by parts, we can rewrite the action as:

$$S[A] = \frac{1}{2} \int d^4x A_\mu D^{\mu\nu} A_\nu, \quad \text{with} \quad D^{\mu\nu} = g^{\mu\nu} \square - \partial^\mu \partial^\nu. \quad (1.133)$$

In the path integral formulation, the vacuum-to-vacuum transition amplitude is given by:

$$\int \mathcal{D}A e^{iS[A]}. \quad (1.134)$$

But the problem the measure $\mathcal{D}A = \mathcal{D}A^0 \mathcal{D}A^1 \mathcal{D}A^2 \mathcal{D}A^3$ is **not** well defined. The integral is ill-defined because of the redundancy due to infinite gauge copies. Under a local gauge transformation:

$$A_\mu \rightarrow A_\mu^\alpha = A_\mu + \frac{1}{e} \partial_\mu \alpha(x), \quad (1.135)$$

where $\alpha(x)$ is an arbitrary scalar function. For a pure gauge choice where $A_\mu = \frac{1}{e} \partial_\mu \alpha$, we find that the field strength $F_{\mu\nu} = 0$, leading to:

$$S[A] = 0 \quad \Rightarrow \quad e^{iS[A]} = 1. \quad (1.136)$$

This implies that we have an unbounded path integral because the integrand does not decay along the gauge orbits.

3.1 Faddeev-Popov Quantization

Solution: We need to divide out the gauge degrees of freedom in the measure $\mathcal{D}A$. This procedure is known as the **Faddeev-Popov quantization**.

Let $G(A)$ be a function that fixes the gauge (for example, $G(A) = \partial_\mu A^\mu = 0$ for the Lorentz gauge). We want to define $\int \mathcal{D}A e^{iS[A]}$ with a constraint such that the path integral only sums over field configurations fulfilling $G(A) = 0$.

To motivate this, recall the standard Gauss integration with matrix M (rank N):

$$(\det M)^{-1/2} = (2\pi)^{-N/2} \int_{-\infty}^{+\infty} dx_1 \dots dx_N e^{-\frac{1}{2}x^T M x}. \quad (1.137)$$

Generalizing this to a functional determinant (Path integral):

$$(\det M)^{-1/2} = \int \mathcal{D}\phi e^{-\frac{1}{2} \int d^4x \phi(x) M(x) \phi(x)}, \quad (1.138)$$

where M is some differential operator, e.g., $M(x) = \square + m_\phi^2$.

Next, consider the Dirac δ -distribution on an N -dimensional space for an arbitrary function $\vec{g}(x_1, \dots, x_N)$:

$$\int_{-\infty}^{+\infty} dx_1 \dots dx_N \delta^{(N)}[\vec{g}(x_1, \dots, x_N)] \det \left[\frac{\partial g_i}{\partial x_j} \right] = 1 \quad (1.139)$$

In 1-dimension, this reduces to the property:

$$\int dx \delta(ax) a = \int dx \frac{1}{a} \delta(x) a = 1. \quad (1.140)$$

Extending this to a functional Dirac distribution for gauge fields, we have the Faddeev-Popov identity:

$$\int \mathcal{D}\alpha(x) \underbrace{\delta[G(A^\alpha)] \det \left[\frac{\delta G(A^\alpha)}{\delta \alpha} \right]}_{\text{Faddeev-Popov det.}} = 1. \quad (1.141)$$

Quantization of electrodynamics proceeds by inserting this identity “1” into the path integral:

$$I = \int \mathcal{D}A e^{iS[A]} = \int \mathcal{D}A (1) e^{iS[A]} = \int \mathcal{D}A \left(\int \mathcal{D}\alpha \delta[G(A^\alpha)] \det \left[\frac{\delta G(A^\alpha)}{\delta \alpha} \right] \right) e^{iS[A]}. \quad (1.142)$$

If $G(A^\alpha)$ is the Lorentz gauge condition, we have:

$$G(A^\alpha) = \partial_\mu A^\mu + \frac{1}{e} \square \alpha. \quad (1.143)$$

Then the functional derivative becomes:

$$\det \left[\frac{\delta G(A^\alpha)}{\delta \alpha} \right] = \det \left(\frac{\square}{e} \right). \quad (1.144)$$

Notice that in quantum electrodynamics, this determinant is invariant under gauge transformations and independent of A_μ .

Due to gauge invariance, we have $\mathcal{D}A = \mathcal{D}A^\alpha$ and $S[A] = S[A^\alpha]$. Therefore, the path integral can be rewritten as:

$$\begin{aligned} I &= \int \mathcal{D}\alpha \det \left[\frac{\delta G(A^\alpha)}{\delta \alpha} \right] \int \mathcal{D}A \delta[G(A^\alpha)] e^{iS[A]} \\ &= \int \mathcal{D}\alpha \left(\det \left[\frac{\delta G(A^\alpha)}{\delta \alpha} \right] \int \mathcal{D}A^\alpha \delta[G(A^\alpha)] e^{iS[A^\alpha]} \right) \\ &= \int \mathcal{D}\alpha \tilde{I}. \end{aligned} \quad (1.145)$$

By dropping the infinite volume factor of the gauge group $\int \mathcal{D}\alpha$, we define the physical path integral for QED as:

$$\tilde{I} = \det \left[\frac{\delta G(A^\alpha)}{\delta \alpha} \right] \int \mathcal{D}A^\alpha \delta[G(A^\alpha)] e^{iS[A]}. \quad (1.146)$$

3.2 Gauge Fixing and the Photon Propagator

Our next task is to absorb the Dirac constraint $\delta[G(A^\alpha)]$ into a modified action. Let us generalize the gauge fixing condition by setting:

$$G(A^\alpha) = \partial^\mu A_\mu(x) - \omega(x), \quad (1.147)$$

where $\omega(x)$ is an arbitrary scalar function. We also set the determinant remains independent of ω as the form:

$$\det \left[\frac{\delta G(A^\alpha)}{\delta \alpha} \right] = \det \left(\frac{\square}{e} \right). \quad (1.148)$$

So the functional integration $\tilde{I}$ becomes

$$\int \mathcal{D}A e^{-iS[A]} = \det \left(\frac{\square}{e} \right) \int \mathcal{D}A^\alpha \delta[\partial^\mu A_\mu - \omega(x)] e^{iS[A]}. \quad (1.149)$$

Since this formula is satisfied for any $\omega(x)$, so we can integrating over all possible functions $\omega(x)$ with a Gaussian weight centered at $\omega = 0$:

$$I' = N(\xi) \int \mathcal{D}\omega e^{-i \int d^4x \frac{\omega(x)^2}{2\xi}} \tilde{I}, \quad (1.150)$$

where $N(\xi)$ is a normalization factor. Then we substitute $\tilde{I}$ and explicitly show the delta function:

$$\begin{aligned} I' &= N(\xi) \det \left(\frac{\square}{e} \right) \int \mathcal{D}A e^{iS[A]} \int \mathcal{D}\omega e^{-i \int d^4x \frac{\omega^2}{2\xi}} \underbrace{\delta[\partial^\mu A_\mu - \omega]}_{\text{linear covariant gauge}} \\ &= N(\xi) \det \left(\frac{\square}{e} \right) \int \mathcal{D}A e^{iS[A]} e^{-\frac{i}{2\xi} \int d^4x (\partial^\mu A_\mu)^2} \\ &= N(\xi) \det \left(\frac{\square}{e} \right) \int \mathcal{D}A e^{iS'[A]}. \end{aligned} \quad (1.151)$$

The modified action $S'[A]$ becomes:

$$S'[A] = \frac{1}{2} \int d^4x \left[A_\mu D^{\mu\nu} A_\nu - \frac{1}{\xi} (\partial^\mu A_\mu)^2 \right] = \frac{1}{2} \int d^4x A_\mu \tilde{D}^{\mu\nu} A_\nu, \quad (1.152)$$

which implies the new differential operator:

$$\tilde{D}^{\mu\nu} = g^{\mu\nu} \square - \left(1 - \frac{1}{\xi} \right) \partial^\mu \partial^\nu. \quad (1.153)$$

From this modified action, we can derive the generating functional as

$$Z[J_\mu] = \int \mathcal{D}A \exp \left\{ i \int d^4x \left(\frac{1}{2} A_\mu \tilde{D}^{\mu\nu} A_\nu + J^\mu(x) A_\mu(x) \right) \right\}. \quad (1.154)$$

Then transform the field as $A_\mu \rightarrow A'_\mu = A_\mu + \tilde{D}_\mu^{-1\nu} J_\nu$, we have

$$Z[J_\mu] = \int \mathcal{D}A \exp \left\{ i \int d^4x \left(\frac{1}{2} (A'_\mu - \tilde{D}_\mu^{-1\lambda} J_\lambda) \tilde{D}^{\mu\nu} (A'_\nu - \tilde{D}_\nu^{-1\lambda} J_\lambda) + J^\mu(x) (A'_\mu - \tilde{D}_\mu^{-1\lambda} J_\lambda) \right) \right\} \quad (1.155)$$

$$= \left(\int \mathcal{D}A' \exp \left\{ \frac{i}{2} A'_\mu \tilde{D}^{\mu\nu} A'_\nu \right\} \right) \exp \left\{ -\frac{i}{2} \int d^4x d^4y (J_\mu(x) \tilde{D}^{-1\mu\nu} J_\nu(y)) \right\} \quad (1.156)$$

$$= Z[0] \exp \left\{ -\frac{i}{2} \int d^4x d^4y (J_\mu(x) \tilde{D}^{-1\mu\nu} J_\nu(y)) \right\} \quad (1.157)$$

$$= Z[0] \exp \left\{ -\frac{i}{2} \int d^4x d^4y (J_\mu(x) \tilde{G}^{\mu\nu} J_\nu(y)) \right\}, \quad (1.158)$$

where we have ignored the determinant due to $\det\left(\frac{\delta A'}{\delta A}\right) = 1$. And the photon propagator is the inverse of $\tilde{D}^{\mu\nu}$ in momentum space:

$$G^{\mu\nu}(x-y) = -i \int \frac{d^4k}{(2\pi)^4} e^{-ik(x-y)} \frac{g^{\mu\nu} - (1-\xi) \frac{k^\mu k^\nu}{k^2}}{k^2 + i\epsilon}. \quad (1.159)$$

The effective action then can be constructed as same as (1.124)

$$\Gamma[A_{cl}^\mu] = -iZ[J_\mu] - \int d^4x J_\mu(x) A_{cl}^\mu. \quad (1.160)$$

$S'[A]$ with $\tilde{D}^{\mu\nu}$ now has no problems with the longitudinal degrees of freedom for photons.

3.3 Standard Gauge Choices

Depending on the choice of the parameter ξ and conditions, we have different gauges in the linear covariant gauge family:

- $\xi = 0$: **Landau gauge**, corresponding to the condition $\partial^\mu A_\mu = 0$.
- $\xi = 1$: **Feynman gauge**.
- $i = 1, 2, 3$, $\partial^i A_i = 0$: **Coulomb gauge**.
- $n^\mu A_\mu = 0$: **Axial gauge**.

Finally, the path integral in QED for calculating an n -point correlator is written as:

$$\langle \Omega | T \hat{O}(A) | \Omega \rangle = \lim_{T \rightarrow \infty(1-i\epsilon)} \frac{\int \mathcal{D}A \hat{O}(A) \exp \left\{ i \int_{-T}^T d^4x \left[\mathcal{L}(x) - \frac{1}{2\xi} (\partial^\mu A_\mu)^2 \right] \right\}}{\int \mathcal{D}A \exp \left\{ i \int_{-T}^T d^4x \left[\mathcal{L}(x) - \frac{1}{2\xi} (\partial^\mu A_\mu)^2 \right] \right\}}. \quad (1.161)$$

4. Path Integral for Fermions

To define the path integral for fermions, we consider the transition amplitude:

$$Z = \int \mathcal{D}\psi \mathcal{D}\bar{\psi} \exp \left\{ i \int d^4x \mathcal{L}(\psi, \bar{\psi}) \right\}, \quad \mathcal{L} = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi. \quad (1.162)$$

Our task is to make sense of this integral, which requires treating fermion fields as Grassmann-valued quantities.

4.1 Grassmann Numbers and Algebra

Grassmann numbers θ satisfy the anti-commutation relation:

$$\{\theta_i, \theta_j\} = 0 \quad \Rightarrow \quad \theta^2 = 0. \quad (1.163)$$

A differential operator $\frac{d}{d\theta}$ is defined such that it also anti-commutes:

$$\left\{ \frac{d}{d\theta}, \theta \right\} = \frac{d}{d\theta} \theta + \theta \frac{d}{d\theta} 1 = 1. \quad (1.164)$$

Any Grassmann-valued function $f(\theta)$ can be defined through a power series. Due to $\theta^2 = 0$, the series terminates at the linear term:

$$f(\theta) = a + \beta\theta, \quad (1.165)$$

where a is a regular number (real/complex) and β is another Grassmann-valued quantity. Taking the derivative:

$$\frac{d}{d\theta} f(\theta) = \frac{d}{d\theta} (a + \beta\theta) = 0 + \frac{d}{d\theta} (\beta\theta) = -\beta \frac{d\theta}{d\theta} = -\beta. \quad (1.166)$$

Note the sign change because $\frac{d}{d\theta}$ must pass β . Other properties include:

$$\frac{d^2}{d\theta^2} f(\theta) = 0, \quad \frac{d}{d\theta} a = \left\{ \frac{d}{d\theta}, \beta \right\} = 0. \quad (1.167)$$

4.2 Berezin Integration

Integration over Grassmann variables (Berezin integration) is defined to be identical to differentiation:

$$\int d\theta \, 1 = 0, \quad \int d\theta \, \theta = 1. \quad (1.168)$$

For multiple Grassmann variables θ_i with internal symmetry ($i = 1, \dots, N$):

$$\left\{ \frac{d}{d\theta_i}, \theta_j \right\} = \delta_{ij}, \quad \left\{ \frac{d}{d\theta_i}, \frac{d}{d\theta_j} \right\} = 0, \quad \int d\theta_i = 0, \quad \int d\theta_i \, \theta_j = \delta_{ij}. \quad (1.169)$$

Example: Consider a two-dimensional integral:

$$\int d\theta_1 d\theta_2 \, \theta_1 \theta_2 = \int d\theta_1 \left(\int d\theta_2 \, \theta_1 \theta_2 \right) = \int d\theta_1 \, (-\theta_1) = -1. \quad (1.170)$$

4.3 Gaussian Integrals and Determinants

For a complex Grassmann pair $\eta, \bar{\eta}$, we represent the determinant of a matrix M_{ij} :

$$\int \left[\prod_{j=1}^N d\eta_j d\bar{\eta}_j \right] \exp \left(\sum_{k,l} \bar{\eta}_k M_{kl} \eta_l \right) = \det[M]. \quad (1.171)$$

Auxiliary Calculation (1D Example):

$$\int d\bar{\eta} d\eta \, e^{-\bar{\eta} b \eta} = \int d\bar{\eta} d\eta \, (1 - \bar{\eta} b \eta) = \int d\bar{\eta} d\eta \, (1 + \eta \bar{\eta} b) = \int d\bar{\eta} \, (\bar{\eta} b) = b. \quad (1.172)$$

Similarly, for the insertion of variables:

$$\int d\bar{\eta} d\eta \, \eta \bar{\eta} e^{-\bar{\eta} b \eta} = \int d\bar{\eta} d\eta \, \eta \bar{\eta} (1 - \bar{\eta} b \eta) = \int d\bar{\eta} d\eta \, \eta \bar{\eta} = 1 = \frac{1}{b} \cdot b. \quad (1.173)$$

4.4 Transformation of Variables

Under a linear transformation $\theta_i = U_{ij} \theta'_j$, the product of Grassmann variables transforms with the determinant:

$$\prod_{k=1}^N \theta'_k = \frac{1}{N!} \varepsilon^{i_1 \dots i_N} U_{i_1 j_1} \dots U_{i_N j_N} \theta_{j_1} \dots \theta_{j_N} = \det[U] \prod_{k=1}^N \theta_k. \quad (1.174)$$

This is inverse to the bosonic case where the Jacobian appears in the denominator.

4.5 Comparison: Bosonic vs. Fermionic Case

For a Hermitian matrix M and source w, ξ :

- **Bosonic Case:**

$$F(M, \omega) = \int d^N x \exp \left(-\frac{1}{2} x^T M x + \omega^T x \right) = \pi^{N/2} (\det M)^{-1/2} \exp \left(\frac{1}{2} \omega^T M^{-1} \omega \right). \quad (1.175)$$

- **Fermionic Case:**

$$F(M, \xi, \bar{\xi}) = \int \prod d\eta_i d\bar{\eta}_i \exp (\bar{\eta} M \eta + \bar{\xi} \eta + \bar{\eta} \xi) = \det[M] \exp (-\bar{\xi} M^{-1} \xi). \quad (1.176)$$

5. Path Integral Formulation for Dirac Spinors

A Dirac spinor $\psi_a(x)$ consists of Grassmann-valued quantities:

$$\psi(x) = \sum_{a=1}^4 \psi_a(x), \quad \text{where } \psi_a \text{ are Grassmann variables.} \quad (1.177)$$

The Dirac Lagrangian is $\mathcal{L}_{\text{Dirac}} = \bar{\psi}_a(x) (i\gamma_{ab}^\mu \partial_\mu - m\delta_{ab}) \psi_b(x)$. The vacuum amplitude is:

$$\int \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp \left[i \int d^4x \mathcal{L}(\bar{\psi}, \psi) \right] = \text{const} \cdot \det(i\not{\partial} - m). \quad (1.178)$$

5.1 Fermion Propagator

In the path integral formulation, the propagator is defined as the time-ordered product:

$$\langle 0 | T \{ \psi(x_1) \bar{\psi}(x_2) \} | 0 \rangle = \frac{\int \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp [i \int d^4x \bar{\psi}(i\not{\partial} - m)\psi] \psi(x_1) \bar{\psi}(x_2)}{\int \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp [i \int d^4x \bar{\psi}(i\not{\partial} - m)\psi]}. \quad (1.179)$$

The calculation yields:

$$\langle 0 | T \{ \psi(x_1) \bar{\psi}(x_2) \} | 0 \rangle = \frac{1}{\text{const} \cdot \det(i\not{\partial} - m)} \cdot \frac{\text{const} \cdot \det(i\not{\partial} - m)}{-i(i\not{\partial} - m)_{ab}} = \frac{i}{(i\not{\partial} - m)_{ab}} = S_{F,ab}(x_1 - x_2). \quad (1.180)$$

5.2 Generating Functional for Fermion Fields

Analogous to the scalar case, we introduce Grassmann-valued sources $\eta(x), \bar{\eta}(x)$:

$$Z[\eta, \bar{\eta}] = \int \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp \left\{ i \int d^4x [\mathcal{L}(\bar{\psi}, \psi) + \bar{\psi}(x)\eta(x) + \bar{\eta}(x)\psi(x)] \right\}. \quad (1.181)$$

The functional derivative with respect to these sources follows:

$$\frac{\delta \eta(x)}{\delta \eta(y)} = \delta(x - y). \quad (1.182)$$

By completing the square, we can express $Z[\eta, \bar{\eta}]$ in terms of the propagator:

$$Z[\eta, \bar{\eta}] = Z[\eta = 0, \bar{\eta} = 0] \exp \left[- \int d^4x d^4y \bar{\eta}(x) S_F(x - y) \eta(y) \right]. \quad (1.183)$$

5.3 Derivation of the Propagator

The propagator can finally be extracted via functional derivatives:

$$\langle 0 | T \{ \psi(x_1) \bar{\psi}(x_2) \} | 0 \rangle = \frac{1}{Z[0,0]} \left(-i \frac{\delta}{\delta \bar{\eta}(x_1)} \right) \left(-i \frac{\delta}{\delta \eta(x_2)} \right) Z[\eta, \bar{\eta}] \Big|_{\eta, \bar{\eta}=0} = S_F(x_1 - x_2). \quad (1.184)$$

6. Symmetries in Path Integral Formalism

The path integral approach provides a direct link between symmetries and physical observables through the following hierarchy: **Variational calculus** → **Schwinger-Dyson equations** → **Noether's theorem** → **Ward identities in QED**.

6.1 Schwinger-Dyson Equations

For simplicity, we still consider the scalar field theory as an example. By doing the partial integration, the action can be rewritten as:

$$S[\phi] = \int d^4x \mathcal{L}[\phi] = \int d^4x \left\{ \frac{1}{2} (\partial_\mu \phi)^2 - \frac{1}{2} m^2 \phi^2 \right\} = -\frac{1}{2} \int d^4x \phi (\square + m^2) \phi. \quad (1.185)$$

The n -point correlator is given by:

$$\langle \Omega | T \{ \phi(x_1) \dots \phi(x_n) \} | \Omega \rangle = \frac{\int \mathcal{D}\phi \phi_1 \dots \phi_n e^{iS[\phi]}}{\int \mathcal{D}\phi e^{iS[\phi]}}. \quad (1.186)$$

We derive classical equations of motion from a variation of fields: $\phi(x) \rightarrow \phi'(x) = \phi(x) + \varepsilon(x)$, i.e., $\delta\phi(x) = \varepsilon(x)$. The integration measurement is invariant due to ε is infinitesimal: $\mathcal{D}\phi' = \mathcal{D}\phi$. And the action changes as: $S[\phi'] = S[\phi + \varepsilon] = S[\phi] + \delta S$. So we have

$$\int \mathcal{D}\phi \phi_1 \dots \phi_n e^{iS[\phi]} = \int \mathcal{D}\phi' \phi'_1 \dots \phi'_n e^{iS[\phi + \varepsilon]} \quad (1.187)$$

$$= \int \mathcal{D}\phi e^{i(S[\phi] + \delta S)} (\phi_1 + \varepsilon_1) \dots (\phi_n + \varepsilon_n), \quad (1.188)$$

where we have denoted that $\phi_i \equiv \phi(x_i)$ and $\varepsilon_i \equiv \varepsilon(x_i)$. Then we expand ε to the first order, the exponential is $e^{i(S + \delta S)} \simeq e^{iS} (1 + i\delta S)$. And for δS , we have

$$\delta S = \int d^4y \left(\frac{\partial \mathcal{L}}{\partial \phi} \delta \phi + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \partial_\mu \delta \phi \right) \quad (1.189)$$

$$= \int d^4y \left[-m^2 \phi \varepsilon + \partial^\mu \phi \partial_\mu \varepsilon \right] \quad (1.190)$$

$$= - \int d^4y \left[\varepsilon(y) (\square_y + m^2) \phi(y) \right]. \quad (1.191)$$

Thus, put these result into (1.188), we have:

$$\begin{aligned} & \int \mathcal{D}\phi e^{i(S[\phi] + \delta S)} (\phi_1 + \varepsilon_1) \dots (\phi_n + \varepsilon_n) \\ &= \int \mathcal{D}\phi e^{iS[\phi]} \left[1 - i \int d^4y \varepsilon(y) (\square_y + m^2) \phi(y) \right] [(\phi_1 \dots \phi_n) + (\varepsilon_1 \phi_2 \dots \phi_n) + \dots + (\phi_1 \dots \varepsilon_n)] \end{aligned} \quad (1.192)$$

$$\begin{aligned} &= \int \mathcal{D}\phi e^{iS[\phi]} [(\phi_1 \dots \phi_n) + (\varepsilon_1 \phi_2 \dots \phi_n) + \dots + (\phi_1 \dots \varepsilon_n)] \\ &\quad - \int \mathcal{D}\phi e^{iS[\phi]} \int d^4y i [\varepsilon(y) (\square_y + m^2) \phi(y)] (\phi_1 \dots \phi_n) + \mathcal{O}(\varepsilon^2). \end{aligned} \quad (1.193)$$

Subtracting the first term $\int \mathcal{D}\phi \phi_1 \dots \phi_n e^{iS[\phi]}$ with (1.187), we have

$$0 = -i \int \mathcal{D}\phi e^{iS[\phi]} \left\{ \int d^4y \varepsilon(y) [(\square_y + m^2) \phi(y)] \phi_1 \dots \phi_n + i \sum_{i=1}^n \phi_1 \dots \varepsilon_i \dots \phi_n \right\}, \quad (1.194)$$

where ε_i will replace the corresponding ϕ_j when $i = j$ in the summation. And we can notice that $\varepsilon(x_i) = \int d^4y \varepsilon(y) \delta^{(4)}(x_i - y)$, so we have

$$\int d^4y \varepsilon(y) \left\{ -i \int \mathcal{D}\phi e^{iS[\phi]} \left[[(\square_y + m^2) \phi(y)] \phi_1 \dots \phi_n + i \sum_{i=1}^n \phi_1 \dots \delta^{(4)}(y - x_i) \dots \phi_n \right] \right\} = 0. \quad (1.195)$$

For arbitrary ε holds this relation, the only possible case is

$$\int \mathcal{D}\phi e^{iS[\phi]} \left[[(\square_y + m^2) \phi(y)] \phi_1 \dots \phi_n + i \sum_{i=1}^n \phi_1 \dots \delta^{(4)}(y - x_i) \dots \phi_n \right] = 0. \quad (1.196)$$

Example: Consider two-point correlator, i.e., $n = 1$:

$$(\square_y + m^2) \int \mathcal{D}\phi e^{iS[\phi]} \phi(y) \phi(x_1) = -i \delta^{(4)}(y - x_1) \int \mathcal{D}\phi e^{iS[\phi]} = -i \delta^{(4)}(y - x_1) Z[J = 0] \quad (1.197)$$

Putting it back in the expression for the correlator:

$$(\square_y + m^2) \underbrace{\langle \Omega | T \{ \phi(y) \phi(x_1) \} | \Omega \rangle}_{G_F(y-x_1)} = -i \delta^{(4)}(y - x_1). \quad (1.198)$$

Thus,

$$G_F(y - x_1) = -i \delta^{(4)}(y - x_1) (\square_y + m^2)^{-1}. \quad (1.199)$$

Similarly, we would like to generalize this form into n-point correlator. From (1.196) we have

$$\int \mathcal{D}\phi e^{iS[\phi]} (\square_y + m^2) \phi(y), (\phi_1 \dots \phi_n) = -i \int \mathcal{D}\phi e^{iS[\phi]} \left(\sum_{i=1}^n \phi_1 \dots \delta^{(4)}(y - x_i) \dots \phi_n \right). \quad (1.200)$$

Notice $-(\square_y + m^2) \phi(y)$ in the right-hand side is nothing but the variation of the action:

$$\frac{\delta S[\phi]}{\delta \phi(y)} = \int d^4x \left(\frac{\delta \mathcal{L}}{\delta \phi} \delta(x - y) + \frac{\delta \mathcal{L}}{\delta \partial_\mu \phi} \partial_\mu \delta(x - y) \right) = \frac{\delta \mathcal{L}}{\delta \phi(y)} - \partial_\mu \frac{\delta \mathcal{L}}{\delta \partial_\mu \phi(y)}, \quad (1.201)$$

the result of this variation always the equation of motion (or the Euler-Lagrange equation, if you want). So Without loss of generality, we can replace $-(\square_y + m^2)\phi(y)$ by $\frac{\delta S[\phi]}{\delta \phi(y)}$ in (1.200), i.e.,

$$\frac{\int \mathcal{D}\phi e^{iS[\phi]} \left(\frac{\delta S[\phi]}{\delta \phi(y)} \phi_1 \dots \phi_n \right)}{\int \mathcal{D}\phi e^{iS[\phi]}} = \frac{\int \mathcal{D}\phi e^{iS[\phi]} \left(i \sum_{i=1}^n \phi_1 \dots \delta^{(4)}(y - x_i) \dots \phi_n \right)}{\int \mathcal{D}\phi e^{iS[\phi]}}. \quad (1.202)$$

We then get the **Schwinger-Dyson equation** for n -point functions:

$$\left\langle \Omega \left| \frac{\delta S[\phi]}{\delta \phi(y)} T\{\phi_1 \dots \phi_n\} \right| \Omega \right\rangle = i \sum_{i=1}^n \langle \Omega | T\{\phi_1 \dots \delta^{(4)}(y - x_i) \dots \phi_n\} | \Omega \rangle. \quad (1.203)$$

We can simply denote it as:

$$\left\langle \frac{\delta S[\phi]}{\delta \phi(y)} T\{\phi_1 \dots \phi_n\} \right\rangle = i \sum_{i=1}^n \langle T\{\phi_1 \dots \delta^{(4)}(y - x_i) \dots \phi_n\} \rangle. \quad (1.204)$$

6.2 Conservation Laws and Noether's Theorem

Consider a complex scalar field $\mathcal{L} = \partial_\mu \phi \partial^\mu \phi^* - m^2 \phi \phi^*$. Under the global $U(1)$ transformation $\phi \rightarrow \phi' = e^{i\alpha} \phi$, the variation is:

$$\delta \phi = i\alpha \phi; \quad \text{for } \alpha \in \mathbb{R}, \quad (1.205)$$

with the unchanged Lagrangian. For a local transformation $\alpha \rightarrow \alpha(x)$, we have $\delta \phi(x) = i\alpha(x)\phi(x)$, so the variation for the Lagrangian is

$$\begin{aligned} \delta \mathcal{L} &= \mathcal{L}[\phi'] - \mathcal{L}[\phi] \\ &= \partial_\mu \phi \partial^\mu \phi^* + i\partial_\mu \alpha(x) (\phi \partial^\mu \phi^* - \phi^* \partial^\mu \phi) + \mathcal{O}(\alpha^2) - m^2 \phi \phi^* - \partial_\mu \phi \partial^\mu \phi^* + m^2 \phi \phi^* \\ &= \partial_\mu \alpha(x) j^\mu(x), \end{aligned} \quad (1.206)$$

where $j^\mu(x) = i(\phi \partial^\mu \phi^* - \phi^* \partial^\mu \phi)$. An n -point correlator is invariant under this transformation

$$\int \mathcal{D}\phi \mathcal{D}\phi^* e^{iS[\phi, \phi^*]} (\phi_1 \dots \phi_{n/2}) (\phi_1^* \dots \phi_{n/2}^*) = \int \mathcal{D}\phi' \mathcal{D}\phi'^* e^{iS'[\phi', \phi'^*]} (\phi'_1 \dots \phi'_{n/2}) (\phi'^*_1 \dots \phi'^*_{n/2}), \quad (1.207)$$

Notice that $e^{iS'[\phi', \phi'^*]} = e^{i(S+\delta S)} \simeq e^{iS}(1 + i\delta S)$. Expand the right-hand side to order $\alpha(x)$, and subtracting with the left-hand side, we have

$$0 = \int \mathcal{D}\phi \mathcal{D}\phi^* e^{iS[\phi, \phi^*]} (1 + i\delta S) ((1 + i\alpha)\phi_1 \dots (1 + i\alpha)\phi_n^*) \quad (1.208)$$

$$\begin{aligned} &= \int \mathcal{D}\phi \mathcal{D}\phi^* e^{iS[\phi, \phi^*]} \left\{ i\delta S(\phi_1 \dots \phi_n^*) \right. \\ &\quad \left. + [(i\alpha\phi_1) \dots \phi_n^*] + \dots + [\phi_1 \dots (\pm i\alpha\phi_i^{(*)}) \dots \phi_n^*] + \dots + [\phi_1 \dots (-i\alpha\phi_n^*)] \right\} \end{aligned} \quad (1.209)$$

$$= \int \mathcal{D}\phi \mathcal{D}\phi^* e^{iS[\phi, \phi^*]} \left\{ i \int d^4y \partial_\mu \alpha(y) j^\mu(y) (\phi_1 \dots \phi_n^*) + \sum_{i=1}^n \phi_1 \dots (\pm i\alpha\phi_i^{(*)}) \dots \phi_n^* \right\}, \quad (1.210)$$

where $\pm i\phi^{(*)i}$ denote that for ϕ_i^* taking $-i$ and for ϕ_i taking $+i$. After partial integration to factor out $\alpha(y)$:

$$0 = \int \mathcal{D}\phi \mathcal{D}\phi^* e^{iS[\phi, \phi^*]} \left\{ \int d^4y \alpha(y) \partial_\mu j^\mu(y) \phi_1 \dots \phi_n + i \sum_{i=1}^n \phi_1 \dots (\pm i) \left[\int d^4y \delta(y - x_i) \alpha(y) \right] \phi_i^{(*)} \dots \phi_n \right\} \quad (1.211)$$

$$= \int d^4y \alpha(y) \int \mathcal{D}\phi \mathcal{D}\phi^* e^{iS[\phi, \phi^*]} \left\{ -i \partial_\mu j^\mu(y) \phi_1 \dots \phi_n + i \sum_{i=1}^n \phi_1 \dots (\pm i) \delta(y - x_i) \phi_i^{(*)} \dots \phi_n \right\}. \quad (1.212)$$

Similarly, for any infinitesimally α holding this relation, so the only possible way is

$$\int \mathcal{D}\phi \mathcal{D}\phi^* e^{iS[\phi, \phi^*]} \left\{ -i \partial_\mu j^\mu(y) \phi_1 \dots \phi_n + i \sum_{i=1}^n \phi_1 \dots (\pm i) \delta(y - x_i) \phi_i^{(*)} \dots \phi_n \right\} = 0. \quad (1.213)$$

We can also use the correlator notation:

$$\langle \partial_\mu j^\mu(y) \phi(x_1) \dots \phi(x_n) \rangle = -i \sum_{i=1}^n \langle \phi(x_1) \dots (\pm i) \delta(y - x_i) \phi_i^{(*)} \dots \phi(x_n) \rangle. \quad (1.214)$$

Now we generalize this relation to a set of real fields ϕ_a . For a local transformation $\phi_a(x) \rightarrow \phi'_a(x) = \phi_a(x) + \varepsilon \Delta \phi_a(x)$, if the global transformation gives $\mathcal{L} \rightarrow \mathcal{L} + \varepsilon \partial_\mu J^\mu$, then for a local $\varepsilon(x)$ we have

$$\partial_\mu \phi'_a = \partial_\mu (\phi_a + \varepsilon \Delta \phi_a) = \partial_\mu \phi_a + \varepsilon \partial_\mu (\Delta \phi_a) + (\partial_\mu \varepsilon) \Delta \phi_a. \quad (1.215)$$

So when we expand the Lagrangian, we have:

$$\mathcal{L}(\phi', \partial\phi') = \mathcal{L}(\phi, \partial\phi) + \sum_a \left(\frac{\partial \mathcal{L}}{\partial \phi_a} \delta \phi_a + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \delta (\partial_\mu \phi_a) \right), \quad (1.216)$$

then plug the variation $\delta \phi_a = \varepsilon \Delta \phi_a$ into it and using the Euler-Lagrange equation:

$$\delta \mathcal{L} = \sum_a \left[\frac{\partial \mathcal{L}}{\partial \phi_a} (\varepsilon \Delta \phi_a) + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} (\varepsilon \partial_\mu (\Delta \phi_a) + (\partial_\mu \varepsilon) \Delta \phi_a) \right] \quad (1.217)$$

$$= \sum_a \left[\partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} (\varepsilon \Delta \phi_a) + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} (\varepsilon \partial_\mu (\Delta \phi_a) + (\partial_\mu \varepsilon) \Delta \phi_a) \right] \quad (1.218)$$

$$= \sum_a \left[\varepsilon \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \Delta \phi_a \right) + (\partial_\mu \varepsilon) \Delta \phi_a \right]. \quad (1.219)$$

Then we get the variation for the Lagrangian:

$$\mathcal{L}(\phi_a) \rightarrow \mathcal{L}(\phi'_a) = \mathcal{L}(\phi_a) + \varepsilon(x) \partial_\mu J^\mu + \partial_\mu \varepsilon(x) \sum_a \Delta \phi_a \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \quad (1.220)$$

$$= \mathcal{L}(\phi_a) + \varepsilon(x) \partial_\mu J^\mu - \varepsilon(x) \partial_\mu \left(\sum_a \Delta \phi_a \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} \right) \quad (1.221)$$

$$= \mathcal{L}(\phi_a) - \varepsilon(x) \partial_\mu \left(\sum_a \Delta \phi_a \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_a)} - J^\mu \right), \quad (1.222)$$

which appears an extra term due to the infinitesimal transformation rely on the coordinate. Therefore, the Noether current j^μ can be defined from the variation for the action

$$S[\phi'_a] = \int d^4y [\mathcal{L}[\phi_a] - \varepsilon(y)\partial_\mu j^\mu(y)] = S[\phi_a] - \int d^4y \varepsilon(y)\partial_\mu j^\mu(y). \quad (1.223)$$

Thus, the Noether current j^μ is

$$j^\mu = \sum_a \Delta\phi_a \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} - J^\mu. \quad (1.224)$$

Put this current back to (1.214), we get the conserved symmetry Schwinger-Dyson equation as:

$$\langle \partial_\mu j^\mu(y) \phi_{a_1}(x_1) \dots \phi_{a_n}(x_n) \rangle = -i \sum_{i=1}^n \langle \phi_{a_1}(x_1) \dots \delta(y - x_i) \Delta\phi_{a_i} \dots \phi_{a_n}(x_n) \rangle \quad (1.225)$$

6.3 Ward-Takahashi Identity in QED

Consider the QED Lagrangian with fermions:

$$\mathcal{L}[\psi, \bar{\psi}, A_\mu] = \bar{\psi}(i\gamma^\mu D_\mu - m)\psi + A_\mu j^\mu, \quad (1.226)$$

where $j^\mu = e\bar{\psi}\gamma^\mu\psi$. The global $U(1)$ transformation gives

$$\psi \rightarrow e^{i\alpha}\psi \implies \delta\psi = i\alpha\psi. \quad (1.227)$$

So the local transformation gives

$$\mathcal{L}[\psi', \bar{\psi}'] = \mathcal{L}[\psi, \bar{\psi}] - (\partial_\mu \alpha) j^\mu. \quad (1.228)$$

Applying Schwinger-Dyson equation with two fields, we have

$$\partial_\mu \langle 0|T\{j^\mu(y)\psi(x_1)\bar{\psi}(x_2)\}|0\rangle = \delta^{(4)}(y-x_1)\langle 0|T\{\psi(x_1)\bar{\psi}(x_2)\}|0\rangle - \delta^{(4)}(y-x_2)\langle 0|T\{\psi(x_1)\bar{\psi}(x_2)\}|0\rangle. \quad (1.229)$$

The minus sign shows that ψ follows **Fermi statistics**. Take a Fourier transform to momentum space:

$$k_\mu \langle 0|T\{j^\mu(k)\psi(q)\bar{\psi}(p)\}|0\rangle = e (\langle 0|T\{\psi(q-k)\bar{\psi}(p)\}|0\rangle - \langle 0|T\{\psi(q)\bar{\psi}(p+k)\}|0\rangle), \quad (1.230)$$

which is nothing but the **Ward-Takahashi identity** in QED that we are familiar with:

$$k_\mu \mathcal{M}^\mu(k; q, p) = e\mathcal{M}_0(p, q-k) - e\mathcal{M}_0(q, p+k). \quad (1.231)$$

It can also be expressed by the Feynman diagram:

$$k_\mu \left(\begin{array}{c} \text{Diagram 1: A shaded circle with an incoming wavy line from the left labeled } \mu \text{ and } k, \text{ and two outgoing fermion lines labeled } q \text{ and } p. \end{array} \right) = e \left(\begin{array}{c} \text{Diagram 2: A shaded circle with an incoming fermion line from the bottom labeled } p, \text{ and two outgoing lines labeled } q-k=p \text{ and } q. \end{array} \right) - \left(\begin{array}{c} \text{Diagram 3: A shaded circle with an incoming fermion line from the bottom labeled } p+k=q, \text{ and two outgoing lines labeled } q \text{ and } p. \end{array} \right). \quad (1.232)$$

Especially for the one loop case:

$$k_\mu \left(\begin{array}{c} \text{Diagram 4: A shaded circle with an incoming wavy line from the left labeled } \mu \text{ and } k, \text{ and two outgoing fermion lines labeled } q \text{ and } p. \end{array} \right) = e \left(\begin{array}{c} \text{Diagram 5: A shaded circle with an incoming fermion line from the bottom labeled } p, \text{ and two outgoing lines labeled } q-k=p \text{ and } q. \end{array} \right) - \left(\begin{array}{c} \text{Diagram 6: A shaded circle with an incoming fermion line from the bottom labeled } p, \text{ and two outgoing lines labeled } q \text{ and } q=k+p. \end{array} \right). \quad (1.233)$$

Renormalization

1. Renormalization in ϕ^4 -theory

Firstly, we briefly recall some basic idea about renormalization. For counting the divergent of a theory, one may use the superficial degree of divergence:

$$D = \text{power of } k \text{ in numerator} - \text{power of } k \text{ in denominator.} \quad (2.1)$$

Suppose the Loop integrals in $d = 4$ space-time dimension have superficial degree of divergence D , which can be easily seen from the directly calculating:

$$\int^{\Lambda} d^4k \frac{k^{a-4}}{k^b} = \int^{\Lambda} k^3 dk d\Omega \frac{k^{a-b}}{k^4} \sim \int^{\Lambda} dk \frac{k^{a-b}}{k} = \begin{cases} \Lambda^D & \text{for } D = a - b \neq 0 \\ \ln \Lambda & \text{for } D = 0 \end{cases}, \quad (2.2)$$

where Λ is the cut-off. But we would like to use some more general form to think about it. As an example, considering the ϕ^n -theory

$$\mathcal{L} = \frac{1}{2}(\partial_{\mu}\phi)(\partial^{\mu}\phi) - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{n!}\phi^n, \quad (2.3)$$

in which the propagators have the form of $(k^2 - m^2)^{-1}$, vertices are $\sim \lambda$ that connect n lines. So we can define So in this case, we can

1. $L = P - V + 1$

- L : loops (loop integral with d^4k)
- P : internal propagators (propagator with $(k^2)^{-1}$)
- V : vertices (vertices with energy-momentum conservation $\delta(\sum p_i)$)
- $+1$: global conservation of energy-momentum

2. $nV = 2P + N$

- n : number of lines at each vertex
- P : internal propagators
- N : external lines (or external propagator)

So for each loop, the power of k in numerator is 4, which is contributed by an integral measurement $\sim d^4k$, and the power of k in denominator is contributed by the propagator. Therefore, the superficial degree of divergence is

$$D = dL - 2P; \quad d = 4. \quad (2.4)$$

Thus, base on the relation we introduced above, we have

$$\begin{aligned} D &= 4(P - V + 1) - 2P \\ &= 2P - 4V + 4 \\ &= (n - 4)V - N + 4. \end{aligned} \quad (2.5)$$

Thus, we find D is independent of P , but only depends on N for the case $n = 4$. But for a more general n in $\frac{\lambda}{n!}\phi^n$, we have

- $n = 2$: $\implies$ free theory (included in the mass term);
- $n = 3$: $D = 4 - N - V \implies$ **super renormalizable**
- $n = 4$: $D = -N + 4 \implies$ **renormalizable**
- $n > 4$: $D = (n - 4)V - N + 4 \implies$ **non-renormalizable**

For a theory that the signature in front of V is “ $-$ ”, we call it as **super renormalizable theory** because the divergence only appear in some leading order terms, with respect the order getting higher, the diagram will finally be convergent, in which $D < 0$. *So in the super renormalizable theory only contains finite number of divergent degree*, which means only need finite counterterms. Similarly, for a theory that calls **renormalizable**, it is because it has no relation with V , which means *no matter how many orders we are counting, the divergent degrees are the same!* Although in this case, it contains infinite divergent diagrams, we can still by re-define the “physical” parameter to include the divergence. In the end, it is quite obvious to understand why we call the last situation as **non-renormalizable**. Because the pre-signature of V is positive, which means with the order getting higher, the divergent will be keep increasing, *so we need infinite parameter to cancel the divergence*, which in most of cases are technically impossible.

Another case is QED, we have two type of propagators, but they all contribute $\sim k^{-2}$

- N_γ : external photon lines;
- N_e : external electron lines;
- P_e : electron propagators;
- P_γ : photon propagators;
- $V = 2P_\gamma + N_\gamma = \frac{1}{2}(2P_e + N_e)$: number of vertices;
- $L = P_e + P_\gamma - V + 1$: number of the loops.

$$\begin{aligned}
 D &= 4L - P_e - 2P_\gamma \\
 &= 4(P_e + P_\gamma - V + 1) - P_e - 2P_\gamma \\
 &= 3P_e + 2P_\gamma - 4V + 4 \\
 &= 4 - N_\gamma - \frac{3}{2}N_e.
 \end{aligned} \tag{2.6}$$

So as we discussed above, since D is non-related with V , so QED is a renormalizable theory.

1.1 Analysis of dimensions in view of renormalization

Now we consider another way to analyze the renormalizability. We use the dimension of mass as a standard. So

$$[\text{Length}] = [\text{Time}] = [\text{Energy}]^{-1} = [\text{Mass}]^{-1}. \tag{2.7}$$

We require the dimension for the action $[S] = 0$, thus

$$[d^4x] = -4, \quad [\mathcal{L}] = 4, \quad [\partial_\mu] = 1, \quad [\phi] = 1. \tag{2.8}$$

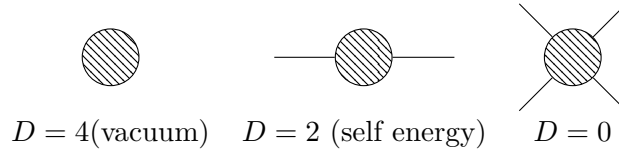
The last one is because $[\partial_\mu \phi \partial^\mu \phi] = 4$. The coupling constant $[\lambda]$ in $\frac{\lambda}{n!} \phi^n$ then has the dimension

$$[\lambda] = 4 - n. \quad (2.9)$$

Take a sharp look! This number, $4 - n$, is nothing but the prefactor of superficial degree of divergence in ϕ^n theory, i.e.,

$$D = 4 - N - \underbrace{(4 - n)}_{[\lambda]} V. \quad (2.10)$$

Thus, base on the discussion we did above, we find $[\lambda] < 0$ corresponds to a non-renormalizable theory. As an example, we consider the ϕ^4 -theory. So $n = 4$ brings the superficial degree of divergence is $D = 4 - N \geq 0$ for degree up to $N = 4$, which is a renormalizable theory.



$$D = 4(\text{vacuum}) \quad D = 2 \text{ (self energy)} \quad D = 0$$

We can also do the same thing for QED. For convenience, we rewrite the QED Lagrangian:

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \bar{\psi}(i\not{\partial} - m)\psi + e A_\mu \bar{\psi} \gamma^\mu \psi. \quad (2.11)$$

Similarly, since the action S is dimensionless, so

$$[\psi] = \frac{3}{2}, \quad [A_\mu] = 1. \quad (2.12)$$

Thus again, we can then see the coupling constant e is dimensionless, i.e., $[e] = 0$, which just corresponds to a renormalizable theory.

1.2 Renormalized perturbation theory

Now we still use ϕ^4 -theory as an example to show the renormalization procedure. The bare Lagrangian is

$$\mathcal{L}_{\text{bare}} = \frac{1}{2}(\partial_\mu \phi_0)(\partial^\mu \phi_0) - \frac{1}{2}m_0^2 \phi_0^2 + \frac{1}{4!}\lambda_0 \phi_0^4 \quad (2.13)$$

where the corner mark “0” represent the bare parameters, i.e., λ_0, m_0 , and ϕ_0 . This Lagrangian is nothing special, it just the original Lagrangian that we used in perturbation theory. Now we just give it a different name to separate with the renormalized Lagrangian that we will discuss later. The name “bare” represent this quantity is not actually the physical quantity that we observe because the bare parameters are actually infinity. These parameters appears only for cancel the infinity in the loop integral. The rest of this canceling is then the physical quantity, which is the renormalized quantity. This explanation shows the core idea of why do we need renormalization, i.e., the physical part of graphs needs to be observable, so we have to find a well-defined quantity that suitable for any order. But in this case, we may have to accept that many “constants” we thought unchanged will start to running with the higher energy scale being considered. And this will be described by the renormalization group equation.

The relation of the renormalized quantities with the bare quantities are

$$\phi_0 = \sqrt{Z} \phi_r, \quad Z m_0^2 = m^2 + \delta m^2, \quad Z^2 \lambda_0 = \lambda + \delta \lambda, \quad (2.14)$$

where $Z = 1 + \delta Z$. And δZ , δm^2 and $\delta\lambda$ are called the **counterterms**. So the bare Lagrangian can be written as

$$\begin{aligned}\mathcal{L}_{bare} &= \mathcal{L}_{ren} + \mathcal{L}_{count} \\ &= \underbrace{\frac{1}{2}(\partial_\mu\phi_r)(\partial^\mu\phi_r) - \frac{1}{2}m^2\phi_r^2 + \frac{\lambda}{4!}\phi_r^4}_{\mathcal{L}_{ren}} + \underbrace{\frac{1}{2}\delta Z(\partial_\mu\phi_r)(\partial^\mu\phi_r) - \frac{1}{2}\delta m^2\phi_r^2 + \frac{1}{4!}\delta\lambda\phi_r^4}_{\mathcal{L}_{counter\ term}}.\end{aligned}\quad (2.15)$$

The counterterms also have corresponding Feynman rules

$$\longrightarrow \otimes \longrightarrow = i(p^2\delta Z - \delta m^2); \quad \bigotimes = -i\delta\lambda. \quad (2.16)$$

And the Feynman rules of renormalized term is

$$\longrightarrow_p = \frac{i}{p^2 - m^2}; \quad \bigtimes = -mi\lambda. \quad (2.17)$$

The renormalization condition is usually called the **scheme**. The on-shell scheme for ϕ_r, m in two-point function is

$$\longrightarrow_p \text{---} \text{---} \text{---} \text{---} \longrightarrow_p = \frac{i}{p^2 - m_0^2 - \Sigma(p^2)} \quad (2.18)$$

$$= \frac{i}{p^2 - m_{\text{phy}}^2} + \underbrace{A + B(p^2 - m_{\text{phy}}^2) + C(p^2 - m_{\text{phy}}^2)^2 + \dots}_{\text{Regular Terms}}, \quad (2.19)$$

where the $\Sigma(p^2)$ is the self-energy correction. The main point of this scheme is requiring the residue at the pole $p^2 = m_{\text{phys}}^2$ to be normalized to 1. The mission of the counterterms is to absorb UV divergences and ensure the self-energy $\Sigma(p^2)$ satisfies the specific renormalization conditions, rather than cancelling the regular terms.

Now for the four-point functions, we impose a similar renormalization condition on the scattering amplitude:

$$\left(\bigotimes \right)_{\text{Amp}} = i\lambda, \quad (2.20)$$

at the threshold $s = 4m^2$ and $t = u = 0$, which corresponds to the case of zero momentum transfer. The Mandelstam variables are defined as usual: $s = (p_1 + p_2)^2$, $t = (p_1 - p_3)^2$, and $u = (p_1 - p_4)^2$. This condition, together with the two-point function conditions mentioned earlier, defines the on-shell renormalization scheme. To implement this in practice, we follow these steps:

1. Calculate the divergent diagrams: Evaluate all Feynman diagrams contributing to the n -point functions at a given order in perturbation theory. A suitable regularization scheme (e.g., dimensional regularization) must be chosen to handle the UV divergences.
2. Determine the counterterms: Extract the specific values of the counterterms δm^2 , δZ , and $\delta\lambda$ by requiring that the renormalized Green's functions satisfy the chosen renormalization conditions (such as the one in (2.20)).

To conclude this section, we will briefly do the renormalization process for the 1-loop for ϕ^4 -theory. Firstly, we expand the two point function until λ^2 :

$$\text{---} \text{---} \text{---} \text{---} \text{---} = \text{---} \text{---} \text{---} + \left(\text{---} \text{---} \text{---} + \text{---} \otimes \text{---} \right) + O(\lambda^2), \quad (2.21)$$

where the first term is just the normal propagator, the second and the third is the 1-loop graph and the counterterm. Thus, we have

$$\text{---} \text{---} \text{---} \text{---} \text{---} = \frac{i}{p^2 - m^2} - \frac{i}{2} \lambda \int \frac{d^d k}{(2\pi)^d} \frac{i}{k^2 - m^2} + i(p^2 \delta Z - \delta m^2) + O(\lambda^2). \quad (2.22)$$

We can by directly calculating the tadpole integral to separate the divergent part, such that to determine the expression for the counterterms. As a result, we only shows the final expression that you should get:

$$\delta Z = 0 \quad (2.23)$$

$$\delta m^2 = -\frac{\lambda}{2} \frac{\Gamma(1 - d/2)}{(4\pi)^{d/2} (m^2)^{1-d/2}}. \quad (2.24)$$

If we take $d = 4 - 2\epsilon$, the theory is renormalizable, so the form of the counterterms are suitable for any order.

The 4-point function also can be expanded into 1-loop order, but since there are two vertices, so the next to next leading order we have is λ^3 :

$$\text{---} \otimes \text{---} = \text{---} \times \text{---} + \left(\text{---} \text{---} \text{---} + \text{---} \text{---} \text{---} + \text{---} \text{---} \text{---} \right) + \text{---} \otimes \text{---} + O(\lambda^3) \quad (2.25)$$

$$= -i\lambda + (-i\lambda)^2 (iV(s) + iV(t) + iV(u)) - i\delta\lambda + O(\lambda^3) \quad (2.26)$$

The counterterm $\delta\lambda$ is determined by requiring the total scattering amplitude to satisfy the renormalization condition at the chosen subtraction point. According to the Feynman rules, the total amplitude at 1-loop order is the sum of the tree-level vertex, the loop corrections (s, t, and u channels), and the counterterm vertex. For each $iV(p^2)$, we have

$$iV(p^2) = \frac{i}{2} \int \frac{d^d k}{(2\pi)^d} \frac{i}{k^2 - m^2} \frac{i}{(k+p)^2 - m^2} \quad (2.27)$$

$$= -\frac{i}{2} \frac{1}{(4\pi)^{d/2}} \int_0^1 dx \frac{\Gamma(2 - d/2)}{(m^2 - x(1-x)p^2)^{2-d/2}}, \quad (2.28)$$

where x is the Feynman parameter. By taking $d = 4 - 2\epsilon$, we can separate the divergent part too:

$$iV(p^2) = -\frac{i}{32\pi^2} \int_0^1 dx \left(\frac{1}{\epsilon} - \gamma_E + \ln(4\pi) - \ln(m^2 - x(1-x)p^2) + O(\epsilon) \right). \quad (2.29)$$

To satisfy the on-shell condition defined in Eq. (1.20), we require the amplitude to be exactly $-i\lambda$ at the threshold $s = 4m^2, t = u = 0$. This implies that the loop contributions and the counterterm must cancel each other out at this specific kinematic point:

$$(-i\lambda)^2 [iV(4m^2) + iV(0) + iV(0)] - i\delta\lambda = 0 \quad (2.30)$$

Solving for $\delta\lambda$ yields the expression for the 1-loop counterterm in ϕ^4 theory:

$$\delta\lambda = -\lambda^2 [V(4m^2) + 2V(0)], \quad (2.31)$$

Therefore, the divergence $1/\epsilon$ pole contained in $V(p^2)$ is exactly absorbed by $\delta\lambda$, rendering the final physical amplitude finite. This ensures that the coupling constant λ corresponds to the strength of the interaction measured at the specific energy scale $s = 4m^2$.

2. Wilson's approach to renormalization

The main idea of Wilson's renormalization is introducing a cut-off Λ and compute how the theory changes at different length scales. For that, we need to working with the generating functional with cut-off and successive integration over large momenta/small length scales:

$$Z = \int \mathcal{D}\phi_\Lambda e^{-S[\phi]}, \quad (2.32)$$

where we used Wick rotation changing to Euclidean space-time:

$$x^0 \rightarrow ix^4, \quad d^4x \rightarrow id^4x_E. \quad (2.33)$$

Thus, the measurement is

$$\mathcal{D}\phi_\Lambda = \prod_{|k| < \Lambda} d\phi(k). \quad (2.34)$$

In this case, the modes of ϕ are well defined in Euclidean space-time such that we can make sure that when $k \rightarrow \infty$ all the components of momentum also go to infinity. This step is to avoid in Minkowski space-time $k^2 = (k^0)^2 - (k^i)^2$. So when $k^2 \rightarrow \infty$, it could be $k^0 \rightarrow \infty$ and $k^i \rightarrow 0$, which doesn't suitable for our analysis.

Now, the action and Lagrangian become

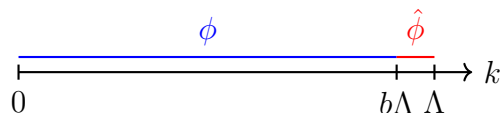
$$S[\phi] = \int d^4x \mathcal{L}, \quad (2.35)$$

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi + \frac{1}{2} m^2 \phi^2 + \frac{\lambda}{4!} \phi^4. \quad (2.36)$$

Suppose the mass $m \ll \Lambda$, and the coupling constant $\lambda \ll 1$, which means the last two terms in the Lagrangian can be treated as perturbation. For going to the perturbation theory, it's useful to split up fields $\phi \rightarrow \phi + \hat{\phi}$ as

$$\hat{\phi} = \begin{cases} \phi(k) & \text{for } b\Lambda \leq |k| \leq \Lambda \\ 0 & \text{for } |k| \leq b\Lambda \end{cases}; \quad \phi = \begin{cases} 0 & \text{for } b\Lambda \leq |k| \leq \Lambda \\ \phi(k) & \text{for } |k| \leq b\Lambda \end{cases}, \quad (2.37)$$

where $(b\Lambda, \Lambda)$ is a very small interval controlled by parameter $b \leq 1$



Then we can also split up the action as $S[\phi] \rightarrow S[\phi] + \hat{S}[\phi, \hat{\phi}]$, so we have

$$S[\phi] = \int d^4x \frac{1}{2} \partial_\mu (\phi + \hat{\phi}) \partial^\mu (\phi + \hat{\phi}) + \frac{1}{2} m^2 (\phi + \hat{\phi})^2 + \frac{\lambda}{4!} (\phi + \hat{\phi})^4 \quad (2.38)$$

$$= \int d^4x \frac{1}{2} \left(\partial_\mu \phi \partial^\mu \phi + 2 \partial_\mu \phi \partial^\mu \hat{\phi} + \partial_\mu \hat{\phi} \partial^\mu \hat{\phi} \right) + \frac{1}{2} m^2 (\phi^2 + \hat{\phi}^2 + 2\phi\hat{\phi}) \quad (2.39)$$

$$+ \frac{\lambda}{4!} \left(\phi^4 + \phi^2 \hat{\phi}^2 + 2\phi^3 \hat{\phi} + \hat{\phi}^2 \phi^2 + \hat{\phi}^4 + 2\phi \hat{\phi}^3 + 2\phi^3 \hat{\phi} + 2\phi \hat{\phi}^3 + 4\phi^2 \hat{\phi}^2 \right) \quad (2.40)$$

$$= \int d^4x \left[\underbrace{\frac{1}{2} (\partial_\mu \phi)^2 + \frac{1}{2} m^2 \phi^2 + \frac{\lambda}{4!} \phi^4}_{\mathcal{L}[\phi]} + \underbrace{\frac{1}{2} (\partial_\mu \hat{\phi})^2 + \frac{1}{2} m^2 \hat{\phi}^2 + \frac{\lambda}{4!} (6\phi^2 \hat{\phi}^2 + 4\phi^3 \hat{\phi} + 4\phi \hat{\phi}^3 + \hat{\phi}^4)}_{\hat{\mathcal{L}}[\phi, \hat{\phi}]} \right], \quad (2.41)$$

where we have dropped terms containing the products $\phi \hat{\phi}$ because Fourier components are orthogonal:

$$\int \frac{d^4k}{(2\pi)^4} \frac{d^4k'}{(2\pi)^4} \phi(k) \hat{\phi}(k') e^{i(k+k') \cdot x} \sim \delta^{(4)}(k+k') = 0 \quad (2.42)$$

Thus, we have

$$\hat{S}[\phi, \hat{\phi}] = \int d^4x (\hat{\mathcal{L}}_0[\hat{\phi}] + \hat{\mathcal{L}}_{\text{int}}[\phi, \hat{\phi}]), \quad (2.43)$$

and

$$\hat{\mathcal{L}}_0[\hat{\phi}] = \frac{1}{2} \partial_\mu \hat{\phi} \partial^\mu \hat{\phi}, \quad (2.44)$$

$$\hat{\mathcal{L}}_{\text{int}}[\phi, \hat{\phi}] = \frac{1}{2} m^2 \hat{\phi}^2 + \lambda \left(\frac{1}{3!} \phi \hat{\phi}^3 + \frac{1}{4} \phi^2 \hat{\phi}^2 + \frac{1}{3!} \phi^3 \hat{\phi} + \frac{1}{4!} \hat{\phi}^4 \right). \quad (2.45)$$

Thus, the first part in $\hat{S}$ can be rewritten by the Fourier transformation as

$$\int d^4x \hat{\mathcal{L}}_0[\hat{\phi}] = \int \frac{d^4k d^4p}{(2\pi)^8} (ik_\mu \hat{\phi}(k)) (ip^\mu \hat{\phi}(p)) \int d^4x e^{i(k+p) \cdot x} \quad (2.46)$$

$$= \int \frac{d^4k d^4p}{(2\pi)^8} (ik_\mu) (ip^\mu) \hat{\phi}(k) \hat{\phi}(p) (2\pi)^4 \delta^{(4)}(k+p) \quad (2.47)$$

$$= \frac{1}{2} \int_{b\Lambda < |k| < \Lambda} \frac{d^4k}{(2\pi)^4} \hat{\phi}(-k) k^2 \hat{\phi}(k) \quad (2.48)$$

$$= \frac{1}{2} \int_{b\Lambda < |k| < \Lambda} \frac{d^4k}{(2\pi)^4} \hat{\phi}^*(k) k^2 \hat{\phi}(k). \quad (2.49)$$

This Gaussian integral form remind us to write a propagator for $\hat{\phi}$ from Wick contraction of $\hat{\phi}$ fields:

$$\overline{\hat{\phi}(k) \hat{\phi}(p)} = \langle \hat{\phi}(k) \hat{\phi}(p) \rangle = \frac{\int \mathcal{D}\hat{\phi} \hat{\phi}(k) \hat{\phi}(p) e^{-S_0[\hat{\phi}]}}{\int \mathcal{D}\hat{\phi} e^{-S_0[\hat{\phi}]}} = \frac{1}{k^2} (2\pi)^4 \delta^{(4)}(k+p) \Theta(k), \quad (2.50)$$

where

$$\Theta(k) = \begin{cases} 1 & \text{for } b\Lambda < |k| < \Lambda \\ 0 & \text{else} \end{cases}. \quad (2.51)$$

Therefore, we can separate the generating functional measurement with the **high energy modes** $\hat{\phi}$ and **low energy modes** $\phi_{b\Lambda}$ as $\mathcal{D}\phi_\Lambda = \mathcal{D}\phi_{b\Lambda}\mathcal{D}\hat{\phi}_\Lambda$. And we can integrate the high energy modes

$$Z = \int \mathcal{D}\phi_\Lambda e^{-S[\phi]} = \int \mathcal{D}\phi_{b\Lambda}\mathcal{D}\hat{\phi}_\Lambda e^{-S[\phi]-\hat{S}[\phi,\hat{\phi}]} = \int \mathcal{D}\phi_{b\Lambda} e^{-S[\phi]} \left(\int \mathcal{D}\hat{\phi}_\Lambda e^{-\hat{S}[\phi,\hat{\phi}]} \right) \quad (2.52)$$

$$= \int \mathcal{D}\phi_{b\Lambda} e^{-S_{\text{eff}}[\phi]}, \quad (2.53)$$

where left a **effective action** defined as

$$S_{\text{eff}}[\phi] = S[\phi] + \delta S[\phi], \quad (2.54)$$

in which $\delta S[\phi]$ can be treated as perturbation, thus it can be expanded as

$$e^{-\delta S[\phi]} = \int \mathcal{D}\hat{\phi}_\Lambda e^{-\hat{S}[\phi,\hat{\phi}]} = \int \mathcal{D}\hat{\phi}_\Lambda e^{-\int d^4x \hat{\mathcal{L}}[\phi,\hat{\phi}]} \quad (2.55)$$

$$\simeq \int \mathcal{D}\hat{\phi}_\Lambda \int d^4x \left[1 - \left(\hat{\mathcal{L}}_0 + \frac{1}{2}m^2\hat{\phi}^2 + \frac{\lambda}{3!}\phi\hat{\phi}^3 + \frac{\lambda}{4}\phi^2\hat{\phi}^2 + \frac{\lambda}{3!}\phi^3\hat{\phi} + \frac{\lambda}{4!}\hat{\phi}^4 \right) + \mathcal{O}(\lambda^2) \right] \quad (2.56)$$

$$= \int \mathcal{D}\hat{\phi}_\Lambda \int d^4x \left[1 - \underbrace{\left(\hat{\mathcal{L}}_0 + \frac{1}{2}m^2\hat{\phi}^2 + \frac{\lambda}{4!}\hat{\phi}^4 \right)}_{\substack{\text{Doesn't contain } \phi, \\ \text{which has no contribution for running}}} - \frac{\lambda}{4}\phi^2\hat{\phi}^2 + \mathcal{O}(\lambda^2) \right], \quad (2.57)$$

where in the third step we have dropped all the odd terms which don't contribute to the integral. The introducing of $\delta S[\phi]$ will change the parameters in the original action $S[\phi]$, which will finally leading the parameters start running. To show this, we now need to work in the perturbation scenario. Since we have assumed that $m \ll \Lambda$ and $\lambda \ll 1$, so we can expand $e^{-S_{\text{eff}}[\phi]}$ with Dyson series in terms of λ . For that, we can generally write the effective Lagrangian $\mathcal{L}_{\text{eff}}$ for effective action $S_{\text{eff}}[\phi]$ as:

$$\mathcal{L}_{\text{eff}}[\phi] = \frac{1}{2}(1 + \Delta Z)\partial_\mu\phi\partial^\mu\phi + \frac{1}{2}(m^2 + \Delta m^2)\phi^2 + \frac{1}{4!}(\lambda + \Delta\lambda)\phi^4 + (\text{higher dimensional operators}). \quad (2.58)$$

Now the mission is to determine ΔZ , Δm^2 and $\Delta\lambda$. We firstly consider the 1-loop order $\mathcal{O}(\lambda)$ correction for the propagator, which corresponds to the term $\frac{1}{4}\lambda\phi^2\hat{\phi}^2$, the graph is :

$$I_1 = \phi \text{ --- } \text{ (loop) } \text{ --- } \phi, \quad \mathcal{O}(\lambda) \quad (2.59)$$

where the double line represent the contraction of $\hat{\phi}$, which doesn't contribute the real physical observable, thus it needs to be integrated. Thus, we have

$$I_1 = -\frac{\lambda}{4} \int d^4x \phi^2 \hat{\phi} \hat{\phi} \quad (2.60)$$

$$= -\frac{\lambda}{4} \int d^4x \int \frac{d^4k_1}{(2\pi)^4} \cdots \frac{d^4k_4}{(2\pi)^4} e^{i(k_1+\cdots+k_4)\cdot x} \phi(k_1)\phi(k_2) \frac{1}{k_3^2} (2\pi)^4 \delta^{(4)}(k_3+k_4) \Theta(k_3), \quad (2.61)$$

integrate for x, k_4 :

$$I_1 = -\frac{\lambda}{4} \int \frac{d^4 k_1}{(2\pi)^4} \cdots \frac{d^4 k_3}{(2\pi)^4} \phi(k_1) \phi(k_2) \frac{1}{k_3^2} (2\pi)^4 \delta^{(4)}(k_1 + k_2) \Theta(k_3) \quad (2.62)$$

$$= -\frac{1}{2} \mu \int \frac{d^4 k_1}{(2\pi)^4} \phi(k_1) \phi(-k_1), \quad (2.63)$$

where

$$\mu = \frac{\lambda}{2} \int_{b\Lambda < |k| < \Lambda} \frac{d^4 k}{(2\pi)^4} \frac{\Theta(k)}{k^2} \quad (2.64)$$

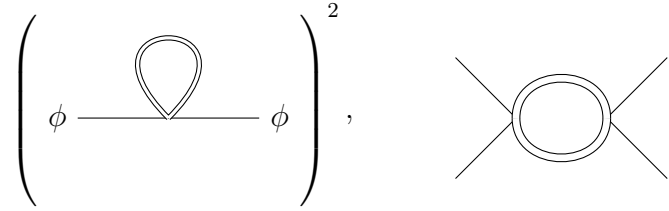
$$= \frac{\lambda}{2} \frac{1}{(2\pi)^4} \int d\Omega \int_{b\Lambda}^{\Lambda} k dk \quad (2.65)$$

$$= \frac{\lambda}{32\pi^2} (1 - b^2) \Lambda^2. \quad (2.66)$$

I_1 has structure of mass correction, which can be put in exponent of effective action as

$$\exp \left\{ - \int d^4 x \frac{1}{2} \mu \phi^2 \right\}. \quad (2.67)$$

And the $\mathcal{O}(\lambda^2)$ order graphs therefore are:



For the first one, it's nothing but the disconnected graph $\sim \frac{1}{2} I_1^2$, which contribute $\mathcal{O}(\lambda^2)$ in the exponent. The second graph represent the correction for the ϕ^4 interaction vertex:

$$I_2 = -\frac{1}{4!} \xi \int d^4 x \phi^4(x), \quad (2.68)$$

factorizing the loop integral, then we have

$$\xi = -4! \cdot \underbrace{2}_{\substack{\text{2 possible} \\ \text{contractions} \\ \text{of } \phi^2(\phi\phi)^2}} \cdot \underbrace{\frac{1}{2!}}_{\substack{\text{symmetry factor} \\ \text{of diagram}}} \cdot \left(\frac{\lambda}{4}\right)^2 \int_{b\Lambda < |k| < \Lambda} \frac{d^4 k}{(2\pi)^4} \left(\frac{1}{k^2}\right)^2 \Theta(k) = -\frac{3\lambda^2}{16\pi^2} \ln \frac{1}{b} \quad (2.69)$$

Thus, we find μ, ξ can be interpreted as combinations to $\Delta m^2, \Delta\lambda$.

Therefore, the generating functional $Z = \int \mathcal{D}\phi_\Lambda e^{-S[\phi]}$ changing to $Z = \int \mathcal{D}\phi_{b\Lambda} e^{-S_{\text{eff}}[\phi]}$ is by rescaling S to effective action S_{eff} by changing the coordinates and momenta:

$$k' = \frac{k}{b}, \quad x' = bx \quad (2.70)$$

Thus,

$$S_{\text{eff}} = \int d^4 x' \frac{1}{b^4} \left(\frac{1}{2} (1 + \Delta Z) b^2 \partial'_\mu \phi \partial'^\mu \phi + \frac{1}{2} (m^2 + \Delta m^2) \phi^2 + \frac{1}{4!} (\lambda + \Delta\lambda) \phi^4 + \cdots + \underbrace{\frac{1}{n!} (\lambda_n + \Delta\lambda_n) \phi^n + \cdots}_{\text{higher dim. operators}} \right) \quad (2.71)$$

$$= \int d^4 x' \left(\frac{1}{2} \partial'_\mu \phi' \partial'^\mu \phi' + \frac{1}{2} m'^2 \phi'^2 + \frac{1}{4!} \lambda' \phi'^4 + \cdots + \frac{1}{n!} \lambda'_n (\phi')^n + \cdots \right), \quad (2.72)$$

where

$$\phi' = \sqrt{(1 + \Delta Z)b^{-2}}\phi \quad (2.73)$$

$$m'^2 = (m^2 + \Delta m^2)(1 + \Delta Z)^{-1}b^{-2} \quad (2.74)$$

$$\lambda' = (\lambda + \Delta\lambda)(1 + \Delta Z)^{-2}b^0 \quad (2.75)$$

$$\lambda'_n = (\lambda_n + \Delta\lambda_n)(1 + \Delta Z)^{-n/2}b^{n-4}. \quad (2.76)$$

This process, mapping $\phi' \rightarrow \phi$, $m'^2 \rightarrow m^2$, $\lambda' \rightarrow \lambda$, is exactly the **renormalization group transformations**. We can also explicitly show the relations for rescaling of parameters for mapping $x' = bx$ on $D = 4$ dimension:

$$m'^2 = b^{-2}\Delta m^2 \quad (2.77)$$

$$\lambda' = b^0\Delta\lambda \quad (2.78)$$

$$\lambda'_n = b^{n-4}\Delta\lambda_n \quad (2.79)$$

Generally, in D dimensions, we have:

$$m'^2 = b^{-2}\Delta m^2 \quad (2.80)$$

$$\lambda' = b^{D-4}\Delta\lambda \quad (2.81)$$

$$\lambda'_n = b^{D-4+n-4}\Delta\lambda_n \quad (2.82)$$

for $b < 1$ on $D = 4$:

- Super-renormalizable m^2 grows as $b^{-2} \longleftrightarrow$ relevant operator ϕ^2 .
- Renormalizable λ scales as $b^0 \longleftrightarrow$ marginal operators $\lambda\phi^4$.
- Non-renormalizable λ_n vanishes as $b^{n-4} \longleftrightarrow$ irrelevant operators $\lambda_n\phi^n, n > 4$.

$$\lambda' = \lambda - \frac{3}{16\pi^2}\lambda^2 \ln\left(\frac{1}{b}\right) + O(\lambda^3) \quad (2.83)$$

3. Callan-Symanzik Equation

Goal: Quantify scale dependence of Green's functions $G^{(n)}(x_1 \dots x_n, M)$, where M sets the scale of external momenta, which leads to the Callan-Symanzik equation.

3.1 Alternative choice for renormalization in ϕ^4 -theory

In ϕ^4 -theory, the 1-PI graphs of propagator can be renormalized by taking the renormalization scheme. We simply recall this process again, firstly, taking

$$\text{---} \bigcirc \text{1PI} \text{---} = -i\Sigma(p^2), \quad (2.84)$$

for all 1-PI contributions to propagators. Thus, the full propagator goes like:

$$\begin{aligned} \text{---} \xrightarrow{p} \text{---} \bigcirc \text{---} \xrightarrow{p} \text{---} &= \text{---} \xrightarrow{p} \text{---} + \text{---} \xrightarrow{p} \bigcirc \text{1PI} \xrightarrow{p} \text{---} + \text{---} \xrightarrow{p} \bigcirc \text{1PI} \xrightarrow{p} \bigcirc \text{1PI} \xrightarrow{p} \text{---} + \dots \\ &= \frac{i}{p^2 - m^2 - \Sigma(p^2)}. \end{aligned} \quad (2.85)$$

where $\delta\eta = \frac{-\delta Z}{2}$. The latter gives the way of Green's function changing:

$$G_{\text{ren}}^{(n)} \rightarrow Z^{-n/2}(1 + \delta Z)^{-n/2} G_{\text{bare}}^{(n)} \quad (2.91)$$

$$\simeq Z^{-n/2} \left(1 - \frac{n}{2} \delta Z\right) G_{\text{bare}}^{(n)} \quad (2.92)$$

$$= Z^{-n/2} (1 + n\delta\eta) G_{\text{bare}}^{(n)} \quad (2.93)$$

$$= (1 + n\delta\eta) G_{\text{ren}}^{(n)}, \quad (2.94)$$

We will ignore the corner index “ren” in $G_{\text{ren}}^{(n)}$, **namely the renormalized Green's function denotes as $G^{(n)}$** . Thus from $\delta G_{\text{bare}} = 0$, we have

$$0 = \delta(Z^{n/2} G^{(n)}) = \underbrace{(\delta Z^{n/2}) G^{(n)}}_{n\delta\eta Z^{n/2}} + Z^{n/2} \frac{\partial}{\partial M} G^{(n)} \delta M + Z^{n/2} \frac{\partial}{\partial \lambda} G^{(n)} \delta \lambda. \quad (2.95)$$

Introducing the new variables:

$$\beta = \frac{M}{\delta M} \delta \lambda, \quad \gamma = \frac{M}{\delta M} \delta \eta \quad (2.96)$$

Thus, dropping the arbitrary δM , we obtain **the Callan-Symanzik equation**:

$$\left(M \frac{\partial}{\partial M} + \beta \frac{\partial}{\partial \lambda} + n\gamma \right) G^{(n)}(x_1 \dots x_n, \lambda, M) = 0, \quad (2.97)$$

where β and γ are universal for a given theory, which is independent for particular Green's function. This also means that these parameters are independent of any UV-cut off (dimensionless).

$$\beta = \beta(\lambda), \quad \gamma = \gamma(\lambda). \quad (2.98)$$

We can also directly compute $dG^{(n)}$ as

$$dG^{(n)} = (1 + n\delta\eta) G^{(n)} - G^{(n)} = n\delta\eta G^{(n)} = n \frac{dM}{M} \left(M \frac{d\eta}{dM} \right) G^{(n)} \quad (2.99)$$

$$= \frac{\partial G^{(n)}}{\partial M} dM + \frac{\partial G^{(n)}}{\partial \lambda} d\lambda \quad (2.100)$$

$$= \frac{\partial G^{(n)}}{\partial M} dM + \left(M \frac{d\lambda}{dM} \right) \frac{dM}{M} \frac{\partial G^{(n)}}{\partial \lambda} \quad (2.101)$$

$$= \frac{dM}{M} \left[M \frac{\partial}{\partial M} + \left(M \frac{d\lambda}{dM} \right) \frac{\partial}{\partial \lambda} \right] G^{(n)}, \quad (2.102)$$

from this construction, we can also obtain the Callan-Symanzik equation.

3.2 β -function in massless ϕ^4 -theory

Now let's focus some actual cases to find the *beta*-function and the anomalous dimension $\gamma(\lambda)$. For simplicity, we consider the 1-loop correction for **massless ϕ^4 -theory**. The Callan-Symanzik equation for $G^{(2)}$ gives us $\gamma = 0 + O(\lambda^2)$, because in 1-loop order, there's no

contribution to Z , but only to δm^2 . Only until 2-loops or higher orders which contribute Z and also of course δm^2 .

$$\begin{aligned}
 \text{---} \bigcirc \text{1PI} \text{---} &= \text{---} + \text{---} \bigcirc \text{---} + \frac{1}{2} \left(\text{---} \bigcirc \text{---} \right)^2 + \text{---} \bigcirc \text{---} + \text{---} \bigcirc \text{---} + O(\lambda^3). \\
 &\quad (2.103)
 \end{aligned}$$

For $\beta(\lambda)$ at $O(\lambda)$, we need to consider the 4-point function correction:

$$G^{(4)}(p) = \text{---} \times \text{---} + \left(\text{---} \bigcirc \text{---} + \text{---} \bigcirc \text{---} + \text{---} \bigcirc \text{---} \right) + \text{---} \times \text{---} + O(\lambda^3) \quad (2.104)$$

$$= -i\lambda + (-i\lambda)^2(iV(t) + iV(s) + iV(u)) - i\delta\lambda + O(\lambda^3), \quad (2.105)$$

where the renormalization condition are taken in $t = s = u = -M^2$. So for each diagram vertex correction, for $m = 0$ we have:

$$V(p^2) = \frac{1}{2} \int \frac{d^d k}{(2\pi)^d} \frac{i}{k^2} \frac{i}{(p-k)^2}. \quad (2.106)$$

Counterterm $\delta\lambda$ determined from amputated Green's function

$$G_{\text{amp}}^{(4)}|_{s=t=u=-M^2} \rightarrow \delta\lambda = (-i\lambda)^2 3V(-M^2). \quad (2.107)$$

Take $D = 4 - 2\epsilon$:

$$V(-M^2) = \lim_{\epsilon \rightarrow 0} \frac{1}{2} \frac{-1}{(4\pi)^{d/2}} \int_0^1 dx \frac{\Gamma(\epsilon)}{(x(1-x)M^2)^\epsilon} \quad (2.108)$$

$$= -\frac{1}{32\pi^2} \left(\frac{1}{\epsilon} - \ln M^2 + O(1) \right). \quad (2.109)$$

Thus

$$\delta\lambda = \frac{3\lambda^2}{32\pi^2} \left(\frac{1}{\epsilon} - \ln M^2 + O(1) \right). \quad (2.110)$$

This will lead to

$$G^{(4)} = -i\lambda + 3i(i\lambda)^2 V(p^2) - i\delta\lambda + O(\lambda^3) \quad (2.111)$$

$$= -i\lambda + \frac{3i\lambda^2}{32\pi^2} \left(\frac{1}{\epsilon} - \ln p^2 - \frac{1}{\epsilon} + \ln M^2 \right) + O(\lambda^3) \quad (2.112)$$

$$= -i\lambda + \frac{3i\lambda^2}{32\pi^2} \ln \left(\frac{M^2}{p^2} \right) + O(\lambda^3) \quad (2.113)$$

Thus, we have

$$M \frac{\partial}{\partial M} G^{(4)} = \frac{3i\lambda^2}{16\pi^2} + O(\lambda^3). \quad (2.114)$$

Thus, bring it into the Callan-Symanzik equation, we have

$$-\beta(\lambda) \frac{\partial}{\partial \lambda} G^{(4)} = i \frac{3\lambda^2}{16\pi^2} + O(\lambda^3) \quad (2.115)$$

where $\gamma = 0 + O(\lambda^2)$. Since we only consider order $O(\lambda^2)$, therefore we only need to expand $G^{(4)}$ into $O(\lambda)$, i.e.,

$$\frac{\partial}{\partial \lambda} G^{(4)} = -i + \frac{3i\lambda}{16\pi^2} \ln \left(\frac{M^2}{p^2} \right) + O(\lambda^2) \simeq -i + O(\lambda), \quad (2.116)$$

thus the β -function at 1-loop is:

$$\beta(\lambda) = \frac{3}{16\pi^2} \lambda^2 + O(\lambda^3). \quad (2.117)$$

3.3 Analogous analysis for massless QED

Now we consider the QED, which as a good example to calculate the anomalous dimension $\gamma(\lambda)$. In QED, we need to renormalize the vertex(Z_1) and the propagator for electron(Z_2) and photon(Z_3). These three parameters defined the electron charge renormalization:

$$e_0 = Z_1^{-1} Z_2 Z_3^{1/2} e. \quad (2.118)$$

The Green's function $G^{(n_e, n_A)}(x_1 \dots x_n, e, M)$ now contains two fields, so the Callan-Symanzik equation is

$$\left(M \frac{\partial}{\partial M} + \beta(e) \frac{\partial}{\partial e} + n_e \gamma_2 + n_A \gamma_3 \right) G^{(n_e, n_A)}(x_1 \dots x_n, e, M) = 0, \quad (2.119)$$

where n_e denote for the electron fields, n_A for photon fields. And γ_2, γ_3 are anomalous dimensions of electron and photon field respectively

$$\gamma_i = \frac{1}{2} M \frac{\partial}{\partial M} \delta_i, \quad i = 2, 3. \quad (2.120)$$

The γ_i comes from the field renormalization, which related with Z_2 and Z_3 . And $Z_i = 1 + \delta_i$, where δ_i is the counterterms of fields. So for $i = 2$, i.e., the electron field

$$\begin{aligned} \delta_2 &= \frac{d}{d\bar{p}} \Sigma(p) \Big|_{p^2 = -M^2} = \frac{d}{d\bar{p}} \left(\text{---} \text{---} \text{---} \right) \\ &= \frac{-e^2}{16\pi^2} \left(\frac{\Gamma(\epsilon)}{(M^2)^{\epsilon/2}} + O(1) \right) \simeq \frac{-e^2}{16\pi^2} \left(\frac{2}{\epsilon} - \gamma_E \right) (1 - \epsilon \ln M) \\ &= \frac{-e^2}{16\pi^2} \left(\frac{2}{\epsilon} - \gamma_E - 2 \ln M \right), \end{aligned} \quad (2.121)$$

which leads to

$$\gamma_2 = \frac{e^2}{16\pi^2} \equiv \frac{\alpha}{4\pi}. \quad (2.122)$$

Also $i = 3$ is for the photon field

$$\begin{aligned} \delta_3 &= \Pi(p^2) \Big|_{p^2 = -M^2} = \left(\text{---} \text{---} \text{---} \right) = \left(g^{\mu\nu} - \frac{p^\mu p^\nu}{p^2} \right) \Pi(p^2) \\ &= \frac{-e^2}{12\pi^2} \left(\frac{\Gamma(\epsilon)}{(M^2)^{\epsilon/2}} + O(1) \right) \simeq \frac{-e^2}{12\pi^2} \left(\frac{2}{\epsilon} - \gamma_E - 2 \ln M \right). \end{aligned} \quad (2.123)$$

Thus,

$$\gamma_3 = \frac{e^2}{12\pi^2} \equiv \frac{\alpha}{4\pi} \left(\frac{4}{3} \right). \quad (2.124)$$

For calculating the β -function, we first briefly review a very important solution in QED renormalization. Consider the Ward-Takahashi identity

$$(p - q)_\mu \Gamma^\mu(p, q) = S_F^{-1}(p) - S_F^{-1}(q), \quad (2.125)$$

when we take the limit that $p \rightarrow q$ and move it to the right, thus we have

$$\Gamma^\mu(p, p) = \frac{\partial S_F^{-1}(p)}{\partial p_\mu}. \quad (2.126)$$

Notice $S_F^{-1}(p) = \gamma^\mu p_\mu - m - \Sigma(p)$, thus we have

$$\Gamma^\mu(p, p) = \gamma^\mu - \frac{\partial \Sigma(p)}{\partial p_\mu}. \quad (2.127)$$

Beside the momentum p , the propagator can be renormalized as

$$S_F^{-1}(p) = Z_2^{-1}(\gamma \cdot p - m_{\text{ph}}) + \dots, \quad (2.128)$$

which means

$$\frac{\partial S_F^{-1}}{\partial p_\mu} \approx Z_2^{-1} \gamma^\mu. \quad (2.129)$$

And for the renormalized vertex, we have

$$\Gamma^\mu = Z_1^{-1} \Gamma_R^\mu. \quad (2.130)$$

In low energy approximation, the vertex just $\Gamma_R^\mu \rightarrow \gamma^\mu$. Thus, comparing (2.127) and (2.130), we have

$$Z_1 = Z_2. \quad (2.131)$$

Thus, $\delta_1 = \delta_2$. Then we immediately have

$$0 = \frac{\partial e_0}{\partial M} = \frac{\partial}{\partial M} \left[e \left(1 - \delta_1 + \delta_2 + \frac{1}{2} \delta_3 \right) \right] \quad (2.132)$$

$$= e \frac{\partial}{\partial M} \left(1 - \delta_1 + \delta_2 + \frac{1}{2} \delta_3 \right) + \left(1 - \delta_1 + \delta_2 + \frac{1}{2} \delta_3 \right) \frac{\partial e}{\partial M} \quad (2.133)$$

$$\simeq e \frac{\partial}{\partial M} \left(-\delta_1 + \delta_2 + \frac{1}{2} \delta_3 \right) + \frac{\partial e}{\partial M}. \quad (2.134)$$

Therefore, we get the final result for the β -function:

$$\begin{aligned} \beta(e) &= M \frac{\partial}{\partial M} e = -e M \frac{\partial}{\partial M} \left(\underbrace{-\delta_1 + \delta_2}_{\delta_1 = \delta_2} + \frac{1}{2} \delta_3 \right) \\ &= \frac{e^3}{12\pi^2} \equiv e \frac{\alpha}{4\pi} \left(\frac{4}{3} \right). \end{aligned} \quad (2.135)$$

4. Evolution of coupling

4.1 Solution of Callan-Symanzik equation

Now we would like to solve the solutions of Callan-Symanzik equation. We firstly start with the 2-point function $G^{(2)}(p)$, which generally can be written as:

$$G^{(2)}(p) = \frac{i}{p^2} f\left(\frac{-p^2}{M^2}\right), \quad (2.136)$$

where $f\left(\frac{-p^2}{M^2}\right)$ is a dimensionless function, which only depends on $\frac{-p^2}{M^2}$. In momentum space, without the amputation, which means we add the external lines to contribute the momentum p in the propagator. We have:

$$M \frac{\partial}{\partial M} G^{(2)}(p) = \frac{i}{p^2} \left(\frac{2p^2}{M^2}\right) f'\left(\frac{-p^2}{M^2}\right), \quad (2.137)$$

$$p \frac{\partial}{\partial p} G^{(2)}(p) = -2G^{(2)}(p) - \frac{i}{p^2} \left(\frac{2p^2}{M^2}\right) f'\left(\frac{-p^2}{M^2}\right). \quad (2.138)$$

Therefore, we have:

$$M \frac{\partial}{\partial M} G^{(2)}(p) = \left(-p \frac{\partial}{\partial p} - 2\right) G^{(2)}(p). \quad (2.139)$$

Putting this results into the Callan-Symanzik equation for 2-point function, we then have:

$$\left(p \frac{\partial}{\partial p} - \beta \frac{\partial}{\partial \lambda} + 2 - 2\gamma(\lambda)\right) G^{(2)}(p^2, \lambda, M) = 0. \quad (2.140)$$

This is an equation for a current. We can firstly write the solution in free theory, i.e. $\lambda = 0$ which means $\beta = \gamma = 0$:

$$G^{(2)}(p) = \frac{i}{p^2} \bar{G}_0, \quad (2.141)$$

where $\bar{G}_0$ is a constant function in p^2 . But one can not happy if we only have the free theory. Therefore, we need to solve the renormalization group equation with a more general form. For that, we firstly introduce some new variables to make the PDE simpler:

$$t = \ln\left(\frac{p}{M}\right), \quad (2.142)$$

$$v(x) = -\beta(\lambda), \quad (2.143)$$

$$\rho(x) = 2\gamma(\lambda) - 2, \quad (2.144)$$

$$x = \lambda. \quad (2.145)$$

Now we call the new function as $D(t, x)$, which is nothing but the $G^{(2)}$. So (2.140) becomes

$$\left(\frac{\partial}{\partial t} + v(x) \frac{\partial}{\partial x} - \rho(x)\right) D(t, x) = 0. \quad (2.146)$$

The solution of $D(t, x)$ is somehow like a current with the current density

$$j(t, x) = \rho(t, x) v(t, x). \quad (2.147)$$

So the solution of a current can be expressed as

$$D(t, x) = D_0(\bar{x}(t, x)) \exp\left[\int_0^t dt' \rho(\bar{x}(t', x))\right], \quad (2.148)$$

with the boundary condition and the initial condition:

$$\frac{\partial}{\partial t'} \bar{x}(t', x) = -v(\bar{x}), \quad \bar{x}(0, x) = x, \quad (2.149)$$

where $\bar{x}(t, x)$ is the position of current volum element at t and x . So the Callan-Symanzik equation has the solution:

$$G^{(2)}(p^2, \lambda, M) = \bar{G}_0(\bar{\lambda}(p, \lambda)) \exp \left[\int_{p'=M}^{p'=p} d \ln \left(\frac{p'}{M} \right) 2 [1 - \bar{\gamma}(\bar{\lambda}(p', \lambda))] \right]. \quad (2.150)$$

It is important to note that $\bar{G}_0(\bar{\lambda}(p, \lambda))$ is an unknown function, which does not directly corresponds to the propagator. To fix this function, we firstly put (2.136) into (2.140):

$$0 = \left(p \frac{\partial}{\partial p} - \beta \frac{\partial}{\partial \lambda} + 2 - 2\gamma(\lambda) \right) \left[\frac{i}{p^2} f \left(\frac{p^2}{M^2}, \lambda \right) \right] \quad (2.151)$$

$$= \frac{i}{p^2} \left(-2 + p \frac{\partial}{\partial p} - \beta \frac{\partial}{\partial \lambda} + 2 - 2\gamma(\lambda) \right) f \left(\frac{p^2}{M^2}, \lambda \right) \quad (2.152)$$

Thus, we have

$$\left(p \frac{\partial}{\partial p} - \beta \frac{\partial}{\partial \lambda} - 2\gamma(\lambda) \right) f \left(\frac{p^2}{M^2}, \lambda \right) = 0. \quad (2.153)$$

So the solution is similar with $G^{(2)}$:

$$f \left(\frac{p^2}{M^2}, \lambda \right) = \bar{G}_0(\bar{\lambda}(p, \lambda)) \exp \left[-2 \int_M^p d \ln \left(\frac{p'}{M} \right) \gamma(\bar{\lambda}) \right]. \quad (2.154)$$

Therefore, we can take the another form of the solution by combine the solution of f and (2.136)

$$G^{(2)}(p^2, \lambda, M) = \frac{i}{p^2} \bar{G}_0(\bar{\lambda}(p, \lambda)) \exp \left[-2 \int_M^p d \ln \left(\frac{p'}{M} \right) \gamma(\bar{\lambda}(p', \lambda)) \right]. \quad (2.155)$$

Now, we can take the boundary $p = M$, which all the integration part will vanish. So we have

$$\bar{G}_0(\lambda) = -iM^2 \cdot G^{(2)}(M^2, \lambda, M). \quad (2.156)$$

We can then go to the specific case and use the perturbative expansion of propagator to get each order of $\bar{G}_0(\lambda)$.

4.2 Renormalization group equation

Now the boundary condition and the initial condition in (2.149) are changed to the so-called **renormalization group equation**:

$$\frac{d}{d \ln(p/M)} \bar{\lambda}(p, \lambda) = \beta(\bar{\lambda}), \quad \bar{\lambda}(M, \lambda) = \lambda. \quad (2.157)$$

This equation shows that the flow of the running coupling constant $\bar{\lambda}(p)$ with the speed $\beta(\bar{\lambda})$. And we set the initial energy scale is M , and then set the value at this point is the original coupling constant λ . Now we expand $\bar{\lambda}$ to the first order and use (2.117):

$$\frac{d}{d \ln(p/M)} \bar{\lambda}(p, \lambda) = \beta(\bar{\lambda}) = \frac{3}{16\pi^2} \bar{\lambda}^2 + O(\bar{\lambda}^3). \quad (2.158)$$

So we actually are solving the ODE

$$\frac{d\bar{\lambda}}{dt} = A\bar{\lambda}^2 \quad (2.159)$$

It is easy to solve it by integrating

$$\int_{\bar{\lambda}(M)=\lambda}^{\bar{\lambda}(p)} \bar{\lambda}'^{-2} d\bar{\lambda}' = \int_0^{\ln(p/M)} A dt' \quad (2.160)$$

Thus, we have

$$\frac{1}{\bar{\lambda}(p)} = \frac{1}{\bar{\lambda}} - A \ln(p/M) \quad (2.161)$$

$$= \frac{1 - A\bar{\lambda}(M) \ln(p/M)}{\bar{\lambda}(M)}. \quad (2.162)$$

Put all the parameters back, we have

$$\ln\left(\frac{p}{M}\right) = \left(\frac{3}{16\pi^2}\right)^{-1} \left(\frac{1}{\bar{\lambda}(M)} - \frac{1}{\bar{\lambda}(p)}\right). \quad (2.163)$$

Take $\bar{\lambda}(M) = \lambda$, which leads to the expression for the running coupling $\bar{\lambda}$:

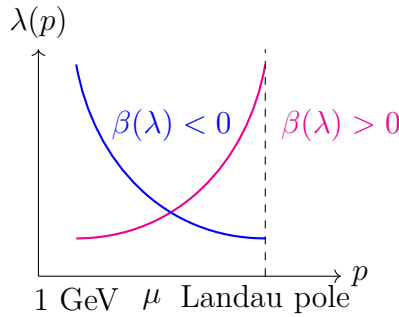
$$\bar{\lambda}(p) = \frac{\lambda}{1 - \frac{3}{16\pi^2} \lambda \ln(p/M)}. \quad (2.164)$$

So $\bar{\lambda}(p)$ grows for increasing scales p . And there is a pole when

$$1 - \frac{3}{16\pi^2} \lambda \ln\left(\frac{p}{M}\right) = 0, \quad (2.165)$$

which is called the **Landau pole**:

$$p = M \exp\left(\frac{16\pi^2}{3\lambda}\right). \quad (2.166)$$



For $p^2 \ll M^2$, theory becomes trivial:

$$\bar{\lambda}(p) \sim \frac{\bar{\lambda}(M)}{\lim_{x \rightarrow 0} (1 - \ln x)} \rightarrow 0_+ \quad (2.167)$$

which corresponds to the free theory.

Generally, we have different cases for the β function, and that corresponds to the different background theory

$$\frac{d}{d \ln(p/M)} \bar{\lambda}(p, \lambda) = \beta(\bar{\lambda}) \begin{cases} > 0 & \text{IR free } (\phi^4\text{-theory, QED, ...}) \\ = 0 & \text{conformal theory } (N = 4 \text{ SYM}) \\ < 0 & \text{UV free/asymptotically free (QCD, SU(N) gauge theory).} \end{cases} \quad (2.168)$$

4.3 Higgs mass and fate of universe

As the second example, we consider the Higgs mechanism. The story begin with the Lagrangian:

$$\mathcal{L}_{\text{unbroken}} = \bar{\psi}(i\gamma^\mu \partial_\mu)\psi + \frac{1}{2}(\partial^\mu \phi)(\partial_\mu \phi) - V(\phi) + y_\psi \phi \bar{\psi}\psi. \quad (2.169)$$

where $y_\psi \phi \bar{\psi}\psi$ is the Higgs-Yukawa interaction, and we dropped all the gauge terms. To get the spontaneous symmetry breaking, we take the potential as

$$V(\phi) = \mu^2 \phi^2 + \lambda \phi^4. \quad (2.170)$$

The spontaneous symmetry breaking happens when

$$\phi = H + v. \quad (2.171)$$

where H is the Higgs field, v is the vacuum expectation value, which in experiment measure are

$$v \simeq \frac{1}{\sqrt{\sqrt{2}G_F}} \simeq 246 \text{ GeV} \quad (2.172)$$

So plug (2.171) into the Lagrangian, we have the spontaneous symmetry breaking Lagrangian:

$$\mathcal{L}_{\text{broken}} = \bar{\psi}(i\not{\partial})\psi + \frac{1}{2}(\partial_\mu H)(\partial^\mu H) - \underbrace{\frac{\lambda v^2}{\frac{1}{2}M_H^2}}_{\frac{1}{2}M_H^2} H^2 - \lambda v H^3 - \frac{\lambda}{4} H^4 - \underbrace{\frac{1}{\sqrt{2}} y_\psi v (1 + H/v)}_{m_\psi} \bar{\psi}\psi \quad (2.173)$$

This mechanism turns out to the particle mass by the Higgs interact with itself and the other fields. So the coupling constant renormalization is very similar with ϕ^4 theory and Yukawa theory:

$$\underbrace{\text{diagram 1}}_{-i\lambda} + \underbrace{\text{diagram 2} + \text{diagram 3} + \text{diagram 4}}_{O(\lambda^2)} + \underbrace{\text{diagram 5}}_{O(y_\psi^4)} \quad (2.174)$$

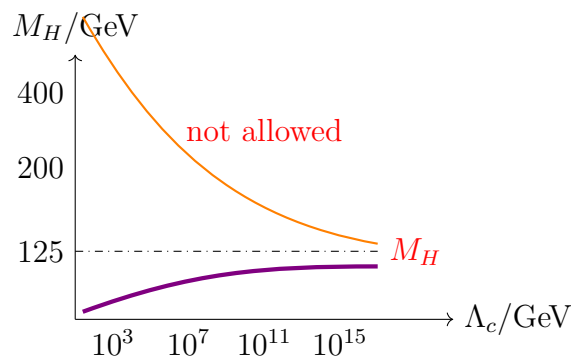
So the solution for $\lambda(Q)$ at scale $Q^2 \gg v$ is:

$$\lambda(Q) = \lambda(v) \frac{1}{1 - \frac{3}{16\pi^2} \lambda(v) \ln(Q/v)} \quad (2.175)$$

Cut off Λ_c corresponds to the Landau pole, up to which perturbative Higgs sector holds:

$$\Lambda_c = v \exp\left(\frac{16\pi^2}{3\lambda}\right) \simeq v \exp\left(\frac{16\pi^2 v^2}{M_H^2}\right) \quad (2.176)$$

where $M_H^2 = 2\lambda v^2$. So the area above the orange line are forbidden.



Now we consider fermion as the top quark, which is the heaviest fermion the standard model. It has contributions to $\beta(\lambda)$ from Higgs-Yukawa interaction, which the box diagram gives

$$Q \frac{d}{dQ} \lambda = \frac{1}{64\pi^2} (-3y_t^4 + 6\lambda y_t^2 + 12\lambda^2) + \dots \quad (2.177)$$

It is interesting that the first term is negative, which means it evolves in the opposite way with the Higgs. Solving this equation for small λ ($\lambda \ll 1$), we have

$$\lambda(Q) = \lambda(v) + \frac{1}{32\pi^2} \left(-\frac{3m_t^4}{v^4} \right) \ln \left(\frac{Q}{v} \right) \quad (2.178)$$

since $m_t = y_t \frac{v}{\sqrt{2}}$, $m_t \sim 172$ GeV, so in principle, $\lambda(Q)$ can be negative at large Q ! That means the vacuum will be super unstable. So to avoid that, we require a bound on Higgs mass:

$$M_H^2 \geq \frac{v^2}{16\pi^2} \left(-\frac{3m_t^4}{v^4} \right) \ln \left(\frac{Q}{v} \right), \quad (2.179)$$

such that we can keep $\lambda(Q) > 0$.

5. Renormalization for local operators and evolution of mass parameter

We first introduce the idea of local operator. The local operator is multiple field operator compositing in the same space-time position, namely the production of fields. This will happen when the energy scale is much lower than the high energy scattering, e.g. weak decay. So to understand this, we can recall the beta decay in electroweak interaction.

The electroweak interaction between left fermions and W-boson is:

$$\delta\mathcal{L} = \frac{g_{ew}}{\sqrt{2}} W_\mu \bar{\psi} \gamma^\mu (1 - \gamma_5) \psi. \quad (2.180)$$

And the propagator of W-boson is

$$\mu \text{ --- } W^- \text{ --- } \nu = \frac{-ig^{\mu\nu}}{q^2 - M_W^2 + i\epsilon} \quad (2.181)$$

In the low energy limit $q^2 \ll M_W^2$, the vertex and propagator of β -decay reduce to the 4-Fermi interaction vertex:

$$\begin{array}{ccc} \begin{array}{c} d \\ \swarrow \\ \bar{u} \end{array} & \begin{array}{c} W^- \\ \text{---} \\ \end{array} & \begin{array}{c} e^- \\ \swarrow \\ \bar{\nu}_e \end{array} \\ & \xrightarrow{q^2 \ll M_W^2} & \\ \begin{array}{c} d \\ \swarrow \\ \bar{u} \end{array} & \begin{array}{c} \bullet \\ \text{---} \\ \end{array} & \begin{array}{c} e^- \\ \swarrow \\ \bar{\nu}_e \end{array} \end{array} \quad (2.182)$$

So we can reduce a process propagate from μ to ν to the process happens in a fixed point due to the reaction happening so fast, i.e.:

$$\begin{array}{c} d \\ \swarrow \\ \bar{u} \end{array} \begin{array}{c} \bullet \\ \text{---} \\ \end{array} \begin{array}{c} e^- \\ \swarrow \\ \bar{\nu}_e \end{array} = \frac{g_{ew}^2}{2M_W^2} (\bar{\psi} \gamma^\mu (1 - \gamma_5) \psi)_{\text{quark}} (\bar{\psi} \gamma_\mu (1 - \gamma_5) \psi)_{\text{lep}} = \frac{g_{ew}^2}{2M_W^2} \mathcal{O}(x). \quad (2.183)$$

Generally, this special vertex can be seen as a Green's function:

$$\langle \psi(p_1) \bar{\psi}(p_2) \psi(p_3) \bar{\psi}(p_4) \mathcal{O}(0) \rangle = \begin{array}{c} p_1 \quad p_3 \\ \swarrow \quad \searrow \\ \bullet \quad \mathcal{O}(0) \\ \nearrow \quad \nwarrow \\ p_2 \quad p_4 \end{array} \quad (2.184)$$

5.1 Renormalized Green's functions with local operator $\mathcal{O}$ in ϕ^4 -theory

Now we consider the renormalization of the local operator. Since the vertex is described by the Green's function, we can then consider a general form:

$$G^{(n,\mathcal{O})}(p_1 \dots p_n, k) = \langle \phi(p_1) \dots \phi(p_n) \mathcal{O}(k) \rangle \quad (2.185)$$

$$= Z^{-n/2}(\mu) \underbrace{Z_{\mathcal{O}}^{-1}(M)}_{\text{operator renormalization}} \underbrace{\langle \phi_0(p_1) \dots \phi_0(p_n) \rangle}_{\text{bare fields}} \underbrace{\mathcal{O}_0(k)}_{\text{bare operator}} \quad (2.186)$$

$$= Z^{-n/2}(\mu) Z_{\mathcal{O}}^{-1}(M) G_0^{(n,\mathcal{O})}(p_1 \dots p_n, k) \quad (2.187)$$

where $\mathcal{O} = \phi^2$ or ϕ^4 or $\dots$. The lower index "0" represent the bare quantity, and M is the renormalization scale. The bare operator and the renormalized operator have the relation:

$$\mathcal{O}_{\text{bare}} = Z_{\mathcal{O}}(M) \mathcal{O}_{\text{ren}}. \quad (2.188)$$

So we can derive the Callan-Symanzik equation by using

$$0 = M \frac{d}{dM} G_0^{(n,\mathcal{O}_0)} \quad (2.189)$$

$$= M \frac{d}{dM} (Z^{n/2} Z_{\mathcal{O}} G^{(n,\mathcal{O})}) \quad (2.190)$$

$$= \left(M \frac{\partial}{\partial M} Z^{n/2} \right) Z_{\mathcal{O}} G^{(n,\mathcal{O})} + Z^{n/2} \left(M \frac{\partial}{\partial M} Z_{\mathcal{O}} \right) G^{(n,\mathcal{O})} + Z^{n/2} Z_{\mathcal{O}} \left(M \frac{\partial}{\partial M} G^{(n,\mathcal{O})} \right), \quad (2.191)$$

dividing $Z^{n/2} Z_{\mathcal{O}}$ on both side, we have

$$0 = \left(\frac{M}{Z^{n/2}} \frac{\partial Z^{n/2}}{\partial M} \right) G^{(n,\mathcal{O})} + \left(\frac{M}{Z_{\mathcal{O}}} \frac{\partial Z_{\mathcal{O}}}{\partial M} \right) G^{(n,\mathcal{O})} + M \frac{\partial}{\partial M} G^{(n,\mathcal{O})} \quad (2.192)$$

$$= \left(M \frac{\partial}{\partial M} \ln Z^{n/2} \right) G^{(n,\mathcal{O})} + \left(M \frac{\partial}{\partial M} \ln Z_{\mathcal{O}} \right) G^{(n,\mathcal{O})} + M \frac{\partial}{\partial M} G^{(n,\mathcal{O})}, \quad (2.193)$$

where we can define the anomalous dimension respectively for field and local operator as:

$$\gamma = \frac{M}{2} \frac{\partial}{\partial M} \ln Z_{\lambda}(M) \quad (2.194)$$

$$\gamma_{\mathcal{O}} = M \frac{\partial}{\partial M} \ln Z_{\mathcal{O}}(M). \quad (2.195)$$

Then we have the Callan-Symanzik equation:

$$\left(M \frac{\partial}{\partial M} + \beta(\lambda) \frac{\partial}{\partial \lambda} + n\gamma(\lambda) + \gamma_{\mathcal{O}}(\lambda) \right) G^{(n,\mathcal{O})} = 0, \quad (2.196)$$

Now we compute $\gamma_{\mathcal{O}}$ for N-point Green's function. The most general case can be expressed as

$$\text{Diagram with red dot and cross} + \sum_i \left(\text{Diagram with red dot and circle} + \text{Diagram with red dot and cross} \right) + \text{Diagram with red cross}, \quad (2.197)$$

where the red point and red cross represent the operator vertex and the operator counterterm.

5.2 Renormalization for local operators in massless ϕ^4 -theory

As an example to help us understanding this. We consider $\mathcal{O} = \phi^2$ inserted in two-point function:

$$\text{Diagram with red dot and cross} = \langle \Omega | \phi(q) \phi(p) \phi^2(k) | \Omega \rangle = G^{(2, \phi^2)}. \quad (2.198)$$

Thus, we can expand it as the normal Dyson series:

$$G^{(2, \phi^2)}(q, p; k) = \sum_{m=0}^{\infty} \frac{i^m}{m!} \int \left[\prod_{a=1}^m d^d x_a \right] \left(-\frac{\lambda}{4!} \right)^m \langle 0 | T \{ \phi(q) \phi(p) \phi^2(k) [\phi^4(x_1) \dots \phi^4(x_m)] \} | 0 \rangle_{\text{connect}}. \quad (2.199)$$

The leading order, namely the tree-level non-amputated Green's function is just simply the contraction of the field and the operator:

$$G_{\text{tree}}^{(2, \phi^2)}(q, p; k) = \text{Diagram with red dot and cross} = \int d^d x e^{-ik \cdot x} \langle 0 | T \{ \phi(q) \phi(p) \phi_A(x) \phi_B(x) \} | 0 \rangle \quad (2.200)$$

$$= \int d^d x e^{-ik \cdot x} \left[\overbrace{\phi(q) \phi(p) \phi_A(x) \phi_B(x)} + \overbrace{\phi(q) \phi(p) \phi_A(x) \phi_B(x)} \right] \quad (2.201)$$

$$= \left(\frac{i}{q^2} \right) \left(\frac{i}{p^2} \right) \cdot 2. \quad (2.202)$$

To consider the renormalization for the operator, we need to compute 1-PI contribution at 1-loop (non-amputated):

$$G_{1\text{-loop}}^{(2,\phi^2)}(q, p; k) = \text{u+v} \quad \text{Diagram: a circle with an incoming line from the top labeled } ik, \text{ and two outgoing lines at the bottom labeled } q \text{ and } p. \quad = \int d^d x d^d z e^{-ik \cdot x} \left\langle 0 \left| T \left\{ \phi(q) \phi(p) \left[\frac{-i\lambda}{4!} \phi^4(z) \right] \left[\frac{1}{2} \phi_A(x) \phi_B(x) \right] \right\} \right| 0 \right\rangle \quad (2.203)$$

$$= \frac{-i\lambda}{2} \int d^d x d^d z e^{-ik \cdot x} \left[\overbrace{\phi(q) \phi(p)} \quad \overbrace{\phi(z) \phi(z)} \quad \overbrace{\phi(z) \phi(z)} \quad \overbrace{\phi_A(x) \phi_B(x)} \right] \quad (2.204)$$

$$= \left(\frac{i}{q^2} \right) \left(\frac{i}{p^2} \right) \int \frac{d^d v}{(2\pi)^d} (-i\lambda) \frac{i}{v^2} \frac{i}{(k+v)^2}, \quad (2.205)$$

with some calculation, one has:

$$G_{1\text{-loop}}^{(2,\phi^2)}(q, p; k) = \frac{i}{p^2} \frac{i}{q^2} \left(-\frac{\lambda}{(4\pi)^{d/2}} \frac{\Gamma(2 - \frac{d}{2})}{\Delta^{2-d/2}} \right) \quad (2.206)$$

with the counterterm:

$$\text{Diagram: a vertex with two outgoing lines labeled } q \text{ and } p. \quad = \frac{i}{p^2} \frac{i}{q^2} 2\delta\phi^2 \quad (2.207)$$

Comparing of coefficients in these two graphs, we then can find the counterterm at scale $p^2 = -M^2$ should be:

$$\delta\phi^2 = \frac{\lambda}{2(4\pi)^{d/2}} \frac{\Gamma(2 - \frac{d}{2})}{(M^2)^{2-d/2}}. \quad (2.208)$$

This implies the operator renormalization is

$$Z_{\phi^2} = 1 - \delta\phi^2 = 1 - \frac{\lambda}{2(4\pi)^{d/2}} \frac{\Gamma(2 - \frac{d}{2})}{(M^2)^{2-d/2}} \quad (2.209)$$

$$\stackrel{d=4-\epsilon}{=} 1 - \frac{\lambda}{(4\pi)^2} \frac{1}{\epsilon} + \text{finite terms}. \quad (2.210)$$

Using $\ln(1+x) \approx x$, we can obtain the anomalous dimension of the operator in $d = 4 - \epsilon$:

$$\gamma_{\phi^2} = M \frac{\partial}{\partial M} \ln Z_{\phi^2} = \frac{-\lambda}{2(4\pi)^{d/2}} \Gamma\left(2 - \frac{d}{2}\right) \cdot \left[-(4-d) M^{-(4-d)} \right] \quad (2.211)$$

$$\stackrel{d=4-\epsilon}{=} \frac{-\lambda}{2(4\pi)^{2-\epsilon/2}} \Gamma\left(\frac{\epsilon}{2}\right) \cdot \left[-\epsilon M^{-\epsilon} \right] \quad (2.212)$$

$$= \frac{\lambda}{16\pi^2}. \quad (2.213)$$

More generally, if we insert the operator into an n-point Green's function, we then need to consider the field renormalization with the operator renormalization $Z_\phi^{n/2} Z_{\mathcal{O}}$, therefore the anomalous dimension becomes:

$$\gamma_{\mathcal{O}}(\lambda) = M \frac{\partial}{\partial M} \left(-\delta_{\mathcal{O}} + \frac{n}{2} \delta_Z \right), \quad (2.214)$$

with the approximation:

$$\ln(Z_\phi^{n/2} Z_\mathcal{O}) = \frac{n}{2} \ln Z_\phi + \ln Z_\mathcal{O} \approx \frac{n}{2} \delta_Z - \delta_\mathcal{O}. \quad (2.215)$$

This case can be treated as renormalized ϕ^4 -theory with effective operator $-\frac{1}{2}m^2\phi^2$, where ϕ^2 is local operator inserted in Green's function. So the Callan-Symanzik equation with mass anomalous dimension $m(\mu)$ is

$$\left(M \frac{\partial}{\partial M} + \beta(\lambda) \frac{\partial}{\partial \lambda} + n\gamma(\lambda) + \gamma_{\phi^2} m^2 \frac{\partial}{\partial m^2} \right) G^{(n, \phi^2)}(\{p_i\}, M, \lambda, m) = 0 \quad (2.216)$$

5.3 Evolution of mass parameter

Moreover, one can write the generalization to local operators $\mathcal{O}_i$ in ϕ^4 -theory as:

$$\mathcal{L}(C_i) = \mathcal{L}_{\text{eff}} + \sum_i C_i \mathcal{O}_i(x) \quad (2.217)$$

where C_i is the coupling of local operators. So the Callan-Symanzik equation is

$$\left(M \frac{\partial}{\partial M} + \beta(\lambda) \frac{\partial}{\partial \lambda} + n\gamma(\lambda) + \sum_i \gamma_i(\lambda) C_i \frac{\partial}{\partial C_i} \right) G^{(n)}(M, \lambda, \{C_i\}) = 0 \quad (2.218)$$

To guarantee the dimension of Lagrangian is $[\text{Mass}]^4$ in 4-dimension spacetime, so we need to explicit mass dimension in coupling C_i as:

$$C_i = \rho_i M^{4-d_i} \quad (2.219)$$

where ρ_i is dimensionless, d_i is dimension of $\mathcal{O}_i$. Therefore, one can see that the variables change to $(M, \lambda, \{\rho_i\})$. To make the partial derivative independent to ρ_i , we have

$$M \frac{\partial}{\partial M} \Big|_{\rho_i} = M \frac{\partial}{\partial M} \Big|_{C_i} + \sum_i \left(M \frac{\partial C_i}{\partial M} \Big|_{\rho_i} \right) \frac{\partial}{\partial C_i} \quad (2.220)$$

Notice that

$$\rho_i \frac{\partial}{\partial \rho_i} = \rho_i M^{4-d_i} \frac{\partial}{\partial C_i} = C_i \frac{\partial}{\partial C_i} \quad (2.221)$$

so we have

$$M \frac{\partial}{\partial M} \Big|_{C_i} = M \frac{\partial}{\partial M} \Big|_{\rho_i} - \sum_i (4 - d_i) \rho_i \frac{\partial}{\partial \rho_i} \quad (2.222)$$

Then the Callan-Symanzik equation becomes:

$$\left(M \frac{\partial}{\partial M} + \beta(\lambda) \frac{\partial}{\partial \lambda} + n\gamma(\lambda) + \sum_i (\gamma_i(\lambda) + d_i - 4) \rho_i \frac{\partial}{\partial \rho_i} \right) G^{(n)} = 0, \quad (2.223)$$

where we can define the β function for new coupling ρ_i :

$$\beta_i(\lambda) = (\gamma_i(\lambda) + d_i - 4) \rho_i, \quad (2.224)$$

then the Callan-Symanzik equation can be reduced as:

$$\left(M \frac{\partial}{\partial M} + \beta(\lambda) \frac{\partial}{\partial \lambda} + n\gamma(\lambda) + \sum_i \beta_i \rho_i \frac{\partial}{\partial \rho_i} \right) G^{(n)} = 0. \quad (2.225)$$

With this, we can easily solve the running coupling ρ_i from the renormalization group equation for β_i

$$\frac{d}{d \ln(p/M)} \bar{\rho}_i = \beta_i(\bar{\rho}_i, \lambda). \quad (2.226)$$

Since β_i in the low energy limit, namely $\gamma_i(\lambda) \ll 1$, is dominated by the classical dimension $d_i - 4$, we then have

$$\frac{d}{d \ln(p/M)} \bar{\rho}_i \simeq (d_i - 4 + \dots) \bar{\rho}_i \quad (2.227)$$

Therefore, the solution in low energy limit reads:

$$\bar{\rho}_i = \rho \left(\frac{p}{M} \right)^{d_i - 4}. \quad (2.228)$$

Now we can clearly see that for the operators with mass dimension $d_i > 4$ will vanish as $p \rightarrow 0$, e.g., ϕ^6 theory with $d_i = 6$. These operators are called the **irrelevant operators**, which usually decrease fast until vanish in low energy limit, namely one can never feel these operators' contribution. On the contrary, the **relevant operators** are the operators with mass dimension $d_i < 4$, e.g., the mass term $\frac{1}{2}m^2\phi^2$, which will exponentially increase in low energy limit. And the operators like ϕ^4 are called the **marginal operators**, which do not changing with the energy scale due to $d_i = 4$. This type operators are only determined by the anomalous dimension $\gamma_i(\lambda)$.

This reminds us that the effective field theory truly works even though we don't know all the details in high energy level. But we still can derive the precise renormalizable Lagrangian in low energy level.

Non-Abelian gauge theories

1. Recap of gauge theory and brief look at Lie group theory

1.1 Recap: Abelian gauge theory

We firstly start with QED, which as we know is an Abelian $U(1)$ gauge theory. It start from the free fermion Lagrangian:

$$\mathcal{L} = i\bar{\psi}\gamma^\mu\partial_\mu\psi - m\bar{\psi}\psi. \quad (3.1)$$

The Noether current is: $j^\mu = \bar{\psi}\gamma^\mu\psi$. To approach the QED Lagrangian, we introduce the (global) gauge transformation as:

$$\psi \rightarrow \psi' = e^{i\alpha}\psi \quad (3.2)$$

$$\bar{\psi} \rightarrow \bar{\psi}' = e^{-i\alpha}\bar{\psi} \quad , \quad \alpha \in \mathbb{R}. \quad (3.3)$$

The local gauge transformation is just changing the parameter α rely on the coordinate:

$$\alpha \rightarrow \alpha(x). \quad (3.4)$$

The gauge symmetry is saying that the Lagrangian is invariant under the gauge transformation. This requires us to introduce the covariant derivative D_μ :

$$D_\mu\psi(x) = (\partial_\mu - igA_\mu)\psi(x). \quad (3.5)$$

where in QED case, the charge $g = e$, and the gauge field A_μ is just the photon field, which satisfies the transformation rule:

$$A_\mu \rightarrow A'_\mu = A_\mu + \frac{1}{g}\partial_\mu\alpha(x). \quad (3.6)$$

Then we immediately have the covariant derivative transfers as

$$D_\mu\psi \rightarrow (D_\mu\psi)' = \partial_\mu\psi' - igA'_\mu\psi' = e^{i\alpha}D_\mu\psi. \quad (3.7)$$

Thus the invariant of $\mathcal{L}$ requires that

$$\mathcal{L}(\psi, A_\mu) = \mathcal{L}(\psi', A'_\mu) = i\bar{\psi}\gamma^\mu D_\mu\psi - m\bar{\psi}\psi \quad (3.8)$$

$$= i\bar{\psi}\gamma^\mu\partial_\mu\psi - m\bar{\psi}\psi + gA_\mu j^\mu. \quad (3.9)$$

For photon dynamical, we need to add kinetic term for A_μ :

$$\mathcal{L}_{\text{Maxwell}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}. \quad (3.10)$$

with $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.

1.2 $SU(N)$ gauge theory

Now we consider the non-Abelian $SU(N)$ gauge theory with N Dirac spinors ψ_j , where $j = 1, \dots, N$. The original Lagrangian is

$$\mathcal{L} = \sum_{j=1}^N i\bar{\psi}_j \gamma^\mu \partial_\mu \psi_j - m\bar{\psi}_j \psi_j. \quad (3.11)$$

Under the global unitary transformation $U_{ij} \in \mathbb{C}$, $UU^\dagger = \mathbb{1}$, the spinors obey

$$\psi_j \rightarrow \psi'_j = \sum_k U_{jk} \psi_k, \quad (3.12)$$

$$\bar{\psi}_j \rightarrow \bar{\psi}'_j = \sum_k \bar{\psi}_k U_{kj}^\dagger, \quad (3.13)$$

thus the mass term transformation under $SU(N)$

$$\sum_j \bar{\psi}_j \psi_j \rightarrow \sum_j \bar{\psi}'_j \psi'_j = \sum_{k,l,j} \bar{\psi}_k \underbrace{U_{kj}^\dagger U_{jl}}_{\delta_{kl}} \psi_l = \sum_k \bar{\psi}_k \psi_k \quad (3.14)$$

is invariant.

Similar with the Abelian case, the local gauge transformation is $U \rightarrow U(x)$. We also require the Lagrangian invariant under the transformation. This gives coupling of fermions to gauge field in covariant derivative analogous to QED:

$$D_\mu \psi_j = \partial_\mu \psi_j - ig \sum_k (A_\mu)_{jk} \psi_k. \quad (3.15)$$

This form of covariant derivative also requires the transformation rule as

$$D_\mu \psi_j \rightarrow (D_\mu \psi_j)' = \partial_\mu \psi'_j - ig \sum_k (A'_\mu)_{jk} \psi'_k \quad (3.16)$$

$$= \sum_k U_{jk} (D_\mu \psi_k) \quad (3.17)$$

For convenient, we will use the short-hand notation (Matrix notation), which basically just ignore the index of the matrix element:

$$D_\mu \psi \rightarrow (D_\mu \psi)' = U D_\mu \psi = e^{i\alpha(x)} D_\mu \psi. \quad (3.18)$$

We claim that the gauge field transforms following

$$A_\mu \rightarrow A'_\mu = U A_\mu U^\dagger - \frac{i}{g} (\partial_\mu U) U^\dagger. \quad (3.19)$$

Although we can directly use the definition to compute this expression, I just don't want to write so much. Instead, we can have a quick check that this claim indeed satisfies our requirement:

$$D_\mu \psi \rightarrow (D_\mu \psi)' = \partial_\mu (U\psi) - ig \left(U A_\mu U^\dagger - \frac{i}{g} (\partial_\mu U) U^\dagger \right) U\psi \quad (3.20)$$

$$= U \partial_\mu \psi + (\partial_\mu U) \psi - ig U A_\mu \psi - (\partial_\mu U) \psi \quad (3.21)$$

$$= U (\partial_\mu \psi - ig A_\mu \psi) = U D_\mu \psi. \quad (3.22)$$

Thus, the Lagrangian is invariant under $SU(N)$ gauge transformation.

Now we need to approach the dynamical of non-Abelian gauge field, which requires a field strength. Since we keep calling it as “non-Abelian”, but until now it’s not quite obvious to see how it is “non-Abelian”, therefore, the field strength should give you some direct idea when we claim:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu]. \quad (3.23)$$

We can also check it is true that this field strength indeed satisfies our requirement, namely keep the Lagrangian invariant. To show this, we firstly have the transformation rule:

$$F_{\mu\nu} \rightarrow F'_{\mu\nu} = \partial_\mu A'_\nu - \partial_\nu A'_\mu - ig[A'_\mu, A'_\nu] = U F_{\mu\nu} U^\dagger. \quad (3.24)$$

The Lagrangian is proportion to the trace over $F_{\mu\nu} F^{\mu\nu} = (F_{\mu\nu})_{jk} (F^{\mu\nu})^{kj} = \text{tr}(F_{\mu\nu} F^{\mu\nu})$, due to the Lagrangian is a scalar, thus the only possible combination is the trace over $F_{\mu\nu}$. Thus $\text{tr} F_{\mu\nu}$ transfers as

$$\begin{aligned} \text{tr}(F_{\mu\nu} F^{\mu\nu}) &\rightarrow \text{tr}(F'_{\mu\nu} F'^{\mu\nu}) = \text{tr}(U F_{\mu\nu} U^\dagger U F^{\mu\nu} U^\dagger) \\ &= \text{tr}(U U^\dagger F_{\mu\nu} F^{\mu\nu}) \\ &= \text{tr}(F_{\mu\nu} F^{\mu\nu}). \end{aligned} \quad (3.25)$$

Then we get **the Lagrangian of Yang-Mills theory**

$$\mathcal{L} = -\kappa \text{tr}(F_{\mu\nu} F^{\mu\nu}) + \sum_{j,k=1}^N [\bar{\psi}_j \gamma^\mu (D_\mu)_{jk} \psi_k - m \bar{\psi}_j \psi_j], \quad (3.26)$$

where κ is normalized constant.

1.3 Recap for essential Lie group theory

To continue our journey, it’s necessary to master some basic knowledge about Lie group and Lie algebra. So in the next, we will recall some essential knowledge about these mathematical tools and show how are they used in gauge theory.

Firstly, we recall the infinitesimal transformation of the gauge operation. For Abelian theory, e.g. QED:

$$e^{i\alpha} \simeq 1 + i\alpha. \quad (3.27)$$

For non-Abelian gauge theory, e.g. Yang Mills theory:

$$U = \mathbb{1} + \hat{H} + \mathcal{O}(\hat{H}^2), \quad (3.28)$$

$$U^\dagger = \mathbb{1} + \hat{H}^\dagger + \mathcal{O}(\hat{H}^2). \quad (3.29)$$

Thus we have

$$U^{-1} = U^\dagger \implies \hat{H} = -\hat{H}^\dagger \quad (\text{anti-hermitian}) \quad (3.30)$$

where we note $\hat{H} = iH$ with $H = H^\dagger$. Expand $\hat{H}$ in parameters of transformation α^a

$$\hat{H}_{jk} = i \sum_a \alpha^a (t^a)_{jk} \quad (3.31)$$

with $\alpha^a \in \mathbb{R}$, $t^a = (t^a)^\dagger$ is hermitian. Thus t^a span a basis in the space of hermitian $N \times N$ matrices such that we can expand $\hat{H}$ in terms of this basis. t^a are called the generators of gauge transformations.

Generally, the unitary $N \times N$ matrix has N^2 linear independent generators due to t^a is $N \times N$. However, in our case, we require $\det U = 1$, which gives one constraint. So there are $\mathbf{N}^2 - \mathbf{1}$ **generators in $\mathbf{SU}(\mathbf{N})$** . Further more, base on $\det U = 1$ and the Jacobi identity $\det(e^M) = e^{\text{tr}(M)}$, we find

$$\det U = \det(e^{\hat{H}}) = e^{\text{tr} \hat{H}} = 1, \quad (3.32)$$

thus, $\text{tr} \hat{H} = 0$ and implies

$$\text{tr}(t^a) = 0. \quad (3.33)$$

So **t^a are traceless generators.**

Now we discuss the algebra structure for generators t^a . The Lie algebra, namely the commutator is given by:

$$[t^a, t^b] = i \sum_c f^{abc} t^c. \quad (3.34)$$

where f^{abc} is the structure constant of Lie algebra. The elements of Lie group can be expanded around $\mathbb{1}$.

$$g(\alpha) = \mathbb{1} + i \sum_{a=1}^d \alpha_a t^a + \frac{1}{2} \sum_{a,b=1}^d \alpha_a \alpha_b T^{ab} + \mathcal{O}(\alpha^3), \quad (3.35)$$

where $T^{ab} = T^{ba}$ is symmetric, which can be expanded in terms of the generator t^a, t^b . Notice $g(0) = \mathbb{1}$ and because of $g(\alpha)\mathbb{1} = \mathbb{1}g(\alpha)$, we then have

$$\gamma_a(\alpha_1, \dots, \alpha_d; 0, \dots, 0) = \gamma_a(0, \dots, 0; \alpha_1, \dots, \alpha_d) = \alpha_a \quad (3.36)$$

where the γ is defined by the group multiplication:

$$g(\alpha_1, \dots, \alpha_d) \cdot g(\beta_1, \dots, \beta_d) = g(\gamma_1, \dots, \gamma_d), \quad (3.37)$$

in which γ is the function of α and β :

$$\gamma_a = \gamma_a(\alpha_1, \dots, \alpha_d; \beta_1, \dots, \beta_d). \quad (3.38)$$

Base on the fact that $\gamma(\alpha, 0) = \alpha$ and $\gamma(0, \beta) = \beta$, then we have only one choice to expand γ_a until the second order to keep these relation, namely:

$$\gamma_a = \alpha_a + \beta_a + \sum_{b,c} C_{bc}^a \alpha_b \beta_c + \dots, \quad (3.39)$$

where C_{bc}^a is a unknown coefficient. Thus, use $g(\alpha)g(\beta) = g(\gamma)$, we have for the left-hand side:

$$g(\alpha)g(\beta) = \mathbb{1} + i \sum_a (\alpha_a + \beta_a) t^a + \frac{1}{2} \sum_{a,b} [(\alpha_a \alpha_b + \beta_a \beta_b) T^{ab} - 2\alpha^a \beta^b t^a t^b] + \dots, \quad (3.40)$$

And for the right-hand side:

$$g(\gamma) = \mathbb{1} + i \sum_a \gamma_a t^a + \frac{1}{2} \sum_{a,b} \gamma_a \gamma_b T^{ab} + \dots \quad (3.41)$$

Thus use (3.39), then the right-hand side becomes

$$g(\gamma(\alpha, \beta)) = \mathbb{1} + i \sum_a (\alpha_a + \beta_a) t^a + i \sum_{a,b,c} C_{bc}^a \alpha_b \beta_c t^a + \frac{1}{2} \sum_{a,b} (\alpha_a + \beta_a)(\alpha_b + \beta_b) T^{ab} + \dots \quad (3.42)$$

Compare (3.40) with (3.42), we find

$$-t^a t^b = i \sum_c C_c^{ab} t^c. \quad (3.43)$$

this gives the structure constant:

$$[t^a, t^b] = i \sum_{c=1}^d (C_c^{ba} - C_c^{ab}) t^c \quad (3.44)$$

$$= i \sum_{c=1}^d f^{abc} t^c \quad (3.45)$$

where $f^{abc} = -f^{bac}$, and d is the dimension of Lie algebra.

Examples of Lie groups

- **GL(N)**: Group of $N \times N$ matrices in $\mathbb{R}^N$ with $\det M \neq 0$.
- **U(N)**: Group of unitary $N \times N$ matrices in $\mathbb{C}^N$, satisfies

$$U^\dagger U = U U^\dagger = \mathbb{1} \quad \text{and} \quad \det U \neq 0. \quad (3.46)$$

Infinitesimal version:

$$U = \mathbb{1} + i \sum_{a=1}^d \alpha_a t^a. \quad (3.47)$$

with $d = N^2$ is the dimension of $U(N)$ Lie algebra/group.

- **SU(N)** := $\{A \in U(N) | \det A = 1\}$. Dimension of Lie algebra $\mathfrak{su}(N)$ is $d = N^2 - 1$. Generators t^a are traceless

$$\det U = \det \left(\mathbb{1} + i \sum_{a=1}^d \alpha_a t^a \right) = 1 + i \sum_{a=1}^d \alpha_a \text{tr } t^a = 1 \quad (3.48)$$

thus

$$\text{tr } t^a = 0 \quad (3.49)$$

- **O(N)**: group of orthogonal $N \times N$ matrices in $\mathbb{R}^N$.

$$O O^T = O^T O = \mathbb{1} \quad (3.50)$$

Infinitesimal version:

$$O = \mathbb{1} + i \sum_{a=1}^d \alpha_a t^a \quad (3.51)$$

Dimension of Lie algebra $\mathfrak{o}(N)$ is $\frac{1}{2}N(N-1)$. And notice:

$$O O^T = \left(\mathbb{1} + i \sum_{a=1}^d \alpha_a t^a \right) \left(\mathbb{1} + i \sum_{b=1}^d \alpha_b (t^b)^T \right) = \mathbb{1}, \quad (3.52)$$

thus,

$$\mathbb{1} = \mathbb{1} + i \sum_{a=1}^d \alpha_a t^a + i \sum_{b=1}^d \alpha_b (t^b)^T + \mathcal{O}(\alpha^2) \quad (3.53)$$

$$= \mathbb{1} + i \sum_{a=1}^d (\alpha_a t^a + \alpha_a (t^a)^T), \quad (3.54)$$

which implies

$$t^a = -(t^a)^T. \quad (3.55)$$

- $\mathbf{SO}(N) := \{A \in O(N) \mid \det A = 1\}$
- $\mathbf{SO}(N, M) = \{\Lambda \in \text{Mat}(N+M, \mathbb{R}) \mid \Lambda^T \eta \Lambda = \eta, \det \Lambda = 1\}$, where

$$\eta = \text{diag}(\underbrace{1, \dots, 1}_N, \underbrace{-1, \dots, -1}_M) \quad (3.56)$$

- **Lorentz group:** $SO(1, 3)$

1.4 Representations of Lie algebra

Def: Representation of group

A **representation of a group G on a d_r -dimensional vector space V** is a group homomorphism

$$\rho : G \rightarrow GL(V), \quad (3.57)$$

which maps each group element $g \in G$ to an invertible linear operator $\rho(g)$ acting on V , such that:

$$\rho(g_i) \cdot \rho(g_j) = \rho(g_i g_j) \quad \forall g_i, g_j \in G. \quad (3.58)$$

The dimension d_r of the vector space V is called **the dimension of the representation**.

- A representation ρ is called **reducible** if there exists a non-trivial proper subspace $U \subset V$ (where $U \neq \{0\}$ and $U \neq V$) that is invariant under the action of all group elements, meaning:

$$\rho(g)u \in U \quad \forall g \in G, \forall u \in U. \quad (3.59)$$

- If no such invariant subspace exists, the representation is called irreducible (irrep).

For a Lie group characterized by continuous parameters α , a representation is a mapping $\rho : g(\alpha) \rightarrow M(\alpha)$, where the matrices (or operators) $M(\alpha)$ preserve the group multiplication:

$$M(\alpha) \cdot M(\beta) = M(\gamma) \quad \text{whenever} \quad g(\alpha) \cdot g(\beta) = g(\gamma). \quad (3.60)$$

In the next, we introduce some useful representations.

• Fundamental(defining) representation

For a matrix Lie group, the **fundamental representation** is simply the group of matrices themselves, acting on their natural vector space.

Example: $SU(2)$ fundamental representation

- **Representation Space (V):** A 2-dimensional complex vector space $\mathbb{C}^2$, physically corresponding to a spin-1/2 state

$$|\psi\rangle = \alpha |\uparrow\rangle + \beta |\downarrow\rangle. \quad (3.61)$$

- **Dimension of the representation:** $d = 2$ (since the matrices are 2×2). Note that the dimension of the $SU(2)$ group/algebra itself is $2^2 - 1 = 3$ (the number of

independent generators). The 3 Hermitian generators for this $d = 2$ fundamental representation are the Pauli matrices:

$$t^a = \frac{1}{2}\sigma^a \quad (a = 1, 2, 3), \quad (3.62)$$

which satisfy the Lie algebra $[t^a, t^b] = i\epsilon^{abc}t^c$, and the Pauli matrix identity:

$$\sigma^k \sigma^l = \delta^{kl} \mathbb{1} + i\epsilon^{klm} \sigma^m \quad (3.63)$$

Physically, we can use these fundamental building blocks to construct higher-dimensional representations via tensor products, such as the 3-dimensional spin-1 triplet state. In fact, $SU(2)$ has spin- j representations with dimension $2j + 1$. For instance, $j = 1/2$ corresponds to the 2-dimensional fundamental representation using 2×2 Pauli matrices.

- **Adjoint representation**

Instead of acting on physical states, the **adjoint representation** is a d -dimensional representation where the group/algebra acts on the Lie algebra space itself (spanned by the generators $\{t^a\}$ as basis vectors). The definition is as follows

Adjoint representation

We first define a automorphism map of group G :

$$\Psi : G \rightarrow \text{Aut}(G) \quad (3.64)$$

$$g \mapsto hgh^{-1} \equiv \Psi_h(g), \quad \forall h \in G, \quad (3.65)$$

where $\text{Aut}(G)$ is the automorphism group of G . Then we can define the derivative of Φ_g at the origin noted as Ad_g :

$$\text{Ad}_g = (d\Phi_g)_e : T_e G \rightarrow T_e G, \quad (3.66)$$

where $T_e G$ is the tangent space in the natural element of G . Thus the **adjoint representation** is a homomorphism:

$$\text{Ad} : G \rightarrow \text{Aut}(T_e G) \quad (3.67)$$

$$g \mapsto \text{Ad}_g. \quad (3.68)$$

To obtain the explicit matrix form of the adjoint representation, we consider an infinitesimal group element $g = e + i\alpha^a t^a \in G$. Acting on a generator $t^b \in T_e G$ via the adjoint map Ad_g , we expand to the linear order of α :

$$\text{Ad}_{e+i\alpha^a t^a}(t^b) = (e + i\alpha^a t^a)t^b(e - i\alpha^a t^a) \simeq t^b + i\alpha^a [t^a, t^b] + \mathcal{O}(\alpha^2) \quad (3.69)$$

This motivates us to define the infinitesimal adjoint map $\text{ad}_{t^a} \equiv d(\text{Ad})_e(t^a)$ on the Lie algebra, which acts purely as a commutator:

$$\text{ad}_{t^a}(t^b) = \lim_{\alpha^a \rightarrow 0} \frac{(\text{Ad}_{e+i\alpha^a t^a}(t^b))_{kl} - (t^b)_{kl}}{i\alpha^a} = [t^a, t^b]_{kl} = i \sum_{c=1}^d f^{abc}(t^c)_{kl}. \quad (3.70)$$

Viewing ad_{t^a} as a linear transformation matrix acting on the basis vector t^b to produce t^c , the expansion coefficients (structure constants) directly define the matrix elements of the adjoint generators:

$$(t_{\text{adj}}^a)_{bc} = i f^{abc} \quad (3.71)$$

Hence, the structure constants f^{abc} are not just algebraic constants, they are precisely **used to construct the adjoint representation**.

In this representation, the group elements act on the Lie algebra itself. The generators are $d \times d$ matrices (where $d = N^2 - 1$), with matrix elements defined by (3.71). We can verify that these matrices satisfy the Lie algebra by using the Jacobi identity:

$$\sum_{A=1}^{d_r} (f^{aAe} f^{bcA} + f^{bAe} f^{caA} + f^{cAe} f^{abA}) = 0. \quad (3.72)$$

Writing the commutation relation

$$[t_{\text{adj}}^a, t_{\text{adj}}^b]_{de} = i \sum_{c=1}^d f^{abc} (t_{\text{adj}}^c)_{de}, \quad (3.73)$$

in terms of structure constants gives:

$$\sum_{A=1}^d [(i f^{aAd})(i f^{bAe}) - (i f^{bAd})(i f^{aAe})] = i \sum_{c=1}^d f^{abc} (i f^{cde}). \quad (3.74)$$

Thus we have:

$$\sum_{A=1}^d (-f^{aAd} f^{bAe} + f^{bAd} f^{aAe}) = - \sum_{A=1}^d f^{abA} f^{Ade}. \quad (3.75)$$

Using the anti-symmetry property of structure constant: $f^{abc} = -f^{bac}$, we can reduce to the standard Jacobi identity form:

$$0 = \sum_{A=1}^d (-f^{aAd} f^{bAe} + f^{bAd} f^{aAe} + f^{Ade} f^{abA}) \quad (3.76)$$

$$= \sum_{A=1}^d (f^{aAd} f^{beA} + f^{bAd} f^{eaA} + f^{eAd} f^{abA}), \quad (3.77)$$

which is nothing but the Jacobi identity (3.72).

1.5 Inner product of generators

Now since the Lie algebra generators are all matrices, so it's natural to consider the inner product of two matrices. The most natural way is by the trace of matrix product. So we define a matrix D_r^{ab} for generators t_r^a as

$$D_r^{ab} = \text{tr}(t_r^a t_r^b). \quad (3.78)$$

For $a, b = 1, \dots, d = N^2 - 1$. Cyclicity of trace doesn't change the matrix, so it's symmetry and real-valued:

$$D_r^{ab} = D_r^{ba}, \quad (3.79)$$

and also Hermitian:

$$(D_r^{ab})^\dagger = \text{tr}((t_r^a)^\dagger (t_r^b)^\dagger) = D_r^{ab}. \quad (3.80)$$

As we know from linear algebra, for any real-valued symmetry matrix, one can always diagonalize it by a orthogonal transformation. Thus, diagonalization of D_r^{ab} gives

$$\text{tr}(t_r^a t_r^b) = C(r) \delta^{ab}. \quad (3.81)$$

Since the generators are diagonal, then the position of each components in the same t_r are equal, which imply that $C(r)$ is the same for any component of t_r . Thus $C(r)$ is called the **index of representation** or **Dynkin index**.

Example: $SU(N)$

The generators for $SU(2)$ is $t_{\text{fund}}^a = \frac{\sigma^a}{2}$ ($a = 1, 2, 3$), where σ^a is the Pauli matrix:

$$\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (3.82)$$

For the fundamental representation matrix, we can check that

$$\text{tr}(t_r^a t_r^a) = \frac{1}{2}. \quad (3.83)$$

So

$$C(r) = \frac{1}{2}. \quad (3.84)$$

More over, for general case $SU(N)$, $C(r) = \frac{1}{2}$ due to the normalization of N -dim matrices are all similar with $SU(2)$. Using the index of representation, we can find the structure constants of $SU(N)$ satisfy:

$$\begin{aligned} \text{tr}([t^a, t^b] t_r^c) &= i f^{abc} \text{tr}(t_r^c t_r^c) \\ &= \text{tr}(t_r^a t_r^b t_r^c - t_r^b t_r^a t_r^c) = i f^{abc} C(r) \\ &= \text{tr}(t_r^a [t_r^b, t_r^c]) = i f^{abc} C(r). \end{aligned} \quad (3.85)$$

This shows that f^{abc} are totally anti-symmetric:

$$f^{abc} = -f^{bac} = f^{bca}. \quad (3.86)$$

As we all learned from quantum mechanics: $[J^2, J_i] = 0$. So similarly, one can define an operator T^2 that satisfies the similar commutation relation. This operator is called the **quadratic Casimir operator**, which is defined as:

$$T_{ij}^2 = \sum_{a=1}^{d(G)} \sum_{k=1}^{d(r)} t_{ik}^a t_{kj}^a, \quad (3.87)$$

where t_{ij}^a is fundamental representation. We can check that for any a , we all have: $[t^a, T^2] = 0$, namely T^2 commutes with all generators:

$$[t^a, T^2] = \left[t^a, \sum_b t^b t^b \right] = \sum_b ([t^a, t^b] t^b + t^b [t^a, t^b]) \quad (3.88)$$

$$= \sum_b \left(\left(i \sum_c f^{abc} t^c \right) t^b + t^b \left(i \sum_c f^{abc} t^c \right) \right) \quad (3.89)$$

$$= i \sum_{b,c} f^{abc} (t^c t^b + t^b t^c) \quad (3.90)$$

$$= i \sum_{c,b} f^{acb} (t^b t^c + t^c t^b) \quad (3.91)$$

$$= i \sum_{b,c} (-f^{abc}) (t^c t^b + t^b t^c) \quad (3.92)$$

$$= -i \sum_{b,c} f^{abc} (t^c t^b + t^b t^c) \quad (3.93)$$

$$= 0 \quad (3.94)$$

This implies that T^2 must be proportional to identity:

$$T_{ij}^2 = C_2(r) \delta_{ij} \quad (3.95)$$

And $C_2(r)$ is called the **quadratic Casimir coefficient**. It can be obtained by taking the trace in both sides:

$$\text{tr}(T^2) = \text{tr} \left(\sum_{a=1}^{d(G)} t^a t^a \right) = \sum_{a=1}^{d(G)} \text{tr}(t^a t^a) = C_2(r) \text{tr} \delta_{ij}, \quad (3.96)$$

thus we have:

$$C(r) d(G) = C_2(r) d(r) \quad (3.97)$$

1.6 Representation of Lorentz group

The Lorentz group, as we mentioned before, is $SO(1, 3)$ with metric $\eta = (1, -1, -1, -1)$. The group element Λ with infinitesimal expansion can be written as

$$\Lambda_\nu^\mu = \delta_\nu^\mu + i \sum_a \omega^a (t^a)_\nu^\mu + \mathcal{O}(\omega^2). \quad (3.98)$$

Defining relation of $SO(N, M)$ groups says

$$\Lambda^T \eta \Lambda = \eta, \quad (3.99)$$

thus the infinitesimal form implies

$$(\mathbb{1} + i\omega^a (t^a)^T) \cdot \eta \cdot (\mathbb{1} + i\omega^a t^a) = \eta, \quad (3.100)$$

after dropping the $\mathcal{O}(\omega^2)$, one obtains the anti-symmetric relation:

$$(t^a)^T \eta = -\eta t^a. \quad (3.101)$$

As we mentioned in the last subsection, $SO(1, 3)$, as a subgroup of $O(1, 3)$, has $\frac{1}{2} \times 4 \times (4-1) = 6$ generators, physically are 3 rotation + 3 boosts, which usually noted as $J^{\mu\nu}$. To limit the number of free parameters in $J^{\mu\nu}$, we require

$$J^{\rho\sigma} = -J^{\sigma\rho}. \quad (3.102)$$

So we change the note as

$$\sum_a \omega^a t^a \rightarrow \sum_{\rho\sigma} \omega_{\rho\sigma} J^{\rho\sigma}, \quad (3.103)$$

where $J^{\rho\sigma}$ fulfill $SO(1, 3)$ algebra. Generally, all $SO(N, M)$ groups have also spinor representations generators

$$S^{\mu\nu} = \frac{i}{4} [\gamma^\mu, \gamma^\nu], \quad (3.104)$$

fulfill $SO(1, 3)$ algebra:

$$[J^{\mu\nu}, J^{\rho\sigma}] = i(\eta^{\nu\rho} J^{\mu\sigma} - \eta^{\mu\rho} J^{\nu\sigma} - \eta^{\nu\sigma} J^{\mu\rho} + \eta^{\mu\sigma} J^{\nu\rho}). \quad (3.105)$$

1.7 Integration over gauge group

Since a Lie group is also a manifold, so we can introduce the integration structure based on the group element g and function $f(g)$, the measurement dg is called Haar measure, which include the left and right invariant measure:

$$\text{left invariant measure : } \int (dg)_L f(g) = \int (dg)_L f(g'g); \quad (3.106)$$

$$\text{right invariant measure : } \int (dg)_R f(g) = \int (dg)_R f(gg'), \quad (3.107)$$

where the normalization is $\int dg \, 1 = 1$, and for most of situation the left and right Haar measure are the same $dg = (dg)_L = (dg)_R$. This measurement is just like the normal integral is invariant under the transformation $x \rightarrow x + C$, where C is a constant:

$$\int dx f(x) = \int dx f(x + C). \quad (3.108)$$

To obtain the computable form of the measure, we need explicit realization of dg with metric tensor M_{ij} :

$$M_{ij} \equiv \text{tr}(g^{-1} \partial_i g \cdot g^{-1} \partial_j g), \quad (3.109)$$

where $\partial_i g = \frac{\partial}{\partial \alpha_i} g(\alpha)$. Then we can connect to the normal integral:

$$\int dg f(g) = N \int d\alpha |\det M|^{1/2} f(g(\alpha)), \quad (3.110)$$

in which the normalization is the Jacobi determinant.

Example

- $U(1) = \{e^{i\theta} \mid -\pi < \theta < \pi\}$: For $g(\theta) = e^{i\theta}$ we have

$$M_{\theta\theta} = \text{tr}(g^{-1} \partial_\theta g \cdot g^{-1} \partial_\theta g) = \text{tr}(i \cdot i) = -1. \quad (3.111)$$

And using the fact

$$1 = N \int_{-\pi}^{\pi} d\theta \cdot 1 = N \cdot 2\pi, \quad (3.112)$$

thus $N = \frac{1}{2\pi}$. So the integral is

$$\int dg f(g) = \frac{1}{2\pi} \int_{-\pi}^{\pi} d\theta f(e^{i\theta}) \quad (3.113)$$

- $SU(2) = \{a_0 + i\vec{\sigma} \cdot \vec{a} \mid a_0^2 + \vec{a}^2 = 1\}$: For the element $g(a)$ we have:

$$\partial_0 g = \frac{\partial g}{\partial a_0} = \mathbb{1}_{2 \times 2} \quad (3.114)$$

$$\partial_k g = \frac{\partial g}{\partial a_k} = i\sigma_k \quad (k = 1, 2, 3) \quad (3.115)$$

Thus, the metric is

$$M = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & -2 \end{pmatrix}, \quad (3.116)$$

therefore, $|\det M|^{1/2} = \sqrt{16} = 4$. The last piece is the normalization factor, for that we need to compute the integral for unity. Since the constrain require a sphere, we then can express the unity by delta function:

$$1 = N \int d^4a \delta(a^2 - 1) = N \int_0^\infty r^3 dr \int d\Omega_3 \delta(r^2 - 1), \quad (3.117)$$

thus $N = \frac{1}{\pi^2}$. And finally:

$$\int dg f(g) = \frac{1}{\pi^2} \int d^4a \delta(a^2 - 1) f(g(a)). \quad (3.118)$$

- More general case, for $SU(N)$, Schur's orthogonality relation requires

$$\int dg g_{ij} (g^{-1})_{kl} = \frac{1}{N} \delta_{il} \delta_{jk}. \quad (3.119)$$

2. Quantization of non-Abelian gauge theory

2.1 Preparation

To quantize the non-Abelian gauge theory, we need to list what do we have now, everything begins with the Lagrangian:

$$\mathcal{L} = \sum_{j=1}^{d(r)} (i\bar{\psi}_j \gamma^\mu (D_\mu)_{jk} \psi_k - m\bar{\psi}_j \psi_j) - \kappa \text{tr}(F_{\mu\nu} F^{\mu\nu}), \quad (3.120)$$

where the covariant derivative is defined as:

$$(D_\mu)_{jk} = \partial_\mu \delta_{jk} - ig(A_\mu)_{jk}, \quad (3.121)$$

and the field strength is:

$$(F_{\mu\nu})_{jk} = [D_\mu, D_\nu]_{jk}. \quad (3.122)$$

Under the gauge transformation, the gauge fields obey the rule:

$$A_\mu \rightarrow A'_\mu = U A_\mu U^\dagger - \frac{i}{g} (\partial_\mu U) U^\dagger. \quad (3.123)$$

To work in the perturbation theory, we need to explicit show the colour indices. Thus we expand A_μ in basis of adjoint representation:

$$(A_\mu)_{jk} = \sum_{a=1}^{d(G)} A_\mu^a t_{jk}^a \quad (3.124)$$

The infinitesimal transformation with $U_{jk} = \mathbb{1}_{jk} + i \sum_a \alpha^a (t^a)_{jk}$ then becomes:

$$\begin{aligned} (\delta A_\mu)_{jk} &= (A'_\mu)_{jk} - (A_\mu)_{jk} \\ &= -(A_\mu)_{jk} + \left(\mathbb{1} + i \sum_a \alpha^a (t^a) \right)_{jm} (A_\mu)_{mn} \left(\mathbb{1} - i \sum_b \alpha^b (t^b) \right)_{nk} + \frac{1}{g} \sum_a (\partial_\mu \alpha^a) (t^a)_{jm} \left(\mathbb{1} - i \sum_b \alpha^b (t^b) \right)_{mk} \\ &\simeq i \sum_{a=1}^{d(G)} \alpha^a A_\mu^b [t^a, t^b]_{jk} + \frac{1}{g} \sum_{a=1}^{d(G)} (\partial_\mu \alpha^a) t_{jk}^a + \mathcal{O}(\alpha^2) \\ &= \frac{1}{g} \left[-g \sum_{b,c=1}^{d(G)} f^{abc} A_\mu^c \alpha^b + \partial_\mu \alpha^a \right] t_{jk}^a, \end{aligned} \quad (3.125)$$

thus we have

$$\delta A_\mu^a = \frac{1}{g} (D_\mu \alpha)^a = \frac{1}{g} \sum_{b=1}^{d(G)} (D_\mu)^{ab} \alpha^b. \quad (3.126)$$

where $(D_\mu)^{ab}$ is the covariant derivative in adjoint representation:

$$(D_\mu)^{ab} = \partial_\mu \delta^{ab} - g \sum_{c=1}^{d(G)} f^{abc} A_\mu^c. \quad (3.127)$$

2.2 Quantization by path integral

Now we are ready to start quantization via path integral method. We have already seen that the action $S[A]$ is gauge invariant. But

$$\int \mathcal{D}A e^{-S[A]} \quad (3.128)$$

is not well defined, because $\mathcal{D}A$ counts infinite gauge redundant. And if one integrate two times over the pure Yang-Mills term $F_{\mu\nu}^a F^{\mu\nu a}$, one can see that the momentum $M_{\mu\nu}$ satisfies

$$M_{\mu\nu} = -k^2 \eta_{\mu\nu} + k_\mu k_\nu, \quad (3.129)$$

which is degenerated (or not invertible):

$$M_{\mu\nu} k^\nu = -k^2 k_\mu + k_\mu k^2 = 0, \quad (3.130)$$

which means the determinate is 0. So we cannot define the propagator due to it's nothing but the inverse of the momentum operator, which we proved in chapter 1.

The idea to solve these problem comes from Faddeev and Popov. To divide out redundant gauge degrees of freedom, we require a gauge condition

$$G^a(A^u) = 0, \quad (3.131)$$

for a particular point A^u in an orbit of the gauge group, where u is a group element of the non-Abelian gauge group (e.g. $SU(3)$). Therefore, A^u is a gauge choice for A , in the other words, A is gauge-less. So like the way we having $\int \delta(x-a)dx = 1$, we insert the gauge condition into the delta function and define a similar quantity called **Faddeev-Popov(FP) determinant** $\Delta_G[\mathbf{A}]$:

$$\Delta_G^{-1}[A] \equiv \int \mathcal{D}u \delta(G^a(A^u)). \quad (3.132)$$

So by this definition, one can get a useful identity:

$$1 = \Delta_G[A] \int \mathcal{D}u \delta(G^a(A^u)). \quad (3.133)$$

One can check that this functional determinant is gauge invariant by continuously doing two times gauge transformation: u and u' . So the fields change to $(A^u)^{u'} = A^{u'u}$. So by definition, the new determinant is given by

$$\Delta_G^{-1}[A^{u'}] = \int \mathcal{D}u \delta(G^a((A^{u'})^u)) = \int \mathcal{D}u \delta(G^a(A^{u'u})), \quad (3.134)$$

Now we introduce a new group element $u'' = u'u$, in which transformation the measure keeps invariant: $\mathcal{D}u = \mathcal{D}u''$, thus

$$\Delta_G^{-1}[A^{u'}] = \int \mathcal{D}u'' \delta(G^a(A^{u''})). \quad (3.135)$$

Changing the integral variable name from u'' to u , we then verify that the FP determinant is indeed invariant under gauge transformation:

$$\Delta_G^{-1}[A^{u'}] = \int \mathcal{D}u \delta(G^a A^u) = \Delta_G^{-1}[A] \quad (3.136)$$

Now we insert (3.133) into path integral

$$\int \mathcal{D}A (1) e^{-S[A]} = \int \mathcal{D}A \Delta_G[A] \int \mathcal{D}u \delta(G^a(A^u)) e^{-S[A]}. \quad (3.137)$$

Since only the delta function in the middle integral is u dependence, and the measure $\mathcal{D}$, the determinant $\Delta_G[A]$ and the action $S[A]$ are all gauge invariant, so we can change the variable (or choose this specific gauge) $A \rightarrow A' = A^u$ without causing any difference:

$$\int \mathcal{D}A \Delta_G[A] \int \mathcal{D}u \delta(G^a(A^u)) e^{-S[A]} = \int \mathcal{D}A'^{u^{-1}} \Delta_G[A'^{u^{-1}}] \int \mathcal{D}u \delta(G^a(A')) e^{-S[A'^{u^{-1}}]} \quad (3.138)$$

$$= \int \mathcal{D}A' \Delta_G[A'] \int \mathcal{D}u \delta(G^a(A')) e^{-S[A']}. \quad (3.139)$$

The trick that we are playing now can be understood with a simple comparison in calculus. For an integral in one dimension

$$I = \int_{-\infty}^{+\infty} dx \int_0^{2\pi} d\theta f(x + \theta)g(x + \theta), \quad (3.140)$$

which obviously is θ dependent. But we can change the variable $y \equiv x + \theta$. Since θ in the second integral is a constant, thus $dy = dx$ with the same integration range. So the integral changes to

$$I = \int_{-\infty}^{+\infty} dy \int_0^{2\pi} d\theta f(y)g(y), \quad (3.141)$$

where the integrand is now θ independent. With the same trick, now the functional integral is u independent, so we can change the integration sequence and rename the variable from A' to A as

$$\left(\int \mathcal{D}u \right) \cdot \int \mathcal{D}A \Delta_G[A] \delta(G^a(A)) e^{-S[A]}, \quad (3.142)$$

and since the integral over $\mathcal{D}u$ represents the “volume” of the gauge group, which is infinite. So we can just consider the rest part, which is physical and removed all the gauge redundancy as the correct definition of path integral for non-Abelian gauge theories:

$$\int \mathcal{D}A \Delta_G[A] \delta(G^a(A)) e^{-S[A]}. \quad (3.143)$$

So far so good, it seems that we successfully eliminated the gauge redundancy. However, if we consider a specific example, say Coulomb gauge $G^a(A) = \nabla \cdot A^a = 0$, and consider the derivative of G with respect to α , so we have

$$\frac{\delta G_\mu^a}{\delta \alpha^b} = \nabla \cdot \frac{\delta A_\mu^a}{\delta \alpha^b} = \frac{1}{g} \nabla \cdot D_\mu^{ab}, \quad (3.144)$$

where δA_μ^a is given by (3.126), and (3.127) gives

$$\vec{\nabla} \cdot \vec{D}^{ab} = \nabla^2 \delta^{ab} - g f^{abc} \vec{A}^c \cdot \vec{\nabla}, \quad (3.145)$$

which is the Coulomb gauge $\vec{\nabla} \cdot \vec{D}^{ab}$ operator. The problem is it can have non-trivial solution for A_i , namely,

$$(\vec{\nabla} \cdot \vec{D}^{ab}) \alpha^b = 0, \quad (3.146)$$

where $\alpha^b(x) \neq 0$. It means we find a new field by the gauge transformation $\alpha^b(x)$ as $A' = A + \delta A$, with the new gauge condition:

$$G^a(A') = G^a(A) + \delta G^a = 0 + \frac{1}{g} (\vec{\nabla} \cdot \vec{D}^{ab}) \alpha^b = 0, \quad (3.147)$$

which also satisfies the Coulomb gauge $\vec{\nabla} \cdot \vec{A}'^a = 0$. This problem is called the **Gribov Ambiguity**, which only troubles in non-Abelian gauge theory, e.g. QCD, due to the Abelian gauge theory have the boundary $\alpha \rightarrow 0$ at infinity and zero structure constant $f^{abc} = 0$. But for non-Abelian theory, with larger enough A in low energy confinement case, it's possible for a non-zero α satisfying

$$\nabla^2 \alpha^a = g f^{abc} \vec{A}^c \cdot \vec{\nabla} \alpha^b. \quad (3.148)$$

But at least we solved the divergent problem due to infinite gauge choice with the price of introducing a delta function and an FP determinant. These two, however also make trouble because we cannot expand perturbation series directly. The latter is limited by the delta function that only valid around $G = 0$. Therefore, to compute FP determinant, we can do the infinitesimal transformation around $G = 0$. We first consider $G(A^u)$ as function of group element $u(\alpha)$, then change the integral measure in (3.132) to: $\mathcal{D}u = \mathcal{D}G \det \left| \frac{\delta \alpha}{\delta G} \right|$, so we have:

$$\Delta_G^{-1}[A] = \int \mathcal{D}G \det \left| \frac{\delta \alpha}{\delta G} \right| \delta(G). \quad (3.149)$$

Integration over gauge group G gives the FP determinant, then we get:

$$\Delta_G[A] = \det \left| \frac{\delta G}{\delta \alpha} \right|_{G=0}. \quad (3.150)$$

So (3.143) becomes:

$$\int \mathcal{D}A \delta(G(A)) \det \left| \frac{\delta G}{\delta \alpha} \right| e^{-S[A]}. \quad (3.151)$$

Next step we are going to put $\delta(G)$ and FP determinant into the exponential to get a pure path integral form. For that, we consider the Gauss integral

$$\int_{-\infty}^{+\infty} dc e^{-\frac{c^2}{2\xi}} \delta(G - c) = e^{-\frac{G^2}{2\xi}}. \quad (3.152)$$

Generating into the functional space, with $a = 1, \dots, d(G)$, and change the gauge $G^a = 0$ to $G^a = C^a$ with constants C^a . We have:

$$\int \mathcal{D}C^a e^{-\frac{1}{2\xi} \int d^4x C^a C^a} \delta(G^a - C^a) = e^{-\frac{1}{2\xi} \int d^4x G^a G^a}, \quad (3.153)$$

where we successfully put the delta function into the exponential. The price is adding a **gauge fixing term** in Lagrangian:

$$S_{\text{gf}} = \int d^4x \frac{1}{2\xi} G^a \cdot G^a. \quad (3.154)$$

Now we left a determinant needs to be eliminated: $\det \left| \frac{\delta G}{\delta \alpha} \right|$. This can be done by introducing the Grassmann number in the path integral:

$$\det \left| \frac{\delta G^a(\alpha(x))}{\delta \alpha^b(y)} \right| = \int \mathcal{D}\bar{\eta} \mathcal{D}\eta e^{+ \int d^4x d^4y \bar{\eta}^a(x) \frac{\delta G^a(x)}{\delta \alpha^b(y)} \eta^b(y)}, \quad (3.155)$$

where these two new field η and $\bar{\eta}$ is the **Faddeev-Popov Ghost fields**:

$$S_{\text{ghost}} = - \int d^4x d^4y \bar{\eta}^a(x) \frac{\delta G^a(x)}{\delta \alpha^b(y)} \eta^b(y). \quad (3.156)$$

Now we are ready to write the final form of Faddeev-Popov quantization:

$$\int \mathcal{D}A \mathcal{D}\bar{\eta} \mathcal{D}\eta e^{-S[A, \bar{\eta}, \eta]}, \quad (3.157)$$

by inserting (3.154) and (3.155) into the Lagrangian:

$$S[A, \bar{\eta}, \eta] = S_{\text{YM}}[A] + S_{\text{gf}}[A] + S_{\text{ghost}}[A, \bar{\eta}, \eta] \quad (3.158)$$

$$= \int d^4x \left[\frac{1}{4} F_{\mu\nu}^a F^{a\mu\nu} - \bar{\psi}(i\not{D} - m)\psi \right] + \frac{1}{2\xi} \int d^4x G^a(x) G^a(x) - \int d^4x d^4y \bar{\eta}^a(x) \frac{\delta G^a(x)}{\delta \alpha^b(y)} \eta^b(y). \quad (3.159)$$

It is worthy to note that we used **Euclidean convention**, which is

$$\int \mathcal{D}A e^{-S}. \quad (3.160)$$

If we use **Minkowski convention** with the sign $(+ - - -)$, which is

$$\int \mathcal{D}A e^{iS}. \quad (3.161)$$

So the action becomes

$$S_M[A, \bar{\eta}, \eta] = \int d^4x \left[\bar{\psi}(i\not{D} - m)\psi - \frac{1}{4}F_{\mu\nu}^a F^{a\mu\nu} \right] - \frac{1}{2\xi} \int d^4x G^a(x)G^a(x) + \int d^4x d^4y \bar{\eta}^a(x) \frac{\delta G^a(x)}{\delta \alpha^b(y)} \eta^b(y). \quad (3.162)$$

2.3 Feynman rules for non-Abelian gauge theory

The gauge condition we choose is the **covariant gauge**(**Lorenz gauge**)

$$G^a = \partial^\mu A_\mu^a = 0. \quad (3.163)$$

The quadratic terms in A_μ from $F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + gf^{abc}A_\mu^b A_\nu^c$ and $G^a = \partial^\mu A_\mu^a$ can be rewritten by the partial integration as:

$$S_{\text{quadratic}} = \int d^4x \left(\frac{1}{4}(\partial^\mu A_\nu^a - \partial^\nu A_\mu^a)(\partial_\mu A_\nu^a - \partial_\nu A_\mu^a) + \frac{1}{2\xi}(\partial^\mu A_\mu^a)(\partial^\nu A_\nu^a) \right) \quad (3.164)$$

$$= \int d^4x \left(\frac{1}{2}\partial^\mu A^{\nu a}\partial_\mu A_\nu^a - \frac{1}{2}\partial^\nu A^{\mu a}\partial_\mu A_\nu^a + \frac{1}{2\xi}(\partial^\mu A_\mu^a)(\partial^\nu A_\nu^a) \right) \quad (3.165)$$

$$= \frac{1}{2} \int d^4x A_\mu^a \left(-\partial_\rho \partial^\rho \delta_{\mu\nu} + \left(1 - \frac{1}{\xi}\right) \partial_\mu \partial_\nu \right) A^{\nu a} + \text{Boundary terms}. \quad (3.166)$$

Now with this gauge fixing term, we can check that the momentum operator becomes:

$$M_{\mu\nu} = -k^2 \eta_{\mu\nu} + \left(1 - \frac{1}{\xi}\right) k_\mu k_\nu, \quad (3.167)$$

satisfying:

$$M_{\mu\nu} k^\nu = -k^2 k_\mu + \left(1 - \frac{1}{\xi}\right) k_\mu k^2 = -\frac{1}{\xi} k^2 k_\mu \neq 0. \quad (3.168)$$

Thus, we can then define the **propagators** legally by take inverse of momentum operator, which can be read from (3.166) as

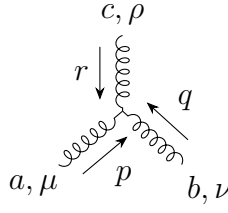
$$a, \mu \xrightarrow{p} b, \nu = \frac{-i\delta^{ab}}{p^2 + i\varepsilon} \left(\eta_{\mu\nu} - (1 - \xi) \frac{p_\mu p_\nu}{p^2} \right). \quad (3.169)$$

And the **gauge boson self-interaction** is vertices is given by the non-Abelian term in $\frac{1}{4}F^2$:

$$\mathcal{L}_{3A} = -gf^{abc}(\partial_\mu A_\nu^a)A^{\mu b}A^{\nu c} \quad (3.170)$$

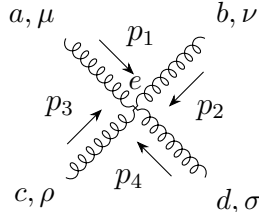
$$\mathcal{L}_{4A} = \frac{1}{4}g^2 f^{abc} f^{ade} A_\mu^b A_\nu^c A^{d\mu} A^{e\nu}. \quad (3.171)$$

The first three gauge boson self interaction gives:



$$= -igf^{abc} [(r_\mu - q_\mu)\delta_{\nu\rho} + (q_\rho - p_\rho)\delta_{\mu\nu} + (p_\nu - r_\nu)\delta_{\mu\rho}] \quad (3.172)$$

The second four gauge boson self interaction gives:



$$= -ig^2 [f^{abe} f^{cde} (\delta_{\mu\rho}\delta_{\nu\sigma} - \delta_{\mu\sigma}\delta_{\nu\rho}) + f^{ace} f^{bde} (\delta_{\mu\nu}\delta_{\rho\sigma} - \delta_{\mu\sigma}\delta_{\nu\rho}) + f^{ade} f^{bce} (\delta_{\mu\nu}\delta_{\rho\sigma} - \delta_{\mu\rho}\delta_{\nu\sigma})]. \quad (3.173)$$

Since we have introduced a new field: ghost field $\bar{\eta}, \eta \rightarrow$, we then can also find its Feynman rule from the Lagrangian. For this, we need to first calculate the derivative by (3.126) and the gauge $G^a = \partial^\mu A_\mu^a$:

$$\frac{\delta G^a(x)}{\delta \alpha^b(y)} = \frac{\delta G^a(x)}{\delta A_\mu^c(x)} \frac{\delta A_\mu^c(x)}{\delta \alpha^b(y)} = \frac{1}{g} \frac{\delta G^a(x)}{\delta A_\mu^c(x)} D_\mu^{bc} \delta(x-y) = \frac{1}{g} \partial^\mu D_\mu^{bc} \delta(x-y), \quad (3.174)$$

with the form of covariant derivative, we have:

$$\frac{\delta G^a(x)}{\delta \alpha^b(y)} = \frac{1}{g} \partial^\mu (\partial_\mu \delta^{ab} - g f^{abc} A_\mu^c) \delta(x-y). \quad (3.175)$$

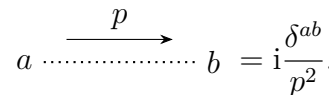
Thus the ghost action becomes:

$$S_{\text{ghost}} = \int d^4x \bar{\eta}^a(x) \left[\frac{1}{g} \partial^\mu (\partial_\mu \delta^{ab} - g f^{abc} A_\mu^c) \right] \eta^b(x) \quad (3.176)$$

$$= \frac{1}{g} \int d^4x \bar{\eta}^a \partial_\mu \partial^\mu \eta^a - \int d^4x f^{abc} \bar{\eta}^a \partial^\mu (A_\mu^c \eta^b) \quad (3.177)$$

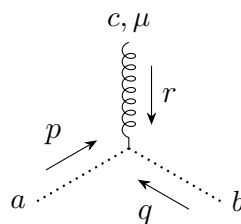
$$= \int d^4x \bar{\eta}^a \partial_\mu \partial^\mu \eta^a - g \int d^4x f^{abc} \bar{\eta}^a \partial^\mu (A_\mu^c \eta^b) \quad (3.178)$$

where we rescaled the field as $\eta \rightarrow \sqrt{g}\eta$ and $\bar{\eta} \rightarrow \sqrt{g}\bar{\eta}$ in the last step. Thus the first term is kinetic term and second is interaction term. So the kinetic term gives the **ghost propagators**:



$$a \cdots \cdots b = i \frac{\delta^{ab}}{p^2}. \quad (3.179)$$

And second interaction term gives the **vertex of ghost-ghost-gluon**:



$$= \frac{g}{2} f^{abc} (r^\mu + p^\mu - q^\mu) = -g f^{abc} q^\mu, \quad (3.180)$$

where the vertex energy-momentum conservation requires $r^\mu + p^\mu + q^\mu = 0$.

3. BRST Symmetry

3.1 Problem of negative norm

In last section, we introduced the gauge fixing term in the action to absorbing the gauge condition $G^a = 0$:

$$S_{\text{gf}} = \int d^4x \frac{1}{2\xi} G^a \cdot G^a, \quad (3.181)$$

which leads to the dynamical term of the gauge field as

$$\frac{1}{2\xi} (\partial^\mu A_\mu^a)(\partial^\nu A_\nu^a). \quad (3.182)$$

However, this operation breaks the local gauge symmetry because it contains $\partial_0 A_0$. The problem of this term is A_0 has a negative norm:

$$\langle 0 | A_\mu(k) A_\nu^\dagger(k') | 0 \rangle \propto -\eta_{\mu\nu} \delta^{(4)}(k - k'), \quad (3.183)$$

thus

$$\langle 0 | A_0(k) A_0^\dagger(k') | 0 \rangle \propto -\delta^{(4)}(k - k') < 0. \quad (3.184)$$

This should not be a problem if local gauge symmetry was not broken, because there wouldn't be the term like $\partial_0 A_0$ get into the kinematic term, i.e., propagator. So the negative norm wasn't shown when we are doing real calculations. Now everything is on the table, the optical theorem tell us we need to count all the contributions including the negative norm final states. So it may cause the sum over all physical final states (positive norm states) is greater than 1. And since the gauge choice can be changed, then the value of the probability may no longer conserved or even depends on the gauge choice, which is unacceptable.

3.2 Infinitesimal BRST transformation

To solve this problem, Becchi, Rouet, Stora and Tyutin(BRST) found that there exist a global symmetry, namely the **BRST symmetry**, which allows us to separate the non-physical states.

To introduce this symmetry, we first consider the Gaussian integral formula:

$$\int_{-\infty}^{\infty} dB \exp \left\{ -\frac{a}{2} B^2 \right\} = \sqrt{\frac{2\pi}{a}}. \quad (3.185)$$

Change the variable $B \rightarrow B + \frac{J}{a}$:

$$\int_{-\infty}^{\infty} dB \exp \left\{ -\frac{a}{2} \left(B + \frac{J}{a} \right)^2 \right\} = \int_{-\infty}^{\infty} dB \exp \left\{ -\frac{a}{2} B^2 - JB - \frac{J^2}{2a} \right\} = \text{Constant}. \quad (3.186)$$

Thus, dividing $\exp\left(\frac{J^2}{2a}\right)$ on both sides, we have

$$\exp \left\{ \frac{J^2}{2a} \right\} \propto \int_{-\infty}^{\infty} dB \exp \left\{ -\frac{a}{2} B^2 - JB \right\}. \quad (3.187)$$

Now we can rewrite the gauge fixing term from the Faddeev-Popov Lagrangian by letting $J = \partial^\mu A_\mu^a$ and $a = -\xi$ as

$$\exp \left\{ i \int d^4x \left(-\frac{1}{2\xi} (\partial^\mu A_\mu^a)^2 \right) \right\} \propto \int \mathcal{D}B \exp \left\{ i \int d^4x \left(\frac{\xi}{2} (B^a)^2 + B^a (\partial^\mu A_\mu^a) \right) \right\}, \quad (3.188)$$

where B^a is an auxiliary field, which means this field is **non-dynamical**, because it doesn't contribute any term including $\partial_\mu B^a$. So the gauge fixing Lagrangian becomes:

$$\mathcal{L}_{\text{gf}} = \frac{\xi}{2}(B^a)^2 + B^a(\partial^\mu A_\mu^a). \quad (3.189)$$

We can check this Lagrangian gives the same form with the Gaussian integration by implying the Euler-Lagrange equation:

$$\frac{\delta \mathcal{L}_{\text{gauge fix}}}{\delta B^a} = \xi B^a + \partial^\mu A_\mu^a = 0, \quad (3.190)$$

$$\implies B^a = -\frac{1}{\xi} \partial^\mu A_\mu^a. \quad (3.191)$$

So the Faddeev-Popov Lagrangian is

$$\mathcal{L}_{\text{FP}} = \bar{\psi}(i\not{D} - m)\psi - \frac{1}{4}F_{\mu\nu}^a F^{\mu\nu a} + \frac{\xi}{2}(B^a)^2 + B^a(\partial^\mu A_\mu^a) + \bar{c}^a(-\partial^\mu D_\mu^{ab})c^b. \quad (3.192)$$

We have already derived variation of A from (3.126):

$$\delta A_\mu^a = D_\mu^{ab} \alpha^b(x), \quad (3.193)$$

where we absorbed the factor $1/g$ into the arbitrary function $\alpha^b(x) \rightarrow g\alpha^b(x)$. But we now want $\alpha^b(x)$ can related to the ghost field:

$$\alpha^b(x) = g\varepsilon c^b(x), \quad (3.194)$$

where ε is a constant that non-related with the space-time. So if we also do the same thing for the other fields' infinitesimal transformations, how will the Lagrangian change? This leads us to the **BRST transformations**. BRST claimed that there is a global symmetry from ε that the transformation keep the Lagrangian invariant

$$\delta_\varepsilon \mathcal{L} = 0. \quad (3.195)$$

And the infinitesimal parameter must be Grassmann-valued:

$$\varepsilon^2 = 0, \quad (3.196)$$

because A_μ^a is a bosonic field and c^b is a fermionic field, so ε must be a fermionic field, which fulfills the Grassmann algebra.

$$\varepsilon c^b = -c^b \varepsilon. \quad (3.197)$$

Thus the BRST transformation for A_μ^a and ψ_j can be derived by implying (3.194) into the normal infinitesimal transformation $\delta A_\mu^a = D_\mu^{ab} \alpha^b(x)$ and $\delta_\varepsilon \psi_j = i \sum_a \alpha^a (t^a \psi)_j$:

$$\delta_\varepsilon A_\mu^a = \varepsilon D_\mu^{ab} c^b = \varepsilon \left(\partial_\mu c^a + g \sum_{b,c} f^{abc} A_\mu^b c^c \right), \quad (3.198)$$

$$\delta_\varepsilon \psi_j = ig\varepsilon \sum_a c^a (t^a \psi)_j. \quad (3.199)$$

These two transformations obviously keep the $\mathcal{L}_{\text{YM}}$ invariant, because it just changes a name of α , and since αc^a is still a local gauge transformation, then $\mathcal{L}_{\text{YM}}$ is automatically invariant under this transformation. One can check that $F_{\mu\nu}^a$ follows

$$\delta_\varepsilon F_{\mu\nu}^a = g\varepsilon f^{abc} F_{\mu\nu}^b c^c, \quad (3.200)$$

thus

$$\delta_\varepsilon \left(-\frac{1}{4} F_{\mu\nu}^a F^{\mu\nu a} \right) = 0. \quad (3.201)$$

And for the fermionic field, the covariant derivative follows:

$$\delta_\varepsilon (D_\mu \psi) = ig\varepsilon c^a t^a (\partial_\mu \psi - igA_\mu^b t^b \psi) = ig\varepsilon c^a t^a (D_\mu \psi), \quad (3.202)$$

thus

$$\delta_\varepsilon (\bar{\psi} (i\gamma^\mu D_\mu - m) \psi) = (\delta_\varepsilon \bar{\psi}) (i\gamma^\mu D_\mu - m) \psi + \bar{\psi} (i\gamma^\mu \delta_\varepsilon (D_\mu \psi) - m \delta_\varepsilon \psi) = 0. \quad (3.203)$$

But it's worthy to mention that **BRST transformation is not a local gauge transformation (transformation parameter ε is global)**. But it indeed **comes from a local gauge transformation (transformation parameter $\alpha^a(x) = g\varepsilon c^a(x)$ is local)**.

Now it comes to the most tedious part, the gauge fixing term and the ghost term. This is because these two terms are not gauge invariant. For convenient, we can just check the following variation of the ghost field and auxiliary field:

$$\delta_\varepsilon c^a = -\frac{1}{2} g\varepsilon \sum_{b,c} f^{abc} c^b c^c \quad (3.204)$$

$$\delta_\varepsilon \bar{c}^a = \varepsilon B^a \quad (3.205)$$

$$\delta_\varepsilon B^a = 0, \quad (3.206)$$

keep the Lagrangian of gauge fixing and ghost invariant:

$$\delta_\varepsilon (\mathcal{L}_{\text{gf}} + \mathcal{L}_{\text{ghost}}) = 0, \quad (3.207)$$

namely to prove:

$$\delta_\varepsilon \left(\frac{\xi}{2} (B^a)^2 \right) + \underbrace{(\delta_\varepsilon B^a)}_{=0} (\partial^\mu A_\mu^a) + B^a \underbrace{\delta_\varepsilon (\partial^\mu A_\mu^a)}_{=\partial^\mu (\varepsilon D_\mu^{ab} c^b)} - \underbrace{(\delta_\varepsilon \bar{c}^a)}_{\varepsilon B^a} (\partial^\mu D_\mu^{ab}) c^b + \underbrace{\bar{c}^a \delta_\varepsilon (\partial^\mu D_\mu^{ab}) c^b + \bar{c}^a (\partial^\mu D_\mu^{ab}) (\delta_\varepsilon c^b)}_{\bar{c}^a \partial^\mu \delta_\varepsilon (D_\mu^{ab} c^b)} = 0. \quad (3.208)$$

The first term is simply 0 because

$$\delta_\varepsilon (B^a)^2 = 2\delta_\varepsilon B^a = 0 \quad (3.209)$$

And for the third term

$$\partial^\mu (\varepsilon D_\mu^{ab} c^b) = \underbrace{(\partial^\mu \varepsilon)}_{=0} D_\mu^{ab} c^b + \varepsilon \partial^\mu (D_\mu^{ab} c^b) = \varepsilon \partial^\mu (D_\mu^{ab} c^b), \quad (3.210)$$

due to ε is a constant. Now the expression to be proved reduces to

$$\cancel{B^a \varepsilon \partial^\mu D_\mu^{ab} c^b} - \varepsilon B^a \partial^\mu D_\mu^{ab} c^b - \bar{c}^a \partial^\mu (\delta_\varepsilon D_\mu^{ab} c^b) \quad (3.211)$$

$$= -\bar{c}^a \partial^\mu \left\{ \partial_\mu (\delta_\varepsilon c^a) + g \sum_{b,c} f^{abc} (\delta_\varepsilon A_\mu^b) c^c + g \sum_{b,c} f^{abc} A_\mu^b (\delta_\varepsilon c^c) \right\} \quad (3.212)$$

$$= -\bar{c}^a \partial^\mu \left\{ \partial_\mu \left(-\frac{g}{2} \varepsilon \sum_{b,c} f^{abc} c^b c^c \right) + g \sum_{b,c} f^{abc} \left[\varepsilon \left(\partial_\mu c^b + g \sum_{d,e} f^{bde} A_\mu^d c^e \right) \right] c^c \right. \\ \left. + g \sum_{b,c} f^{abc} A_\mu^b \left(-\frac{g}{2} \varepsilon \sum_{d,e} f^{cde} c^d c^e \right) \right\} \quad (3.213)$$

$$= -\bar{c}^a \partial^\mu \left\{ g \varepsilon \sum_{b,c} f^{abc} \left[-\frac{1}{2} (\partial_\mu c^b c^c) + (\partial_\mu c^b) c^c \right] + g^2 \varepsilon \sum_{b,c,e,f} f^{abc} \left(f^{bde} A_\mu^d c^e c^c - \frac{1}{2} f^{cde} A_\mu^b c^d c^e \right) \right\} \quad (3.214)$$

The first term will be canceled by using the property of Grassmann number $c^a c^b = -c^b c^a$ and structure constant $f^{abc} = -f^{acb}$:

$$\sum_{b,c} f^{abc} \partial_\mu (c^b c^c) = \sum_{b,c} f^{abc} [(\partial_\mu c^b) c^c + c^b (\partial_\mu c^c)] \quad (3.215)$$

$$= \sum_{b,c} [f^{abc} (\partial_\mu c^b) c^c + f^{acb} c^c (\partial_\mu c^b)] \quad (3.216)$$

$$= \sum_{b,c} [f^{abc} (\partial_\mu c^b) c^c - f^{abc} c^c (\partial_\mu c^b)] \quad (3.217)$$

$$= \sum_{b,c} [f^{abc} (\partial_\mu c^b) c^c + f^{abc} (\partial_\mu c^b) c^c] \quad (3.218)$$

$$= 2 \sum_{b,c} f^{abc} (\partial_\mu c^b) c^c. \quad (3.219)$$

Now the only left term is

$$-\bar{c}^a \partial^\mu g^2 \varepsilon \sum_{b,c,e,f} f^{abc} \left(f^{bde} A_\mu^d c^e c^c - \frac{1}{2} f^{cde} A_\mu^b c^d c^e \right), \quad (3.220)$$

which can be canceled by Jacobi identity

$$f^{ade} f^{bcd} + f^{bde} f^{cad} + f^{cde} f^{abd} = 0. \quad (3.221)$$

For that, we first relabel the indices d, e in the first term:

$$\sum_{b,c,d,e} f^{abc} f^{bde} c^e c^c = \frac{1}{2} \sum_{b,c,d,e} (f^{abc} f^{bde} - f^{abe} f^{bdc}) c^e c^c. \quad (3.222)$$

Thus,

$$\sum_{b,c,d,e} f^{abc} \left(f^{bde} A_\mu^d c^e c^c - \frac{1}{2} f^{cde} A_\mu^b c^d c^e \right) \quad (3.223)$$

$$= \sum_{b,c,d,e} f^{abc} f^{bde} A_\mu^d c^e c^c - \frac{1}{2} f^{abc} f^{cde} A_\mu^b c^d c^e \quad (3.224)$$

$$= \frac{1}{2} \sum_{b,c,d,e} (f^{abc} f^{bde} - f^{abe} f^{bdc}) A_\mu^d c^e c^c - f^{abc} f^{cde} A_\mu^b c^d c^e \quad (3.225)$$

$$= \frac{1}{2} \sum_{b,c,d,e} (f^{ace} f^{cbd} - f^{acd} f^{cbe} - f^{abc} f^{cde}) A_\mu^b c^d c^e \quad (3.226)$$

$$= \frac{1}{2} \sum_{b,c,d,e} (-f^{dbc} f^{ace} + f^{ebc} f^{acd} - f^{abc} f^{cde}) A_\mu^b c^d c^e \quad (3.227)$$

$$= \frac{1}{2} \sum_{b,c,d,e} (f^{dbc} f^{aec} - f^{ebc} f^{adc} - f^{abc} f^{dec}) A_\mu^b c^d c^e \quad (3.228)$$

$$= \frac{1}{2} \sum_{b,c,d,e} (f^{dbc} f^{aec} + f^{ebc} f^{dac} + f^{abc} f^{edc}) A_\mu^b c^d c^e \quad (3.229)$$

$$= 0. \quad (3.230)$$

Finally, with such tedious calculation, we have successfully proved (3.207) being satisfied, which means the gauge fixing term and ghost term together are BRST invariant. Thus, the Faddeev-Popov Lagrangian is BRST invariant:

$$\delta_\epsilon \mathcal{L}_{\text{FP}} = \delta_\epsilon \mathcal{L}_{\text{YM}} + \delta_\epsilon (\mathcal{L}_{\text{gf}} + \mathcal{L}_{\text{ghost}}) = 0. \quad (3.231)$$

3.3 Factorization of Hilbert space via BRST operator

Now since BRST symmetry is a global symmetry, we can in principle define a conservation charge Q , which should upgrade to an operator in Hilbert space since we have already quantized the classical field. This operator thus called the **BRST operator** Q , which can be easily defined from the infinitesimal transformation:

$$\delta_\epsilon \Phi = [\epsilon Q, \Phi] = \epsilon(Q\Phi); \quad \text{for bosonic field } \Phi \quad (3.232)$$

$$\delta_\epsilon \Psi = \{\epsilon Q, \Psi\} = -\epsilon(Q\Psi); \quad \text{for fermionic field } \Psi. \quad (3.233)$$

Thus Q has similar properties with δ_ϵ , which can be directly copy from (3.198),(3.199),(3.204),(3.205),(3.206)

$$QA_\mu^a = D_\mu^{ab} c^b \quad (3.234)$$

$$Q\psi_j = igc^a(t^a\psi_j) \quad (3.235)$$

$$Qc^a = -\frac{1}{2}gf^{abc}c^b c^c \quad (3.236)$$

$$Q\bar{c}^a = B^a \quad (3.237)$$

$$QB^a = 0, \quad (3.238)$$

and (3.196) shows that Q satisfies **Nilpotency**:

$$Q^2 = 0. \quad (3.239)$$

With BRST operator, we can rewrite the Faddeev-Popov Lagrangian as

$$\mathcal{L}_{\text{FP}} = \mathcal{L}_{\text{YM}} + Q \underbrace{\left(\bar{c}^a \partial^\mu A_\mu^a + \frac{\xi}{2} \bar{c}^a B^a \right)}_{\text{BRST ancestors}}. \quad (3.240)$$

With this form, we can easily see that it's BRST invariant by using (3.239) and $\delta_\epsilon \mathcal{L}_{\text{YM}} = 0$:

$$Q\mathcal{L}_{\text{FP}} = 0, \quad Q\mathcal{L}_{\text{YM}} = 0 \quad (3.241)$$

Now we can use BRST operator to show how those “nonphysical” states are separated. Since BRST operator as a conserved “charge”. It naturally commutes with Hamiltonian operator H :

$$[Q, H] = 0, \quad Q = Q^\dagger. \quad (3.242)$$

The Hilbert space of eigenstates of H in Heisenberg picture can be formally expressed as

$$\mathcal{H} = \mathcal{H}_1 \oplus \mathcal{H}_2 \oplus \mathcal{H}_0, \quad (3.243)$$

where the subspace can be defined by the map:

$$Q : \mathcal{H} \rightarrow \mathcal{H}. \quad (3.244)$$

The kernel of Q , i.e., those states are mapped to 0, is

$$\text{Ker}(Q) = \{|\psi\rangle \in \mathcal{H} \mid Q|\psi\rangle = 0\}, \quad (3.245)$$

and the image is

$$\text{Im}(Q) = \{|\psi\rangle \in \mathcal{H} \mid |\psi\rangle = Q|\phi\rangle, \text{ for } |\phi\rangle \in \mathcal{H}, \}. \quad (3.246)$$

And since $Q^2 = 0$, so for any states in $\text{Im}(Q)$ are all in $\text{Ker}(Q)$ too, i.e.,

$$\text{Im}(Q) \subset \text{Ker}(Q). \quad (3.247)$$

Thus, we define the three states as follows:

$$\mathcal{H}_1 = \mathcal{H} \setminus \text{Ker}(Q) = \{|\psi_1\rangle \in \mathcal{H} \mid Q|\psi_1\rangle \neq 0\}, \quad (3.248)$$

$$\mathcal{H}_2 = \text{Im}(Q) = \{|\psi_2\rangle \in \mathcal{H} \mid |\psi_2\rangle = Q|\phi\rangle, |\phi\rangle \in \mathcal{H}_1\}, \quad (3.249)$$

$$\mathcal{H}_0 = \text{Ker}(Q) \setminus \text{Im}(Q) = \{|\psi_0\rangle \in \mathcal{H} \mid Q|\psi_0\rangle = 0 \text{ and } \forall |\phi\rangle \in \mathcal{H}, |\psi\rangle \neq Q|\phi\rangle\}. \quad (3.250)$$

With these definition, we can clearly see that Hilbert space is factorized into the direct sum over three subspace :

$$\mathcal{H} = \underbrace{(\mathcal{H} \setminus \text{Ker}(Q))}_{\mathcal{H}_1} \oplus \underbrace{\text{Im}(Q)}_{\mathcal{H}_2} \oplus \underbrace{(\text{Ker}(Q) \setminus \text{Im}(Q))}_{\mathcal{H}_0}. \quad (3.251)$$

With this, we claim the set of all physical states is the kernel of $\mathcal{H}$ mode the image:

$$\mathcal{H}_{\text{phys}} \cong \mathcal{H}_0 = \text{coh}(Q) = \frac{\text{Ker}(Q)}{\text{Im}(Q)}, \quad (3.252)$$

which is a **cohomology space** satisfying:

$$Q|\psi\rangle = 0. \quad (3.253)$$

To show that it indeed make sense, we need to go back to the free theory, namely $g \rightarrow 0$, so the BRST transformation reduce to

$$QA_\mu^a = \partial_\mu c^a, \quad (3.254)$$

$$Q\bar{c}^a = B^a, \quad (3.255)$$

$$Q\psi_j = Qc^a = QB^a = 0. \quad (3.256)$$

We find this result with the definition of $\mathcal{H}_1$, $\mathcal{H}_2$ and $\mathcal{H}_0$ can help us to classify these asymptotic states:

$$A_\mu^L \in \mathcal{H}_1, \quad \bar{c}^a \in \mathcal{H}_1; \quad (3.257)$$

$$A_\mu^T \in \mathcal{H}_0, \quad \psi_j \in \mathcal{H}_0; \quad (3.258)$$

$$c^a \in \mathcal{H}_2, \quad B^a \in \mathcal{H}_2. \quad (3.259)$$

where A_μ^L represent the longitudinal polarization

$$A_\mu^L \sim k_\mu \phi \implies Q|A_\mu^L\rangle \sim |c^a\rangle \neq 0, \quad (3.260)$$

and A_μ^T represent the transversal polarization

$$k^\mu A_\mu^T = 0 \implies QA_\mu^T \sim k_\mu c^a = 0. \quad (3.261)$$

With this classification, we can see that for the states in $\mathcal{H}_1$, like A_μ^L or $\bar{c}^a$, the norm of time component are **negative**:

$$\langle A_0^L | A_0^L \rangle = \langle 0 | a_0 a_0^\dagger | 0 \rangle = \langle 0 | [a_0, a_0^\dagger] | 0 \rangle = -1 < 0, \quad (3.262)$$

$$\langle \bar{\eta} | \bar{\eta} \rangle = \langle 0 | \bar{c} \bar{c}^\dagger | 0 \rangle = \langle 0 | (-\bar{c}^\dagger)^\dagger \bar{c}^\dagger | 0 \rangle = -\langle 0 | \bar{c} \bar{c}^\dagger | 0 \rangle = \langle 0 | \{c, \bar{c}^\dagger\} | 0 \rangle = -1 < 0. \quad (3.263)$$

For the states in $\mathcal{H}_2$ are **zero norm** due to the nilpotency $Q^2 = 0$:

$$\langle \psi_2 | \psi_2 \rangle = \langle \psi_1 | Q^\dagger Q | \psi_1 \rangle = \langle \psi_1 | Q^2 | \psi_1 \rangle \equiv 0. \quad (3.264)$$

And the states in $\mathcal{H}_0$ are **positive norm**, which is the real physical state space. So we say, physical degrees of freedom are elements of $\mathcal{H}_0$, and unphysical degrees of freedom are in $\mathcal{H}_1, \mathcal{H}_2$.

This result however, one may wondering if it is picture dependent since we assume the Heisenburg picture that the states doesn't evolve with time. The answer is lucky no, picture does not change under time evolution: for any state $|\psi_0, t\rangle$ in $\mathcal{H}$, we have

$$Q|\psi_0, t\rangle = Qe^{iHt}|\psi_0\rangle = e^{iHt}Q|\psi_0\rangle = 0, \quad (3.265)$$

which means $|\psi_0, t\rangle$ must be a linear combination of states from $\mathcal{H}_0$ and $\mathcal{H}_2$, i.e., $|\psi_0, t\rangle \in \mathcal{H}_0 \oplus \mathcal{H}_2$. But if we consider a scattering process from $|\psi_0, t\rangle$ to any $|\psi_2\rangle \in \mathcal{H}_2$ that satisfies $|\psi_2\rangle = Q|\psi_1\rangle$, we have

$$\langle \psi_2 | \psi_0, t \rangle = \langle \psi_1 | Q | \psi_0, t \rangle = 0, \quad (3.266)$$

which means the scalar product vanishes! Namely, $|\psi_0, t\rangle$ has no effective component in $\mathcal{H}_2$. So even though we mix some element of $\mathcal{H}_2$ in the combination of the construction of $|\psi_0, t\rangle$, it doesn't matter because the elements in $\mathcal{H}_2$ won't contribute to the final result.

Thus, we find that **BRST operator factorize the space based on positive norm, negative norm and zero norm**. which helps us to select and drop the negative and zero norm states.